MINDCRAFT

Research Constitution

How do humans become confident mathematical thinkers?

Identity transformation · Evidence · Red Team · Product systems · Experiments

This is not a pitch deck. This is the company's research operating system.
Challenge assumptions. Label uncertainty. Kill weak arguments.
Source corpus: ~48,762 words · multi-chapter living lab

MindCraft Research Lab · ongoing evidence program

MindCraft Research Constitution v1

Status: Living operating document — not a pitch deck

Edition: v1.6 (Synthesizer consolidation of Parts XXXIII–XL into surviving doctrine)

Research question: How do humans become *confident mathematical thinkers*?

Product thesis under audit: The product is identity transformation, not mathematics delivery.

Last updated: 2026-07-30 (Researcher tick — Part XLVII mounted)

Growth model: Core OS (this file) + `chapters/*.md` via `CHAPTER_MANIFEST.txt` → PDF

Scale intent: Multi-month densification toward 150–300 pages of *evidenced* material — never fluff

Epistemic rule: Every claim is labeled FACT / HYPOTHESIS / FOUNDER BELIEF / SPECULATION.

Deep-dive chapters currently mounted

Part	File	Focus
XXIV	`chapters/24_affect_anxiety_belonging.md`	Anxiety, WM, stereotype threat, belonging
XXV	`chapters/25_self_efficacy_narrative_identity.md`	Bandura sources, narrative identity, habit≠identity
XXVI	`chapters/26_cognitive_load_mastery_tutoring.md`	CLT, fading, mastery war, tutoring calibration
XXVII	`chapters/27_markets_parents_competition.md`	WTP, parents, competitive teardown
XXVIII	`chapters/28_history_meaning_math.md`	Invention-story thesis under trial
XXIX	`chapters/29_spacing_retrieval.md`	Spacing, retrieval, memory of competence
XXX	`chapters/30_attribution_helplessness.md`	Weiner attributions, helplessness, feedback language
XXXI	`chapters/31_flow_challenge_skill.md`	Flow, challenge–skill, difficulty design
XXXII	`chapters/32_parent_anxiety_transmission.md`	Maloney/Beilock parent anxiety pathway
XXXIII	`chapters/33_ai_tutors_trust_sycophancy.md`	AI trust / sycophancy; RT: Bastani■PWC, guards≠gain
XXXIV	`chapters/34_formal_causal_dag_identification.md`	FEI causal DAG; L0–L4 claim ladder; confounders
XXXV	`chapters/35_competitive_session_audits.md`	Khan/Duo/Brilliant/ChatGPT session mechanism audits
XXXVI	`chapters/36_equity_audit_story_worlds.md`	Equity audit of story worlds; whose history; tokenism kill

XXXVII	`chapters/37_expectancy_value_eccles.md`	Eccles SEVT; utility/cost/attainment → choice; EVT experiments
XXXVIII	`chapters/38_goal_orientation.md`	Mastery vs performance goal structures; appearance ban; GO experiments
XXXIX	`chapters/39_interleaving_vs_blocking.md`	Interleave vs block; strategy selection; IL experiments
XL	`chapters/40_self_explanation_prompts.md`	Self-explanation prompts; Chi/Renkl → coach UX; SE experiments
XLI	`chapters/41_desirable_difficulties_x_anxiety.md`	Desirable difficulties × anxiety; SAFE-DD; Bjork vs Ashcraft
XLII	`chapters/42_social_comparison_leaderboards.md`	Social comparison & leaderboards; SAFE-COMPARE; ranks help/hurt
XLIII	`chapters/43_habit_formation_science.md`	Habit formation (Wood/Clear); SAFE-HABIT; cue≠colonization
XLIV	`chapters/44_intrinsic_motivation_killers.md`	Intrinsic motivation killers (Deci); SAFE-REWARD; controlling≠informational
XLV	`chapters/45_mathematical_resilience.md`	Mathematical resilience (JW/Lee); SAFE-RESILIENCE; grit theater kill
XLVI	`chapters/46_teacher_tutor_expectancy.md`	Tutor expectancy (Pygmalion/Golem); SAFE-EXPECTANCY; trait-label ban
XLVII	`chapters/47_sleep_stress_learning.md`	Sleep/stress/learning; SAFE-TIMING; all-nighter & sleep-app kill

Queued next: see `NEXT_LAB.md` (Part XLVIII transfer/variation, then QUEUE_EXTENDED).

Synthesizer note (v1.6): Eight researcher chapters (XXXIII–XL) landed without a merge pass. This edition collapses duplicate "motivation / mastery / AI tutor" talk into one commercial doctrine stack (I.4), upgrades Red Team kills (XIV), densifies Experiment A/D into chapter experiment families (IX), and refreshes competitive / metrics language. Deep-dive files remain authoritative for citations; the OS keeps only *surviving* product rules.

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0. How to read this document

This Constitution is the research OS for MindCraft. It exists to prevent the company from building features that feel right and fail quietly.

Rules of engagement:

1. Founder beliefs enter as **FOUNDER BELIEF**, not as law.
2. The Red Team's job is destruction. Unchallenged claims are invalid.
3. Page count is not quality. A shorter true section beats a longer vague one.
4. "AI will make explanations free" is a hypothesis about markets, not a moral slogan.
5. Product implications must map to experiments with falsifiers.

This v1 is dense and incomplete on purpose. Expanding toward 150–300 pages is allowed only when new evidence, frameworks, or contradicted priors justify the ink.

Part I — Executive Summary

I.1 The compressed answer

HYPOTHESIS (confidence: medium–high): Humans become confident mathematical thinkers when three conditions compound:

1. **Affective permission** — the nervous system stops treating math as social/physical threat.
2. **Competence evidence** — the learner accumulates undeniable, self-attributed successes at the right grain of difficulty — including *strategy selection* and *self-accounted why*, not only blocked accuracy.
3. **Identity re-storying** — the person revises the narrative "I am not a math person" into a workable self-story compatible with struggle — via expectancy × value × cost (Eccles), not posters.

Explanations alone rarely produce (1)–(3). Fluent AI explanations can *accelerate wrong confidence* (Bastani unguarded GPT; Part XXXIII). Explanations help (2) only *after* (1) and when the student still does the decisive generative work (Parts XXVI, XL).

I.2 Verdict on the scarcity thesis

FOUNDER BELIEF under audit: AI commoditizes knowledge; scarcity shifts to motivation, confidence, identity, trust, curiosity, persistence, emotional safety, meaning.

Red Team verdict (updated v1.6):

Claim	Status	Notes
Explanations are getting cheaper/faster	**FACT** (directionally)	Generative AI + open content lowers marginal cost of explanation

"Tutoring is free"	**FALSE as stated**	Attention, accountability, diagnosis, calibrated trust remain scarce
Content is free	**MOSTLY FACT** for commodity content	Differentiation = curation, sequencing, trust, outcomes
Motivation/identity become scarce	**HYPOTHESIS**	Product differentiator; not proven sole moat — instrument E×V×Cost separately (XXXVII)
Emotional safety is first-order	**HYPOTHESIS with supportive evidence**	Ashcraft WM tax; segment, don't romanticize softness
Unguarded AI chat improves solo learning	**KILLED**	Bastani: concurrent ↑ then solo exam ↓; guards neutralize harm ≠ prove gain
Bastani "GPT Tutor" ≡ MindCraft ontology wrap	**KILLED**	Category error (XXXIII RT)
Streaks / blocked fluency = mastery	**KILLED as North Star**	HID + XXXVIII/XXXIX; measure delayed mix + transfer

Implication: Do not bet on "better explanations" or "AI tutor warmth." Bet on **FEI + pedagogy wrap**: fear→evidence→identity, with AI as guarded infrastructure and student generation as the spine.

I.3 What to optimize (North Star debate)

Metric	Predicts short-term engagement?	Predicts durable math identity?	Risk
Correct answers (esp. blocked)	Medium	Low–medium	Fluency illusion; gaming
Time spent / chat tokens	High	Low	Engagement / AI-monologue theater
Streaks / DAU	High	Uncertain	Habit ≠ identity (HID)
Confidence (self-report)	Medium	Medium	Cheap talk; overtrust after fluent AI
Challenge-seeking (`challenge_accept` + *why*)	Medium	**High**	Must code mastery vs appearance/normative motive (XXXVIII)
Voluntary return after failure (`retry_120s`)	Medium	**High**	Needs coach content discipline (SE prompts)
Delayed mixed / `transfer_pass` / `solo_transfer_pass`	Medium	**High**	Anti-fake-mastery; anti-AI-crutch
"I am a math person" + behavior	Medium	**Highest target**	Slow, noisy; L3+ claim only

Working North Star (HYPOTHESIS):

Challenge-seeking under safety with transfer — the student chooses a harder problem for a *mastery* reason, returns after a miss, can pick the strategy on a mixed delayed set, and attributes success to strategy — without needing the model to finish the thought.

I.4 Surviving doctrine stack (Synthesizer merge of XXXIII–XL)

Duplicate frameworks collapsed. Deep dives own citations; this table is **company law until killed**.

Layer	Surviving rule	Source parts	Commercial use
FEI / SAFE-CRAFT	Safety → attempt → informative miss → earned win → attribution → transfer	XVI, VIII	Session demo in marketing; not feature matrix
Claim ladder L0–L4	Do not sell identity (L3) from session A/Bs (L1)	XXXIV	Parent/copy hygiene; pre-reg estimands
PWC (pedagogy wrap constraint)	Guarded AI + ontology spine; success = `solo_transfer_pass` ≥ no-AI	XXXIII, XXXV	Ban "we're Bastani's Tutor"; ban unguarded Solver hero
E×V×Cost	Instrument expectancy, value, cost separately; student-authored utility > staff utility dump	XXXVII	Kill single "motivation" KPI

TARGET / mastery climate	Default-ban appearance UX + streak-as-North-Star; soft-wrong = mastery evaluation structure; **SAFE-COMPARE** (XLII): criterion default, peer opt-in, no forced infinite named ranks for Maya; **SAFE-HABIT** (XLIII): cue practice not counter, miss≠identity failure, habit metrics subordinate; **SAFE-REWARD** (XLIV): no expected tokens for mere engagement; feedback > inducement; transfer-gated unlocks; **SAFE-RESILIENCE** (XLV): growth-zone challenge + support recruitment; ban grit NS / danger-zone-as-DD / Resilience Score™; **SAFE-TIMING** (XLVII): encode→sleep→transfer claim; never reward midnight volume; stress/restriction-aware load; pressure≠DD; exam-week retrieval mode; not a sleep clinic	XXXVIII / XLII / XLIII / XLIV / XLV / XLVII	CTA copy audit; leaderboard kill; streak colonization kill; XP-as-love kill; grit-poster kill; all-nighter kill; SC-1…4 / HAB-1…4 / IM-1…4 / RES-1…4 / TIM-1…4
Block → near-miss interleave → spaced mix	Blocking = acquisition scaffold; delayed mixed accuracy = readiness signal	XXIX, XXXIX	Ban “shuffle = science”; ban blocked-accuracy vanity
Student-generated why	Faded examples + structured principle/misconception prompts *before* AI wrap	XXVI, XL	Ban AI-monologue≡SE; ban explain-own-wrong-first default

Competitive wedge	Do not out-content Khan, out-streak Duo, out-delight Brilliant, or out-fluency ChatGPT	XXXV, XX	Sell recoverable struggle + competence evidence + solo transfer
Equity of worlds	Story wrap is identity technology; tokenism and stereotype-cueing copy are kills	XXXVI, XXVIII	HIST-EQ; belonging without "even you can"

Merged / demoted (do not treat as separate products): "AI tutor," "mastery path," "growth mindset," and "engagement" are *not* independent North Stars — they are subordinate UX under FEI + the rows above.

Next research bottleneck (partially addressed): Part XLI landed **SAFE-DD** (equip → destake → dose → frame → measure). Still needs DD-1/DD-2 data before anxiety-segment interleave/SE claims go above L1.

Part II — Key Insights (challenged)

Insight 1 — Fear is a performance bottleneck, not a personality trait

FACT: Math anxiety is associated with reduced working-memory availability during math tasks; anxious learners underperform relative to ability (Ashcraft & Kirk line of work; broader anxiety–WM literature).

HYPOTHESIS: For a large subset of MindCraft's Maya archetype, the binding constraint is not "missing explanation," it is **threat appraisal** ("I will look stupid").

Implication: Soft-wrong UX, hide-correctness diagnostics, wizard coaching, and narrative framing are not polish — they are load-bearing if the thesis holds.

Red Team: Many high performers succeed under stress. Anxiety interventions have heterogeneous effects. Do not medicalize ordinary difficulty.

Insight 2 — Growth mindset is real, small, and context-dependent

FACT: Yeager et al. (2019, *Nature*) — brief online growth-mindset intervention improved grades for lower-achieving students and increased advanced math enrollment in a nationally representative U.S. sample; effects stronger when peer norms supported challenge-seeking.

FACT / REVIEW: Domain-general mindset messaging shows mixed results; math-*specific* mindset interventions more often report positive outcomes (2023 systematic review in *Educational Research Review*).

FACT / CAUTION: "False growth mindset" (praise effort without teaching strategy / changing environment) is harmful (Dweck's own caution; Kohn's critique).

Implication: MindCraft must not ship empty "believe in yourself" copy. Mindset must be embodied in **task design, feedback, and tutor norms**.

Insight 3 — Autonomy, competence, relatedness are the psychological tripod

FACT: Self-Determination Theory (Ryan & Deci) — sustained intrinsic motivation requires autonomy, competence, relatedness.

FACT (meta): SDT-based education interventions show reliable gains for autonomy and competence support; relatedness effects are less consistent (Wang et al. 2024 meta-analysis).

Implication: Product loops must deliver:

- Autonomy: meaningful choice (path, story world, pace)
- Competence: mastery grain that produces earned wins
- Relatedness: tutor presence, parent signal, peer-safe identity (hardest; do not fake with empty social)

Insight 4 — Mastery learning works — until you measure the wrong thing

FACT: Kulik, Kulik & Bangert-Drowns (1990) meta-analysis — mastery programs show positive exam effects (overall ~0.5 SD in their synthesis), stronger for weaker students; can increase time; self-paced college mastery can reduce completion.

FACT / CONTRADICTION: Slavin's "Mastery Learning Reconsidered" (1987) — little support for group-based mastery on *standardized* measures; experimenter-made tests show moderate effects; coverage vs mastery tradeoff is real.

Implication: MindCraft's mastery graph is promising **if** mastery means transferable competence, not local quiz familiarity. Measure far transfer, not only same-item success.

Insight 5 — Habit products ≠ identity products

FACT / INDUSTRY: Duolingo's public method emphasizes gamification for motivation and return (streaks, leagues, variable rewards) (Duolingo Method whitepaper, 2023). Streaks exploit loss aversion; they drive DAU.

HYPOTHESIS: Streaks can increase practice frequency without changing "I am a math person." Habit without identity yields brittle engagement (stop streak → stop learning).

Implication: Use habit mechanics as **on-ramps**, not as the destination. Pair with identity-marking rituals (map fill, story progression, tutor recognition of growth).

Insight 6 — Narrative is not decoration; it is encoding + meaning

FOUNDER BELIEF (WORLD_VISION): Math stripped of invention-story becomes alienating machinery.

HYPOTHESIS: Historical/human invention frames increase curiosity and reduce "arbitrary rule" appraisals, improving persistence.

Evidence status: Strong tradition in math education (history of math pedagogy); fewer large RCTs than mindset literature. Treat as **design prior + research program**, not settled science.

Insight 7 — AI commoditizes *answers*, not *becoming*

HYPOTHESIS: As answer generation approaches free, willingness-to-pay migrates to:

1. Diagnosis of the *actual* break (gap map)
2. Emotional containment during struggle
3. Accountability relationships (tutor)
4. Identity-consistent practice worlds
5. Trusted outcome signaling to parents

SPECULATION: In 30 years, “show me the steps” is ambient. “Help me become the kind of person who stays with hard problems” remains a human+system problem.

Part III — Research Log (v1 seed)

Date	Finding	Type	Action
2026-07-29	Synthesizer v1.6: merge XXXIII–XL into I.4 doctrine stack	Synthesis	Surviving commercial law; next = XLI DD×anxiety
2026-07-29	SE prompts: AI monologue ■ SE; structure before wrap	Evidence	SE-1…4; densify Exp A/D
2026-07-29	Interleaving: block→near-miss→spaced mix; kill shuffle=science	Evidence	IL-1…4; ban blocked vanity
2026-07-29	Goal climate: appearance/streak NS killed; TARGET audit	Evidence	GO-1…4
2026-07-29	Eccles E×V×Cost: kill single motivation KPI	Evidence	EVT-1…4
2026-07-25	Constituted Research Lab + Constitution v1	Process	Create OS
2026-07-25	Scarcity thesis partially false as slogans; true as product wedge	Analysis	Rewrite pitch language
2026-07-25	Yeager 2019 constrains mindset claims	Evidence	Ban empty mindset copy
2026-07-25	Soft-wrong + wizard coach shipped in product	Product	Instrument coach → retry rate
2026-07-25	Mastery literature split (Kulik vs Slavin)	Evidence	Define mastery measurement carefully

2026-07-25	Bloom 2-sigma overstated vs VanLehn / modern tutoring metas	Evidence	Stop marketing 2-sigma
2026-07-25	Founder sequence wounded → SAFE-CRAFT	Synthesis	Attempt earlier; invention-story optional
2026-07-25	Deliberate practice necessary≠sufficient (Hambrick line)	Evidence	Progress-vs-self framing

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Part IV — Evidence Table (selected)

Claim	Best supporting evidence	Contradicting / limiting evidence	Confidence	MindCraft implication
Brief mindset intervention can move grades + advanced math taking	Yeager et al., Nature 2019	Effects heterogeneous; peer norms matter; replication debates in broader mindset meta-analyses	Medium–High	Mindset must be ecological, not banner text
Math-domain mindset > generic	2023 Ed Research Review systematic review	Study quality varies; publication bias possible	Medium	Math-specific identity language
SDT need support raises motivation	Wang et al. 2024 meta	Relatedness effects weaker	Medium–High	Design for autonomy + competence first
Math anxiety impairs performance	Cognitive literature (Ashcraft tradition)	Not all low performers are anxious	Medium–High	Affective onboarding is first-class
Mastery learning raises achievement	Kulik et al. 1990	Slavin 1987 on standardized tests	Medium	Mastery graph + transfer tests
Gamified habits raise return	Duolingo method / industry practice	Habit ≠ deep learning; extrinsic traps (Deci)	Medium	Streaks as on-ramp only

1:1 tutoring is uniquely powerful	Bloom "2 sigma" framing (1984)	Often overstated; quality varies; VanLehn etc. nuance tutoring effects	Medium	Keep human tutors; AI amplifies, not replaces

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Part V — Contradictory Evidence (do not skip)

1. **Mindset skepticism:** Some metas find small average effects; school implementations often degrade into posters.
2. **Grit / resilience hype:** Duckworth-style grit is contested; structural inequality and curriculum quality matter more than character slogans.
3. **Discovery learning failures:** Pure constructivism without guidance often fails (Kirschner, Sweller, Clark critiques). Narrative worlds must still teach.
4. **Engagement ≠ learning:** Time-on-app can rise while understanding falls (edtech's original sin).
5. **AI may *increase* explanation value temporarily:** Novices may trust fluent wrongness; trust becomes the scarce resource, not explanation volume.
6. **Some students want speed and drills:** Identity-through-story is not universal. Offer multiple vessels.

Part VI — Mental Models

VI.1 The Identity Transformation Cascade (ITC)

HYPOTHESIS — MindCraft original synthesis

```
Threat ↓ → Attention available → Micro-success → Attribution ("I figured it")
→ Willingness to re-enter challenge → Accumulated competence evidence
→ Narrative update ("maybe I can") → Identity claim ("I do math")
→ Environment selection (harder courses, peers, tutors)
→ Reinforcing loop
```

Balancing loop: Premature hard problems → threat ↑ → avoidance → identity freeze.

Product translation: Gap scan (no shame) → soft-wrong → tiny earned wins → map fill → tutor recognition → advanced path.

VI.2 The Three Doors (what students actually need in a stuck moment)

Door	Need	Wrong product response	Right response
A	Safety	"Here's a longer explanation"	Contain affect; normalize miss

B	Clarity	Pep talk	Precise model / worked example at right grain
C	Agency	Take over and solve for them	Scaffolded choice; student does the decisive step

FOUNDER BELIEF: Tutors often open Door B first. Many students are still behind Door A.

VI.3 Knowledge commodity stack (post-AI)

```
Layer 0: Facts & procedures → commoditizing fast
Layer 1: Explanations → commoditizing fast
Layer 2: Diagnosis of misconceptions → partially automatable; trust-sensitive
Layer 3: Adaptive sequencing → automatable with good ontology
Layer 4: Emotional co-regulation → scarce (human + careful UX)
Layer 5: Identity & meaning → scarce
Layer 6: Accountability relationship → scarce
```

MindCraft's durable stack lives in **Layers 2–6**, with 0–1 as utilities.

VI.4 Transfer from outside education

Domain	Mechanism	Transfer to MindCraft
Games (Celeste, FromSoftware-lite design)	Fair difficulty + death as information	Soft-wrong as physics, not moral failure
Sports coaching	Film study of *specific* error	Gap map + wrong-choice coach
Music	Scales before repertoire; recital as identity	Concept drills → story quest → public signal
Therapy (exposure)	Graded exposure to feared stimulus	Math anxiety: graded challenge under safety
Aviation checklists	Externalize working memory under stress	Procedure scaffolds when anxious
Martial arts belts	Visible competence ladder	Mastery pips / map regions
Language apps	Habit loops	Daily practice — without Duo's extrinsic ceiling
Religion / ritual	Shared meaning + belonging	Caution: do not cultify; use *light* ritual (map lighting)

Part VII — Universal Learning Principles (provisional)

1. **Emotion gates cognition** in threatened domains.

2. **Earned difficulty** beats random difficulty.
3. **Feedback should be information, not identity judgment.**
4. **Attribution matters:** strategy > talent for growth; luck attributions kill agency.
5. **Spaced retrieval** beats massed re-reading (cognitive science consensus).
6. **Worked examples** → **fading guidance** for novices (cognitive load).
7. **Belonging uncertainty** destroys persistence for stereotyped groups (Walton/Cohen tradition).
8. **Transfer requires varied practice**, not identical clones.
9. **Human accountability** multiplies AI tools.
10. **Identity is a lagging indicator**; design leading indicators (challenge-seeking, return-after-fail).

Part VIII — Product Implications (systems, not features)

VIII.1 Core system: Fear → Evidence → Identity (FEI Loop)

Reinforcing loop R1 (desired):

Safety UX → attempt → informative miss → coach → success → map fill → identity → more attempts.

Balancing loop B1 (threat):

Public shame / red buzz → avoidance → less practice → skill lag → more shame.

MindCraft already partially implements FEI: hide-correctness diagnostic, soft-wrong, wizard coach, story chapters, knowledge map. **Missing instrumentation:** does coach raise retry rate and later challenge-seeking?

VIII.2 What to stop optimizing

- Vanity DAU without learning transfer
- Explanation length as quality proxy
- Points that do not mark identity-relevant milestones

VIII.3 What to instrument next (minimum viable science)

1. **Retry rate within 2 minutes of soft-wrong** (with vs without wizard coach)
2. **Challenge-seeking:** % choosing harder level when offered
3. **Return after fail session** (D1 retention after a loss session)
4. **Self-identity item** (pre/post, 1–2 items, math-specific)
5. **Advanced course intent / enrollment** (slow, gold)

VIII.4 Tutor system role

Tutors are not content pipes. They are:

- Affect regulators
- Attribution coaches
- Bridge diagnosticians (concept A known, link A→B broken)
- Identity witnesses (“I saw you become someone who stays”)

AI should brief tutors, not replace witnessing.

Part IX — Experiments (pre-registered style)

Core A–D remain. Chapter ticks densified them into families — prefer the densified IDs when shipping.

Experiment A — Wizard coach on soft-wrong

- **Question:** Does the under-Graph wizard increase productive retry vs soft-wrong alone?
- **Design:** A/B, chapter + practice
- **Primary:** retry within 120s; secondary: eventual correct without hint binge
- **Falsifier:** no lift; or lift only in engagement with worse accuracy
- **Densified content (v1.6):** Coach turns must be SE-constrained (principle / misconception contrast), not pep talk — see SE-1…4 / SE-2 as operational pairs with Experiment D.

Experiment B — Story-first vs bare stem

- **Question:** Does invention-story framing raise persistence on first miss?
- **Design:** within-concept crossover
- **Primary:** time-to-abandon after first miss
- **Falsifier:** story slows without improving retention/transfer
- **Equity densify:** HIST-EQ / Part XXXVI — whose story, tokenism kill.

Experiment C — North Star validation

- **Question:** Which early metric best predicts 8-week challenge-seeking?
- **Candidates:** streak length, soft-wrong retry, identity item, map mastery Δ, delayed mix accuracy
- **Method:** observational + lagged prediction
- **v1.6 prior:** Expect streaks to lose; `retry_120s` + `transfer_pass` + mastery-motive `challenge_accept` to win.

Experiment D — Explanation timing

- **Question:** Is explanation-before-attempt worse than attempt-with-safety for anxious students?
- **Design:** screen with anxiety pretest; randomize explanation-first vs attempt-first
- **Falsifier:** explanation-first wins for all segments
- **Densified:** Prefer SE-2/SE-3 (contrastive / structured generation) over monologue-first.

Experiment families mounted in chapters (do not orphan)

Family	Identifies (max claim without follow-up)	Parts
AIT-*	Guarded AI → solo transfer / anti-sycophancy	XXXIII
DAG-*	Estimand hygiene; covariate vs mediator	XXXIV
CSA-*	Session preference vs competitors; paste under guards	XXXV
HIST-EQ / equity probes	Story belonging without stereotype cue	XXXVI
EVT-*	Expectancy / value / cost levers	XXXVII
GO-*	Mastery vs performance CTA climate	XXXVIII
IL-*	Schedule → discrimination / delayed mix	XXXIX
SE-*	Prompt regime → transfer / misconception recurrence	XL

Part X — Weekly Priorities (research × product)

1. Instrument FEI loop events in analytics (retry, coach shown, write-mode exit).
2. Run Experiment A for 2 weeks.
3. Red Team the WORLD_VISION narrative claims with one outside learning scientist.
4. Interview 10 Mayas: "When did math stop feeling dangerous?" (qual).
5. Define "mastery" operationally so Slavin's critique cannot gut the graph.

Part XI — Founder Questions (do not dodge)

1. If ChatGPT tutors become "good enough," what *relationship* do parents still buy?
2. Are we building for Maya's identity — or for parents' anxiety about scores? Both? Tradeoff?
3. Would we accept lower short-term accuracy for higher long-term challenge-seeking?
4. Is the story world load-bearing, or is gap+tutor enough?
5. What would falsify the identity-transformation thesis in 90 days?

Part XII — Long-term Roadmap (research horizons)

H1 (now): Prove FEI loop moves retry + challenge-seeking.

H2: Prove identity language + map produce durable self-concept change.

H3: Prove human tutor + AI diagnosis beats AI-alone on persistence and transfer.

H4: Generalize beyond math (science/identity) only after math identity mechanism is solid.

Part XIII — Open Problems

1. Causal path from narrative history-of-math to identity (under-identified).
2. Relatedness at scale without creepy social.
3. Measuring identity without demand effects.
4. Preventing gamification from colonizing meaning.
5. AI fluent nonsense vs trust calibration for teens.
6. Equity: does story-world privilege culturally specific narratives?

Part XIV — Red Team Dossier (destroy weak arguments)

Kill #1: "Explanations are free, therefore MindCraft should not explain"

Destroyed form: Absolutism.

Surviving form: Explanations are *necessary infrastructure* with collapsing *willingness-to-pay*; differentiation is diagnosis + affect + identity + accountability.

Kill #2: "Just fix mindset"

Destroyed: Poster mindset.

Surviving: Ecological mindset (task + feedback + peer/tutor norms), math-specific — TARGET structures (XXXVIII).

Kill #3: "Engagement is the goal"

Destroyed: Engagement without transfer.

Surviving: Engagement as oxygen for the FEI loop, not the fire.

Kill #4: "Mastery graph = Bloom 2-sigma"

Destroyed: Inflated tutoring mythology.

Surviving: Mastery *can* help weaker students if mastery is real (`transfer_pass`) and time costs are managed.

Kill #5: "Identity transformation is unmeasurable, so ship vibes"

Destroyed: Vibes-as-strategy.

Surviving: Leading indicators (retry, challenge-seeking, delayed mix) + lagging identity items; L0–L4 claim ladder (XXXIV).

Kill #6 (v1.6): "AI tutor / Bastani GPT Tutor is our product"

Destroyed: Unguarded chat-as-learning; category-error marketing; guards-as-gain.

Surviving: Pedagogy wrap + `solo_transfer_pass` ≥ no-AI bar (XXXIII RT).

Kill #7 (v1.6): "Blocked accuracy / streaks / shuffle prove learning"

Destroyed: Fluency illusion; streak North Star; random shuffle ≡ interleaving science.

Surviving: Short block → near-miss interleave → spaced mixed check (XXXIX); streaks as on-ramp only (HID).

Kill #8 (v1.6): "Any explain box / AI monologue = self-explanation science"

Destroyed: Untargeted justify-your-wrong-guess; chat length as depth.

Surviving: Structured principle/misconception SE on correct or known-incorrect content before AI wrap (XL).

Kill #9 (v1.6): "One motivation dial"

Destroyed: Collapsing expectancy and value; staff "math is useful" dumps as EVT.

Surviving: E×V×Cost triad with student-authored utility (XXXVII).

Part XV — References (verified starting set)

Do not invent citations. Expand this list only with retrieved sources.

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Part XVI — Mechanism Deep Dive: How identity actually changes

This part answers the research question at mechanism grain. It is synthesis, not settled science.

XVI.1 What “confident mathematical thinker” means (operational)

Do not use: “likes math apps” or “says math is fun once.”

Working definition (HYPOTHESIS): A confident mathematical thinker is a person who, in math-relevant contexts:

1. **Approaches** moderately hard problems without immediate avoidance.
2. **Persists** after a miss long enough to extract information.
3. **Attributes** success primarily to controllable causes (strategy, practice, help-seeking) rather than luck or tutor magic.
4. **Selects** into harder future opportunities when available (courses, contests, careers) at rates above their prior baseline.
5. **Endorses** a math-capable self-concept *and* behaves consistently with that endorsement.

Items 1–4 are leading. Item 5 is lagging. Product should optimize leading indicators while periodically measuring the lagging identity claim.

XVI.2 The threat–competence–narrative triad

Three literatures converge on the same transformation engine:

Mechanism	Core claim	Confidence	Key sources
Threat / affect	Math anxiety transiently reduces working-memory resources available for math	**High (FACT direction)**	Ashcraft & Kirk (2001); Suárez-Pellicioni et al. review (2016); WM mediation meta (2021)
Competence evidence	Motivation requires felt competence; mastery structures can raise achievement when mastery is real	**Medium–High**	SDT (Ryan & Deci); Kulik mastery meta; Slavin critique
Narrative / identity	Self-stories and belonging cues shape challenge-seeking and persistence	**Medium**	Yeager et al. (2019); Walton/Cohen belonging tradition; narrative psychology

HYPOTHESIS — Triad necessity: For students starting from "I am bad at math," improving any one factor alone is usually insufficient:

- Safety without competence → comfort without skill → identity still fragile.
- Competence without safety → possible for some; many never enter the practice volume.
- Narrative without either → empty affirmation (false growth mindset).

XVI.3 Founder tutoring sequence — audit

FOUNDER BELIEF (observed in tutoring):

1. Reduce fear
2. Demystify mathematics
3. Explain why humanity invented it
4. Simplify the narrative
5. Create a tiny success
6. Reinforce success
7. Celebrate progress → identity shifts

Red Team audit:

Step	Universal?	Evidence status	Failure mode
1 Fear ↓	Common for anxious subset; not for all	Strong for anxious learners	Over-soothing that removes challenge
2 Demystify	Often helpful	Medium (cognitive clarity)	Over-simplification that lies

3 Invention story	Founder prior	**Under-evidenced at RCT scale**	Cultural mismatch; time cost
4 Simplify narrative	Often helpful	Medium	Removes necessary difficulty
5 Tiny success	Near-universal lever	High (competence need)	Too-easy wins → hollow identity
6 Reinforce	High if attribution-correct	High (feedback literature)	Praise of talent or praise of empty effort
7 Celebrate	Helpful if genuine	Medium	Performative celebration → cynicism

Improved sequence (HYPOTHESIS — "SAFE-CRAFT"):

1. **S**afety: threat appraisal ↓ (tone, privacy, soft-wrong, no public shame)
2. **A**ttempt: student acts before receiving a full lecture (productive struggle at right grain)
3. **F**eedback-as-information: miss becomes diagnosis, not character
4. **E**arned micro-win: success the student can own
5. **C**ausal story: "what worked" (strategy), optionally "why humans invented this tool"
6. **R**e-entry: immediate second attempt / next slightly harder item
7. **A**ttribution + witness: tutor/system names the growth specifically
8. **F**uture self: map fill / path unlock that marks identity-relevant progress
9. **T**ransfer check: varied item so win was not clone memorization

What changed vs founder sequence: Attempt moves earlier; invention-story becomes optional enrichment inside step 5, not a prerequisite to agency; transfer check prevents fake mastery.

XVI.4 Math anxiety — what is known, what is not

FACT: Higher math anxiety correlates with lower math performance (meta $r \approx -0.17$ in one 2021 synthesis of 57 studies — modest average, not destiny).

FACT: Ashcraft & Kirk (2001) show math anxiety can act like a dual-task load: worry competes for working memory, especially on computation-span / carry-heavy tasks.

HYPOTHESIS: In product UX, anything that increases social-evaluative threat (red buzzers, public leaderboards for novices, tutor contempt, parent hovering UI) will shrink effective WM and look like "they don't get it."

Contradictions / limits:

- Not all underperformance is anxiety; some is missing knowledge, poor curriculum, sleep, language barriers.
- Trait vs state anxiety differ; trait may partly reflect prior failures (reverse causality possible).
- High performers can be anxious and still succeed via avoidance of hard electives — a hidden identity cost.

MindCraft rule: Treat anxiety as a first-class load on the learning system, not as a soft skill sidebar.

XVI.5 Deliberate practice — necessary, not sufficient

FACT / TRADITION: Ericsson's deliberate practice emphasizes structured, feedback-rich, effortful practice aimed at improvement (not mere repetition).

FACT / CRITIQUE: Hambrick and colleagues argue deliberate practice is important but not sufficient; in chess/music reanalyses, deliberate practice often explains roughly ~1/3 of reliable variance, leaving most unexplained (abilities, starting age, opportunity, etc.).

Implication for MindCraft: Do not sell "10,000 hours." Sell **high-quality attempts under feedback**. Also respect individual differences: same practice volume will not equalize outcomes. Identity work must include "progress relative to self," not only absolute rank.

XVI.6 Tutoring effects — demythologize Bloom

FACT: Bloom (1984) popularized the "2 sigma" tutoring + mastery framing.

FACT / CALIBRATION: Broader tutoring metas are far smaller (often ~0.3–0.4 SD in modern RCT syntheses cited by Education Next). VanLehn (2011) estimated human tutoring ~0.79 SD vs no tutoring, with step-based ITS nearly comparable (~0.76).

Implication: Human tutors remain valuable, especially for affect + accountability + diagnosis of weird misconceptions. They are not magic 2-sigma machines. AI step-level feedback can capture much of the *cognitive* tutoring benefit; the scarce remainder is Layers 4–6 (emotion, identity, accountability).

Part XVII — Scarcity Thesis: Full Red Team Trial

XVII.1 The claim under oath

FOUNDER BELIEF: AI commoditizes knowledge; future scarcity is motivation, confidence, identity, trust, curiosity, persistence, emotional safety, meaning.

XVII.2 Cross-examination

Claim A — "Explanations are free"

Verdict: Directionally **FACT** for commodity explanations; **FALSE** for *trusted, personalized, correct, curriculum-aligned* explanations under time pressure.

Evidence: Generative AI + open platforms collapse marginal cost of text explanation.

Contradiction: Hallucinations, shallow fluency, and misalignment create a *trust tax*. Parents may still pay for verified human judgment.

Claim B — "Tutoring is free"

Verdict: FALSE.

AI chat is cheap. High-quality tutoring includes scheduling, relationship, emotional labor, accountability, and reputation. Those remain expensive. Nickow et al.–style tutoring metas still show meaningful effects when tutoring is structured — it is not free at scale.

Claim C — "Content is free"

Verdict: MOSTLY FACT for undifferentiated content; **FALSE** for sequenced, assessed, outcome-linked content systems.

Khan/YouTube prove content abundance. Differentiation is curation + measurement + human wrap.

Claim D — "Motivation/identity become the scarce goods"

Verdict: HYPOTHESIS with strategic plausibility.

As cognitive help approaches free, *willingness to engage difficulty* becomes the binding constraint for many students. This does not mean motivation products automatically win — many habit apps fail to produce identity.

Claim E — "Emotional safety is first-order"

Verdict: HYPOTHESIS — true for anxious / threatened learners; not universal.

For some, speed drills and competition *increase* engagement. Segment, do not romanticize softness.

XVII.3 Surviving thesis (company doctrine)

Adopted doctrine (refined v1.6):

> MindCraft does not compete primarily on explanation volume or AI tutor warmth. It competes on converting threatened learners into challenge-seeking mathematical agents through diagnosis, affective design, mastery-climate evaluation, faded desirable difficulties, student-generated why, narrative meaning under equity constraints, and human accountability — with AI as *guarded* infrastructure that must clear `solo_transfer_pass` ≥ no-AI.

Falsifiers (90-day capable):

1. Anxious segment shows no FEI lift vs explanation-first control.
2. Parents refuse to pay when score gains lag identity metrics.
3. AI-alone matches human+AI on persistence + transfer for target segment.
4. Guarded Solver raises concurrent scores but loses `solo_transfer_pass` vs no-AI (PWC fail).
5. Interleave/SE regimes raise delayed mix but destroy `retry_120s` in high-anxiety segment without sequenced SAFE-DD (XLI; pending DD-1/DD-2).

Part XVIII — Systems Maps (loops, not features)

XVIII.1 FEI reinforcing loop (desired)

```
+------------------ identity claim ("I do math") <---+
| |
v |
safety UX ----> attempt ----> informative miss ----> coach
^ | |
| v v
+------ map fill / witness <---- earned success <+
```

XVIII.2 Shame balancing loop (destroy this)

```
public failure signal -> threat ↑ -> avoidance -> fewer attempts
^ |
+--------- skill lag / worse scores ------+
```

XVIII.3 Extrinsic colonization loop (danger)

```
streak / points ↑ -> return ↑ -> shallow grinding ↑
| |
+-- identity unchanged --------+--> burnout / streak break → exit
```

Design rule: Habit loops may feed FEI but must not replace identity markers.

XVIII.4 Parent trust loop

```
visible diagnosis + honest progress -> parent trust ↑ -> continued purchase
^ |
+---- outcome evidence (scores / challenge-seeking)
```

FOUNDER QUESTION unresolved: Optimize Maya's identity or parent's anxiety? Doctrine: report *both* challenge-seeking and skill evidence; never fake either.

XVIII.5 Network effects (honest)

MindCraft today has weak classic network effects. Do not pretend otherwise.

Possible compounding advantages (HYPOTHESIS):

1. Ontology + misconception memory → better diagnosis over time (data moat if privacy-respecting).
2. Tutor playbooks trained on FEI outcomes → service quality moat.
3. Story worlds with longitudinal character arcs → switching costs of meaning (fragile; easy to cargo-cult).

Part XIX — Cross-Domain Transfer Catalog

Principles that repeatedly change human behavior outside classrooms — filtered for MindCraft use.

Domain	Robust mechanism	Transfer	30-year durable?	AI improve?
Exposure therapy	Graded exposure under safety	Anxiety-graded problem sets	Yes	Yes (personalize gradient)
Sports film study	Specific error review	Wrong-choice coach	Yes	Yes (auto clip of miss type)
Music pedagogy	Technique → repertoire → performance	Drill → quest → “recital” signal	Yes	Partial
Martial arts	Visible ranks + dojo norms	Map regions + tutor dojo culture	Yes	Weak for norms
Aviation	Checklists under stress	Procedure scaffolds when anxious	Yes	Yes
Games (Celeste-like)	Fair failure + retry culture	Soft-wrong physics	Yes	Yes
Chess	Annotated game review	Session postmortems	Yes	Yes
Coaching	Identity + accountability	Tutor as witness	Yes	Assist, not replace
Religion/ritual	Meaning + belonging	Light rituals only; avoid cult dynamics	Yes (human)	Risky
Creator economy	Public craft identity	Optional shareable math artifacts	Uncertain	Yes

Anti-transfer: Casino variable rewards → addiction without skill. Do not copy engagement maxing from social media.

Part XX — Competitive Landscape (mechanism lens)

Product	Primary scarce resource they sell	Identity claim?	Risk
Khan Academy	Free mastery content + reputation	Weak–medium	Explanation commodity; “mastery” without transfer_pass
Duolingo	Habit / streak motivation	Weak (language identity sometimes)	Extrinsic ceiling; appearance of consistency

Brilliant	Aesthetic problem joy + prestige	Medium	Narrow segment; not Maya threat profile
Coursera/edX	Credentials	Medium (career identity)	Completion collapse
Chegg / homework help	Answers under deadline	Anti-identity (outsourcing)	Integrity & AI shock
Private tutors	Human accountability + customization	High when good	Supply constrained
ChatGPT tutors	Instant explanation	Low unless wrapped	Trust / hallucination; Bastani Base harm
MindCraft (target)	FEI conversion + tutor witness + gap diagnosis + solo transfer	**Intended high**	Must prove, not assert

Strategic implication (XXXV densified): Do not out-Khan Khan on content breadth. Do not out-Duo Duo on streaks. Do not out-Brilliant Brilliant on puzzle delight. Do not out-ChatGPT on fluency. Out-compete on the **session**: recoverable struggle, visible competence evidence, strategy selection under mix, student-generated why, and transfer when help is gone — then *say that* in marketing (CSA-2).

Part XXI — Metrics Dictionary (instrumentation)

XXI.1 Banned as North Stars

- Raw DAU / time-on-app without transfer
- Streak length alone
- Explanation open-rate / AI chat token count
- Points / coins
- Blocked-session accuracy alone
- Coach thumbs-up / "felt helpful" without `solo_transfer_pass`

XXI.2 Leading indicators (ship first)

Metric ID	Definition	Why
`retry_120s`	New attempt on same or isomorphic item within 120s of soft-wrong	Persistence under safety
`coach_shown`	Wizard/coach surfaced	Treatment exposure
`write_exit_to_retry`	Leave write mode then attempt	Agency after reflection
`challenge_accept`	Chose harder level when offered	Challenge-seeking (instrument *motive*)

`hint_binge`	≥3 hints without independent solve	Gaming / helplessness
`transfer_pass`	Correct on varied item after mastery mark	Anti-fake-mastery
`solo_transfer_pass`	Transfer with AI/Solver closed or denied	Anti-crutch (XXXIII)
`ai_reveal_rate`	Full-answer / unguarded reveal frequency	Crutch exposure
`strategy_class_error`	Wrong *procedure family* on mixed set	Interleaving discrimination (XXXIX)
`se_principle_hit`	Correct structured principle/ingredient pick when prompted	SE quality proxy (XL)

XXI.3 Lagging indicators

Metric ID	Definition
`math_person_item`	1–7 agreement: "I am a math person" (math-specific)
`anxiety_state_item`	Short state anxiety before session
`advanced_intent`	Intent to take harder course / contest
`tutor_witness_note`	Tutor tagged identity-relevant growth

XXI.4 Decision rule (HYPOTHESIS)

Ship changes that raise `retry_120s` and mastery-motive `challenge_accept` without raising `hint_binge` / `ai_reveal_rate`, and without dropping `transfer_pass` or `solo_transfer_pass`. Prefer delayed mixed accuracy over blocked session accuracy.

Part XXII — Appendices

Appendix A — Glossary

Term	Meaning
FEI Loop	Fear → Evidence → Identity system
ITC	Identity Transformation Cascade
Soft-wrong	Miss treated as information, not shame theater

Gap scan	Low-shame diagnostic of confidence/exposure
Challenge-seeking	Choosing harder math opportunity when optional
False growth mindset	Effort praise without strategy / conditions
SAFE-CRAFT	Improved tutoring sequence (Part XVI.3)
Layers 0–6	Knowledge commodity stack (Part VI.3)
HID	Habit/Identity Divergence — streaks can raise return without identity
MoC	Memory of Competence — spaced retrieval after transfer_pass
PWC	Pedagogy Wrap Constraint — guarded AI + student generation + solo transfer bar
Claim ladder L0–L4	Max defensible causal/marketing claim given design (Part XXXIV)
E×V×Cost	Eccles expectancy × subjective task value × cost triad
TARGET	Classroom/goal-structure dimensions cueing mastery vs performance

Appendix B — Maya interview protocol (qual)

Goal: Map moments when math stopped (or started) feeling dangerous.

1. “Tell me about a time math felt dangerous or humiliating.”
2. “What did your body do? What did you do next?”
3. “Tell me about a time you surprised yourself in math.”
4. “Who saw that moment? Did it matter that someone saw it?”
5. “If an app celebrated you, would that feel real or fake? Why?”
6. “What would make you take a harder math class next year?”
7. “When you get stuck, do you want comfort, a hint, or the answer? In what order?”

Coding: threat markers; attribution language; witness presence; desire for story vs drill.

Appendix C — Causal DAG (provisional)

```
prior failures → math anxiety → WM load → performance ↓
\ \ \→ avoidance → practice ↓ → skill lag
\ \→ fixed identity narrative
\→ missing knowledge ██████████████████████████████→ performance ↓

safety UX → threat ↓ → WM available ↑ → attempt quality ↑
coach + diagnosis → misconception repair → competence evidence ↑
competence evidence + witness → identity update → challenge-seeking ↑
challenge-seeking → harder practice → skill ↑ (reinforcing)
```

Open identification problem: Does identity cause challenge-seeking, or does challenge-seeking cause identity? Likely bidirectional. Measure both.

Appendix D — Experiment pre-registration sketches

A — Wizard coach

- Population: chapter/practice users with soft-wrong enabled
- Arms: coach on / coach off
- Primary: `retry_120s`
- Guardrail: `transfer_pass`, session completion
- Stop rule: clear harm on transfer or spike in hint binge

B — Story frame

- Arms: invention-story wrapper vs bare stem (same math)
- Primary: time-to-abandon after first miss
- Secondary: enjoyment, retention @ 7d

D — Explanation timing × anxiety

- Screen: brief anxiety item
- Arms: explain-first vs attempt-first
- Heterogeneity: high vs low anxiety

Appendix E — Red Team standing orders

1. Attack any claim that cannot name a falsifier.
2. Attack any metric that can be gamed without learning.
3. Attack any narrative feature without a learning outcome.
4. Attack Bloom 2-sigma mythology on sight.
5. Attack “AI makes tutoring free” slogans on sight.

Appendix F — Future chapters (queued, not padded)

1. Neuroscience of math anxiety (amygdala/WM models; cautious translation)
2. History of mathematics as meaning technology — under trial via XXVIII + HIST
3. Parent trust economics & willingness-to-pay — EVT-4 / Part XXVII
4. Equity audit of story worlds — **DONE** (Part XXXVI)
5. Competitive teardown — **DONE** session audits (Part XXXV); usage telemetry still open
6. Formal Bayesian update process for Constitution claims
7. **DONE (2026-07-30):** Desirable difficulties × anxiety (Part XLI) — SAFE-DD stack
8. **DONE (2026-07-30):** Social comparison & leaderboards (Part XLII) — SAFE-COMPARE stack
9. **DONE (2026-07-30):** Habit formation science without identity colonization (Part XLIII) — SAFE-HABIT stack

10. **DONE (2026-07-30):** Intrinsic motivation killers / Deci caveats (Part XLIV) — SAFE-REWARD stack
11. **DONE (2026-07-30):** Mathematical resilience (Part XLV / id 45) — SAFE-RESILIENCE stack
12. **DONE (2026-07-30):** Teacher/tutor expectancy effects (Part XLVI / id 46) — SAFE-EXPECTANCY stack
13. **DONE (2026-07-30):** Sleep, stress, and learning (Part XLVII / id 47) — SAFE-TIMING stack
14. **NEXT:** Transfer & variation theory (Part XLVIII / id 48)

Part XXIII — Research Lab Operating Cadence

Weekly (60–90 min):

1. One claim enters trial (evidence + contradiction).
2. Red Team attempts kill.
3. Surviving claim updates Constitution with confidence tag.
4. One experiment metric reviewed.

Synthesizer (~every 8 researcher ticks): Merge duplicate frameworks into Part I.4 / XIV; refresh Executive Summary; regenerate PDF. Do not write a new chapter on synthesizer ticks.

Monthly:

- Rewrite one weak chapter.
- External reader (learning scientist or skeptical founder).
- Regenerate PDF.

Quarterly:

- Kill or promote North Star metric.
- Publish internal “what we believed that was wrong” memo.

Closing stance

MindCraft’s deepest risk is not technical failure. It is **winning the wrong game**: becoming the best explanation engine in a world where explanations are cheap — or the warmest AI tutor in a world where warmth without solo transfer is a crutch — while losing the only game that compounds: helping a human revise who they are in the presence of difficulty.

This Constitution exists so the company notices that risk early, and runs experiments that can kill beloved ideas.

v1.6 synthesizer pass folded Parts XXXIII–XL into surviving commercial doctrine (I.4). Part XLI adds SAFE-DD; Part XLII adds SAFE-COMPARE; Part XLIII adds SAFE-HABIT; Part XLIV adds SAFE-REWARD; Part XLV adds SAFE-RESILIENCE; Part XLVI adds SAFE-EXPECTANCY; Part XLVII adds SAFE-TIMING (encode→sleep→transfer; all-nighter/midnight-streak kill; exam-week softening; not a sleep clinic). Next researcher id: XLVIII transfer/variation. Page count is not the finish line — falsifiable truth is.

Part XXIV — Affect, Anxiety, Belonging (deep dive)

Chapter status: Living evidence brief

Primary question: When is emotional safety causally load-bearing for mathematical confidence?

Owners: Cognitive Neuroscientist seat · Ed Psych seat · Red Team

XXIV.1 What is happening, mechanistically?

Three related but non-identical constructs get collapsed in product talk. Separate them.

Construct	Definition (working)	Time scale	Product signal
State math anxiety	Transient dread / arousal in a math moment	Seconds–minutes	Avoid click, freeze, rapid hint binge
Trait math anxiety	Stable tendency to experience math as threat	Months–years	Course avoidance, “I’m not a math person”
Belonging uncertainty	“People like me don’t belong in this domain”	Contextual, can be acute	Withdrawal after mild adversity
Stereotype threat	Extra evaluative pressure from group stereotype relevance	Situational	Underperformance vs ability when stereotype is cued

FACT (direction): Math anxiety is associated with worse math performance; one 2021 meta-analysis reported mean r ≈ –0.17 across 57 studies (modest average association — not destiny).

FACT (mechanism candidate): Ashcraft & Kirk (2001) show high math-anxiety individuals have reduced working-memory span especially on computation-based span tasks; dual-task math + memory load amplifies errors/RT. Anxiety behaves like a competing load on central executive resources.

FACT (related): Reviews (e.g., Suárez-Pellicioni et al., 2016) organize explanations as (a) WM competition, (b) attentional control / inhibition deficits, (c) possible low-level numerical representation differences — not mutually exclusive.

HYPOTHESIS for MindCraft: For the Maya-anxious segment, UX that increases social-evaluative threat (shame theater, public ranking early, contemptuous feedback) will look like a “knowledge gap” while actually being a temporary capacity tax.

XXIV.2 Why does this happen? (first principles)

1. **Working memory is scarce.** Multi-step math is a WM sport. Anything that occupies the central executive (worry, self-monitoring for stereotype confirmation, rumination about looking stupid) steals the resource the task needs.
2. **Humans are status-sensitive learners.** Math is a publicly ranked school subject. Threat is often social ("I will be seen as dumb") more than arithmetic.
3. **Avoidance is rational short-term.** Leaving the room ends the aversive arousal. Long-term, avoidance prevents the mastery experiences that would update efficacy.
4. **Identity consolidates avoidance.** Repeated exits become "I'm not a math person," which then licenses future exits (narrative lock-in).

Human need satisfied by safety design: Need for non-humiliation while competence is still fragile; relatedness/respect; predictability.

XXIV.3 Belonging and stereotype threat — what transfers?

FACT: Steele & Aronson tradition — when a test is framed as ability-diagnostic under a negative group stereotype, performance can drop relative to non-diagnostic framing (classic lab pattern; effect sizes and replication quality vary by context).

FACT: Spencer, Steele & Quinn (1999) line — women/math stereotype relevance can impair performance when threat is cued.

FACT: Walton & Cohen (2007, 2011) — belonging uncertainty makes adversity feel identity-diagnostic; brief belonging interventions framing adversity as common/transient improved long-term outcomes for marginalized college students in landmark trials (e.g., Science 2011 belonging intervention).

Red Team limits:

- Effects are **heterogeneous** and context-dependent (like mindset).
- School-wide scaling often attenuates lab effects.
- MindCraft must not cargo-cult a 45-minute belonging essay into a gamified toast.
- Belonging interventions are not a substitute for teaching.

Surviving transfer:

- Normalize struggle without normalizing low standards.
- Make early adversity non-identity-diagnostic ("misses are how the map learns").
- Avoid stereotype-cueing copy ("even girls can…" is often worse).
- Tutor witnessing is a belonging technology when authentic.

XXIV.4 Neuroscience: what we can say without overclaiming

FACT / REVIEW level: Math anxiety literature includes ERP/fMRI correlates; reviews document affective and cognitive network involvement. Exact "brain region product features" are **SPECULATION** and should not drive roadmap.

Allowed claim: Affective arousal and executive control interact; designing for lower threat is biologically plausible.

Forbidden claim: “Our purple calm mode activates the prefrontal cortex.”

XXIV.5 Contradictory evidence (keep visible)

1. Many students underperform from **missing knowledge**, not anxiety.
2. Some learners seek **competitive arousal** and dislike soft UX.
3. Reverse causality: low skill → anxiety → further avoidance (bidirectional).
4. Trait anxiety interventions that only soothe without competence can create comfort without growth.
5. Meta correlations are modest; anxiety is one bottleneck among many.

XXIV.6 Can this generalize? 30 years? AI?

Question	Answer
Generalize beyond math?	Yes — any high-status evaluative domain (music juries, coding interviews).
Still work in 30 years?	Yes — human threat systems are not a 2026 fad.
Can AI improve it?	Yes for personalization of challenge gradient & detection of freeze patterns; risky for fake empathy.
MindCraft change	Soft-wrong, hide-correctness diagnostics, coach tone, private early practice, tutor witness scripts are load-bearing for anxious segment.

XXIV.7 Experimental validation program

ID	Question	Design	Primary	Falsifier
AFF-1	Does soft-wrong raise retry vs hard-fail UX?	A/B	`retry_120s`	No lift / accuracy down
AFF-2	Does anxiety-state moderate explanation-first harm?	Anxiety screen × timing	transfer + retry	No interaction

AFF-3	Does “misses are information” copy reduce belonging threat items?	Pre/post items	belonging/threat items	Demand effects only
AFF-4	Tutor witness note vs no note after growth moment	Tutor RCT	2-week challenge -seeking	Null

Confidence in chapter thesis: Medium–High that affect matters for a large subset; Medium that MindCraft’s current UX already captures the right mechanisms; Low that we have measured it.

XXIV.8 Product system (not features)

Affective Load Manager (system):

1. Detect freeze / hint-binge / rapid exit (behavioral).
2. Downshift evaluative threat (tone, privacy, soft-wrong).
3. Offer graded re-entry (easier isomorphic item), not pep talk alone.
4. Restore challenge once `retry` and success stabilize.
5. Log whether the student later chooses harder work (identity leading indicator).

Balancing loop: chronic downshift → boredom → exit. Must re-raise difficulty.

Part XXV — Self-Efficacy, Narrative Identity, Habit (deep dive)

Chapter status: Living evidence brief

Primary question: What updates the self-story “I am bad at math,” and what merely increases login frequency?

Owners: Learning Science · Narrative Psychology · Behavioral Econ · Red Team

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XXV.1 Self-efficacy is not “confidence vibes”

FACT (theory): Bandura’s social cognitive theory — self-efficacy is belief in capability to organize and execute actions required for specific attainments. It is **task- and domain-specific**, not a global personality glow.

FACT (sources): Efficacy information is processed from four sources (Bandura, 1997):

1. **Mastery experiences** (usually strongest)
2. **Vicarious experiences** (models)

3. **Social persuasion** (credible others)
4. **Physiological / affective states** (how arousal is interpreted)

FACT (measurement tradition): Usher & Pajares (2009/2010 line) validated source scales for middle-school mathematics; sources relate to math self-efficacy, goals, optimism.

HYPOTHESIS: MindCraft's FEI loop is largely a **self-efficacy engineering system**:

FEI stage	Efficacy source
Safety / soft-wrong	Reinterprets physiological arousal ("alert, not doomed")
Earned micro-win	Mastery experience
Coach naming strategy	Persuasion + attribution
Tutor witness / peer story	Vicarious + persuasion
Map fill after transfer	Mastery that survives variation

XXV.2 Why mastery experiences dominate — and how products fake them

What is happening: A win only updates efficacy if the learner **attributes** it to their capability under meaningful difficulty.

Fake mastery (product sins):

- Too-easy items → "this doesn't count"
- Heavy hinting → "the app did it"
- Identical clones after "mastery" → recognition, not competence
- Points for opening lessons → engagement theater

Real mastery design rules (HYPOTHESIS):

1. Difficulty just above comfort (desirable difficulty, not cruelty).
2. Student performs the decisive step.
3. Immediate informative feedback.
4. Soon after: varied item (transfer_pass).
5. Explicit attribution to strategy ("you isolated the factor") not talent.

Contradicting caution: After strong efficacy exists, occasional failure hurts less (Bandura). Early failures hurt more. Protect early arc; do not permanently infantilize.

XXV.3 Reciprocal loops: efficacy ↔ achievement

FACT / EMERGING: Longitudinal work often finds reciprocal relations — achievement raises efficacy and efficacy raises later achievement (e.g., recent high-school math reciprocity analyses in large-scale assessment literature). Causality is not one-way.

Implication: "Just raise confidence" without skill is unstable. "Just raise skill" while humiliating can stall practice volume for anxious students. Build both.

XXV.4 Narrative identity — the story that persists

FACT / TRADITION: Narrative identity research (McAdams tradition) — people maintain identity through evolving life stories with themes of agency, communion, redemption, contamination.

HYPOTHESIS (MindCraft synthesis): Math identity change is a **redemption narrative rewrite**:

- Contamination story: "I failed → I am bad at math → I avoid → I stay bad."
- Redemption story: "I struggled → I used a strategy → I grew → I am becoming someone who stays."

Product implication: Story worlds matter if they supply **symbols for redemption**, not only costume. The map lighting after a real win is a narrative punctuation mark.

Red Team: Most edtech "stories" are skin. If removing the narrative leaves identical pedagogy and identical emotions, the story was decoration. Test with story-off experiments (Experiment B).

XXV.5 Habit vs identity — the Duolingo trap, precisely

FACT / INDUSTRY: Habit products use cues, streaks, variable rewards, loss aversion. Duolingo's method materials emphasize motivation systems for return.

FACT / THEORY tension: Self-Determination Theory warns that controlling extrinsic rewards can undermine intrinsic motivation when they feel coercive (classic Deci findings; nuance exists for informational rewards).

HYPOTHESIS — Habit/Identity Divergence Model (HID):

```
Habit success: cue → routine → reward → return frequency ↑
Identity success: evidence → attribution → self-story → choice of harder future ↑

Overlap zone: daily practice that produces mastery evidence
Failure mode A: habit without mastery → brittle streak addiction
Failure mode B: identity talk without practice → empty affirmation
```

Metric separation (mandatory):

- Habit health: D1/D7 return, streak
- Identity health: `challenge_accept`, advanced course intent, math-person item + behavior consistency

Optimize for overlap zone. Never promote a habit metric that moves opposite identity metrics.

XXV.6 Vicarious experience and "who is the model?"

HYPOTHESIS: Models work when perceived as **similar yet slightly ahead** (“near-peer who struggled”), not as genius tutors who never miss.

Product:

- Villager / character arcs that show struggle → strategy → win.
- Tutor stories: “I used to blank on fractions.”
- Avoid only showcasing perfect speedruns.

Social persuasion: Credibility matters. Random app toast < trusted tutor sentence. Parents can persuade or destroy (“we’re not math people” is efficacy poison — FOUNDER BELIEF with strong anecdotal support; needs interview confirmation).

XXV.7 Universal principles from this chapter

1. Efficacy is specific; measure math efficacy, not “confidence.”
2. Mastery experiences must be owned and non-trivial.
3. Arousal interpretation is designable (anxiety as signal vs doom).
4. Narrative punctuation should mark real competence events.
5. Habits are oxygen; identity is the fire.

XXV.8 Experiments

ID	Question	Design	Primary
EFF-1	Does requiring decisive student step (vs auto-solve) raise efficacy item?	A/B	math efficacy delta
EFF-2	Strategy attribution copy vs talent copy after win	A/B	challenge_accept @ 7d
EFF-3	Streak on vs identity milestone on (map fill)	A/B	retention vs challenge_accept tradeoff
EFF-4	Near-peer struggle story vs expert-perfect story	A/B	persistence after first miss

Falsifier for identity thesis: Habit metrics rise for 8 weeks while challenge-seeking and transfer stay flat — then MindCraft is a streak app, not an identity company.

Part XXVI — Cognitive Load, Mastery, Tutoring (deep dive)

Chapter status: Living evidence brief

Primary question: How do we create earned competence without drowning working memory — and without mythologizing tutoring?

Owners: Cognitive Science · Learning Science · AI Researcher · Red Team

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XXVI.1 Cognitive Load Theory — the non-negotiable physics

FACT / TRADITION: Cognitive Load Theory (Sweller and colleagues) — working memory is limited; instructional design that forces novices into unguided search can impede schema acquisition.

FACT: Worked-example effect — for novices, studying worked examples often beats equivalent time in conventional problem solving for structurally similar later items (classic Sweller & Cooper algebra studies, 1985 line).

FACT / DESIGN IMPLICATION: Guidance fading — move from full worked examples → completion problems → independent solving as expertise grows (Renkl/Atkinson fading tradition). Continuing full examples after expertise rises can trigger **expertise reversal** (redundancy).

Why this matters to identity: Overloaded novices experience failure as self-indictment. Load mismanagement manufactures fake "I'm bad at math" evidence.

XXVI.2 Reconciling CLT with "attempt before lecture"

SAFE-CRAFT says attempt early. CLT says novices need worked examples. **Is this a contradiction?**

Resolution (HYPOTHESIS — Attempt Grain Principle):

Learner state	Right "attempt"	Wrong "attempt"
True novice, new schema	Attempt a **micro-step** or completion blank inside a worked example	Full novel multi-step problem cold
Partial schema	Isomorphic problem with soft-wrong	Leap to transfer without success
Anxious + some knowledge	Slightly easier isomorphic first	Public hard problem as opener
Advanced	Minimal guidance; generative struggle	Endless examples (expertise reversal)

Product rule: "Struggle first" means **agency at the right grain**, not abandonment into search.

This dissolves a false war between "productive struggle" slogans and cognitive load science.

XXVI.3 Mastery learning — keep the war on the table

Supporting evidence: Kulik, Kulik & Bangert-Drowns (1990) meta — mastery programs associated with positive exam effects (often cited ~0.5 SD overall in their synthesis), often stronger for weaker students; time costs can rise; completion can suffer in some self-paced settings.

Contradicting evidence: Slavin (1987) "Mastery Learning Reconsidered" — limited support for group-based mastery on standardized measures; stronger on experimenter-made tests; coverage vs depth tradeoff is real.

MindCraft doctrine:

1. Mastery means **transfer_pass**, not same-item streak.
2. Track time-to-mastery; do not hide cost.
3. Allow strategic coverage decisions (exam track) without pretending infinite time.
4. Ontology graph is only as good as far-transfer checks.

XXVI.4 Tutoring — calibrate the mythology

Bloom (1984): Tutoring + mastery framing popularized as ~2 SD ("2 sigma problem").

Calibration FACTS:

- Cohen, Kulik & Kulik (1982) tutoring meta — much smaller average effects than 2.0.
- Nickow, Oreopoulos & Quan (2020) tutoring RCT meta — on the order of ~0.37 SD (impressive, not magic).
- VanLehn (2011) — human tutoring ~0.79 SD vs no tutoring; step-based ITS ~0.76; answer-based CAI smaller. Granularity of feedback matters; beyond a point, finer grain plateaus.

Strategic reading for MindCraft:

Tutoring function	Automate?	Keep human?
Step feedback	Yes (ITS-like)	Optional
Misconception diagnosis	Partially (ontology + traces)	Human for weird cases
Affect co-regulation	Partially (UX)	Human strong
Accountability / relationship	Weakly	Human strong
Identity witnessing	Poorly	Human strong

AI implication: Compete on diagnosis quality + FEI wrapping; do not sell "we are 2-sigma."

XXVI.5 Deliberate practice inside sessions

FACT / TRADITION: Ericsson et al. (1993) — structured, feedback-rich practice aimed at improvement.

FACT / CRITIQUE: Hambrick/Macnamara line — deliberate practice explains substantial but incomplete variance; not a sufficient account of expertise differences.

Session design (HYPOTHESIS):

1. Clear target skill (ingredient-level).
2. Immediate feedback at step grain when possible.
3. Successive refinement on weak subskill (not random variety first).
4. Then varied practice for transfer.
5. Short enough to protect affect; frequent enough to compound.

XXVI.6 Minimal guidance failures

FACT: Kirschner, Sweller & Clark (2006) critique pure discovery/minimal guidance for novices — often inferior to guided instruction.

Implication for story worlds: Narrative must not become unguided discovery of math. Story wraps guided cognition. World_VISION is compatible with guidance; it is incompatible with vibes-only exploration.

XXVI.7 Systems: Competence Production Line

```
diagnosis (gap) → grain selection → worked/fade → attempt
→ soft-wrong info → repair → transfer check
→ mastery mark → spaced return → challenge raise
```

Balancing loops:

- Too slow fade → boredom
- Too fast fade → threat + load spike → avoidance
- Mastery without spacing → illusion of competence

XXVI.8 Experiments

ID	Question	Design	Primary
CLT-1	Worked-example-first vs problem-first for true novices	Within concept	near & far transfer
CLT-2	Fade schedule adaptive to mastery graph vs fixed	A/B	time-to-transfer_pass

TUT-1	Human tutor + AI brief vs AI-alone on persistence	RCT	retry + 4-week challenge_accept
MAS-1	Mastery gate with transfer_pass vs same-item 3-in-a-row	A/B	2-week retention

Red Team standing kill: Any claim that MindCraft "replicates Bloom 2-sigma" without citing modern tutoring metas.

Part XXVII — Markets, Parents, Competition (deep dive)

Chapter status: Living strategy brief (evidence + market reasoning)

Primary question: If explanations cheapen, what do families still buy — and what should MindCraft refuse to become?

Owners: Founder Philosopher · Product Strategist · Behavioral Econ · Red Team

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XXVII.1 Re-state the scarcity doctrine under market pressure

Adopted doctrine (from Part XVII): Compete on diagnosis, affect, identity, accountability — with AI as infrastructure.

Market test: Parents often buy **score relief** and **homework peace**, not "identity transformation." Students may buy relief from shame. Schools buy coverage and equity metrics.

FOUNDER QUESTION (unresolved): Can MindCraft sell identity truthfully while packaging score outcomes parents recognize?

Working answer (HYPOTHESIS): Sell the FEI system as the *method*; report scores/challenge-seeking as *evidence*. Never sell vibes without transfer.

XXVII.2 Willingness-to-pay stack (post-AI)

Layer	Marginal cost trend	WTP trend	Notes
Raw answers	→ 0	↓↓	Chegg shock / GPT
Generic explanations	→ 0	↓	Unless trusted
Personalized practice	↓	→ / ↑	Needs outcomes
Diagnosis of *this* student	↓ with AI	↑ if trusted	Ontology advantage possible

Emotional safety during struggle	Hard to automate well	↑ for anxious segment	UX + tutor
Accountability relationship	Scarce	↑↑	Scheduling, human
Trusted outcome signal to parents	Scarce	↑↑	Reports that don't lie

SPECULATION (30-year): Ambient AI tutors everywhere. Differentiator = trusted human+system that changes trajectory, not chat fluency.

XXVII.3 Parent trust economics

What parents fear (hypothesis from tutoring practice — needs interviews):

1. Child is falling behind permanently.
2. They cannot help at home.
3. Child hates learning / shuts down.
4. Money spent on apps that entertain without learning.

Trust builders (HYPOTHESIS):

- Specific diagnosis ("fractions↔ratios bridge weak") over vague "needs practice"
- Visible soft-wrong → retry behavior (effort signal)
- Transfer checks (anti-cheat of fake mastery)
- Honest plateaus ("this week: consolidation, not fireworks")
- Tutor notes that sound human and concrete

Trust destroyers:

- Inflated streaks as "progress"
- Grade promises you can't keep
- Dark-pattern engagement
- AI that hallucinates pedagogy confidently

Experiment PAR-1: Parent report variants — score-only vs FEI dashboard (retry, challenge-seeking, mastery map). Measure retention/WTP intent and comprehension of child's actual bottleneck.

XXVII.4 Competitive teardown (mechanism lens)

Khan Academy

- **Sells:** free mastery content + brand trust
- **Strength:** coverage, accessibility
- **Weakness for Maya:** can still feel lonely/evaluative; identity not core

- **MindCraft response:** do not out-content; out-diagnosis + human wrap

Duolingo

- **Sells:** habit motivation
- **Strength:** return loops
- **Weakness:** habit ≠ deep skill/identity (HID model)
- **MindCraft response:** borrow on-ramps; refuse extrinsic colonization of meaning

Brilliant

- **Sells:** aesthetic intellectual joy + prestige
- **Strength:** delightful problems for already-curious
- **Weakness:** may miss threatened segment
- **MindCraft response:** win the anxious middle, not only the delighted

Tutoring marketplaces

- **Sells:** access to humans
- **Strength:** relationship potential
- **Weakness:** uneven quality; weak diagnosis systems
- **MindCraft response:** FEI-trained tutors + AI briefings

Generic GPT tutors

- **Sells:** instant explanation
- **Strength:** 24/7 cognitive help
- **Weakness:** trust, curriculum alignment, affect, accountability
- **MindCraft response:** wrap models in ontology + FEI + human witness

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XXVII.5 Business model risks tied to research thesis

If thesis true…	Business implication
Identity is the product	Retention via meaning + progress, not only content library
Affect is first-order for segment	CAC messaging should speak shame→agency, not “more videos”
Human witness scarce	Hybrid model: AI scale + tutor moments that matter
Parents buy trust	Reporting quality is a product surface

If thesis false…	Pivot signal
Explanation-first wins all segments	Become best adaptive explainer (crowded)
Streaks drive outcomes without identity	Become math Duolingo (crowded, brittle)

Parents only pay for score spikes	Short-cycle test prep (possible, different company)

XXVII.6 Network effects — honesty clause

FACT about present: MindCraft does not have Facebook-style network effects.

Plausible compounding (HYPOTHESIS):

1. Misconception telemetry → better diagnosis (privacy-constrained data advantage)
2. Tutor playbook quality → service moat
3. Longitudinal identity arcs in worlds → switching costs of meaning (fragile)

Red Team: “Community” features often fake relatedness. SDT metas show relatedness interventions harder to move. Do not ship empty social to check a box.

XXVII.7 Pricing ethics (Founder Philosopher)

HYPOTHESIS: Charging for emotional safety is legitimate if competence and transfer are real; illegitimate if the product only soothes.

Guardrail: Every paid claim in marketing must map to a measured FEI or learning metric. If not measured, do not claim.

XXVII.8 Experiments & interviews

1. **10 parent interviews:** what did you think you bought? what would make you cancel?
2. **10 Maya interviews:** when did math stop feeling dangerous?
3. **PAR-1** report A/B
4. **COMP-1** message test: identity framing vs score framing vs hybrid (conversion + 30-day retention)

Confidence: Medium that hybrid packaging wins; Low on exact price points without data.

Part XXVIII — History, Meaning, and Why Math Exists (deep dive)

Chapter status: Design-research brief (under-evidenced; high founder interest)

Primary question: Does explaining *why humanity invented a mathematical idea* change persistence and identity — or is it a beautiful distraction?

Owners: Historian of Mathematics seat · Founder Philosopher · Learning Science · Red Team

XXVIII.1 The founder prior (label it)

FOUNDER BELIEF: Students disengage partly because math is presented as arbitrary machinery. Reconnecting ideas to human invention, need, and story demystifies the subject and restores meaning.

WORLD_VISION alignment: Story-first questions; math frozen, narrative wraps.

Epistemic status: Strong pedagogical tradition; **weak large-scale RCT base** relative to mindset/SDT. Treat as a research program, not settled science.

XXVIII.2 What "meaning" might be doing (mechanisms)

If invention-story helps, why?

Mechanism	Claim type	Testable implication
Arbitrary-rule appraisal ↓	HYPOTHESIS	Fewer "why do we have to learn this?" exits
Curiosity spike	HYPOTHESIS	Longer voluntary time before first quit
Cognitive schema anchor	HYPOTHESIS	Better retention of structure if story encodes structure (not trivia)
Identity affiliation ("humans who solve")	HYPOTHESIS	Higher math-person endorsement
Seductive details (harm)	FACT tradition in multimedia learning	Stories that add irrelevant color can *hurt* retention

Critical constraint: History must encode the **mathematical structure**, not costume the problem with dragons while leaving the math alien.

XXVIII.3 Seductive details problem (Red Team weapon)

FACT / TRADITION: Multimedia learning research (Mayer and others) — interesting but irrelevant details can impair learning (seductive details effect). Not always replicated equally, but the risk is real.

Kill criterion for MindCraft stories: If a narrative element can be removed without changing the mathematical mental model, it is decoration. Decoration can remain for motivation **only if** Experiment B shows persistence/retention gains that justify time cost.

Survive criterion: Narrative that makes the *invariant* memorable (e.g., why we need a common measure; why zero; why coordinates) is structural meaning.

XXVIII.4 History of math as demystification technology

Examples of structural meaning (illustrative — not a curriculum):

1. **Fractions / ratios:** Fair sharing, measurement, comparison when wholes differ.
2. **Negative numbers:** Debt, direction, opposite change — not “fake numbers.”
3. **Coordinates:** Descartes’ move — marry algebra to geometry to locate.
4. **Functions:** Input–output machines for prediction and dependency.
5. **Probability:** Reasoning under incomplete information; insurance, games, risk.
6. **Quadratics / optimization:** Paths of projectiles, areas, tradeoffs — tools for extrema.

HYPOTHESIS: One crisp invention sentence + one visual beat beats a museum lecture.

XXVIII.5 Cross-domain analogy: martial arts lineages & music history

Domain	Meaning practice	Transfer
Martial arts	Lineage stories + why a form exists	Respect + purpose; still drill techniques
Music	Composer context + etudes	Meaning without skipping scales
Religion	Origin myths	Caution: do not sacralize math cultishly
Architecture	“Why this arch holds”	Structure-first stories

Principle: Meaning wraps discipline; it does not replace repetition.

XXVIII.6 Equity and whose history?

Open problem: Whose invention stories? Eurocentric timelines can alienate. Global histories of mathematics (India, Islamic golden age, China, Africa, Indigenous measurement systems) matter for belonging **and** accuracy.

HYPOTHESIS: Inclusive structural histories can support belonging for students who feel math is “not for people like me” — but only if not tokenistic.

Experiment HIST-EQ: Same math, Euro-only story vs plural invention story; measure belonging items + learning — watch for seductive-detail harm.

XXVIII.7 Integration with SAFE-CRAFT

Invention story belongs in **step C (Causal story)** after an attempt grain, or as a 10–20s prelude that encodes structure — not as a gate that delays agency for anxious students who need a micro-win first.

Default order for threatened learners: Safety → micro-attempt → feedback → meaning beat → next attempt.

Default order for curious/low-anxiety: Short meaning beat → attempt → fade guidance.

Segment, don't dogmatize.

XXVIII.8 Experimental program (this chapter lives or dies here)

ID	Question	Design	Primary	Kill condition
HIST-1	Structure-encoding story vs bare stem	Crossover	retention @ 7d + retry	Story slows, no learning lift
HIST-2	Structure story vs seductive skin (dragons, no structure)	3-arm	transfer	Skin ≥ structure
HIST-3	Pre-attempt story vs post-miss story	Timing	time-to-abandon	Either timing harms transfer
HIST-4	Plural history vs single tradition	A/B	belonging + learning	Belonging up, learning down badly

Current confidence that story is load-bearing for identity: Low–Medium.

Current confidence that bad story can harm: Medium–High.

Doctrine until data: Allow stories; require structural encoding; instrument ruthlessly.

XXVIII.9 What would make this chapter graduate from "founder belief"?

Graduation criteria (all required):

1. HIST-1 or HIST-3 shows persistence or retention lift in target segment.
2. No material drop in `transfer_pass`.
3. Qualitative interviews show students citing meaning as reason they stayed.
4. Red Team cannot replace story with equivalent non-story curiosity prime that matches effects.

Until then: **FOUNDER BELIEF under active trial.**

Part XXIX — Spacing, Retrieval, and the Memory of Competence

Chapter status: Living evidence brief

Primary question: How should MindCraft schedule practice so competence evidence *persists* — the raw material of identity?

Owners: Cognitive Science · Data Scientist · Product Strategist · Red Team

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XXIX.1 Why memory science belongs in an identity Constitution

Identity (“I am someone who can do math”) requires **autobiographical evidence**. If skills evaporate between sessions, the self-story collapses into “I used to be able to when the app held my hand.”

Retention is not a separate KPI from identity. It is the substrate.

XXIX.2 Spacing / distributed practice

FACT: Cepeda, Pashler, Vul, Wixted & Rohrer (2006) — quantitative synthesis of distributed practice (317 experiments / 184 articles; 839 assessments). Spacing and lag effects are robust; the inter-study interval that maximizes retention **increases as the desired retention interval increases**.

FACT: Cepeda et al. (2008, *Psychological Science*) — “temporal ridgeline” of optimal retention: gap between study episodes should scale with how long you need the knowledge to last.

Product implication: “Do 40 problems tonight” (massing) can produce local mastery marks that fail a week later. Mastery graph must schedule **returns**, not only first clears.

HYPOTHESIS — Identity Spacing Rule: After `transfer_pass`, schedule a retrieval at a lag matched to the student’s likely next exam/use horizon (days for homework, weeks for ACT track).

XXIX.3 Retrieval practice / testing effect

FACT / TRADITION: Roediger & Karpicke (2006) — retrieval practice (testing) often beats restudy for long-term retention (“power of testing memory”).

FACT / nuance: Expanding vs equal spacing of retrieval — expanding can help short-term; equal spacing often competitive or better for long-term (Karpicke & Roediger, 2007 line).

MindCraft translation:

- Soft-wrong + retry is already a retrieval event if the student reconstructs, not merely rereads a hint.
- “Review mode” that only re-shows worked examples without retrieval is weaker than asking for the decisive step again.

• Gap scan / practice should prefer **generation** over passive video when the goal is durable competence.

XXIX.4 Desirable difficulties (Bjork)

FACT / TRADITION: Conditions that slow acquisition can improve retention/transfer (spacing, interleaving, generation) — “desirable difficulties.”

Red Team / affect conflict: Desirable difficulties raise short-term failure rates → can spike threat for anxious learners.

Resolution (HYPOTHESIS — Sequenced Difficulty):

1. Early FEI: protect safety; use micro-grain success.
2. Then introduce spacing/interleaving once `retry_120s` is healthy.
3. Never introduce desirable difficulty as shame (“you got this wrong because you’re weak”). Frame as training physics.

XXIX.5 Interleaving (brief)

FACT / TRADITION: Mixed practice of related problem types often improves discrimination/transfer vs blocked practice (Rohrer/Taylor line in math practice research — effects depend on similarity structure).

HYPOTHESIS: Interleave *after* initial schema formation (CLT), not before. Block → fade → interleave → spaced retrieval.

XXIX.6 System: Memory of Competence (MoC)

```
learn → transfer_pass → schedule spaced retrieval
↑ |
+---- fail retrieval -------+--> diagnose: forgot vs never had
```

Forgot vs never-had is a product-critical distinction:

• Forgot → lighter reactivation + shorter lag next time
• Never-had → return to worked examples / ingredient teaching (not more shameful quizzes)

XXIX.7 Experiments

ID	Question	Design	Primary
SPC-1	Spaced return vs massed cram after mastery mark	A/B	retention @ 7d / 28d

RET-1	Retrieval prompt vs restudy after soft-wrong	A/B	retention + retry quality
INT-1	Interleave after fade vs blocked-only	Within concept	far transfer

Falsifier for "identity without memory": Challenge-seeking rises while 28-day retention collapses — then celebration is theater.

Confidence: High that spacing/retrieval are real; Medium that MindCraft currently schedules them well; product audit needed.

Part XXX — Attribution, Helplessness, and the Language of Feedback

Chapter status: Living evidence brief

Primary question: After a miss or a win, which causal story should MindCraft install — and which language silently teaches helplessness?

Owners: Ed Psych · UX Researcher · Tutor Ops · Red Team

XXX.1 Weiner's dimensions (the grammar of cause)

FACT / THEORY: Weiner's attributional theory of achievement motivation (e.g., 1979 classroom theory; 1985 *Psychological Review*) — people explain success/failure along dimensions that shape emotion and future expectancy:

Dimension	Poles (simplified)	Linked consequence (Weiner tradition)
Locus	Internal ↔ External	Pride/shame vs externalization
Stability	Stable ↔ Unstable	Expectancy of future same outcome
Controllability	Controllable ↔ Uncontrollable	Guilt/responsibility vs helplessness / pity

HYPOTHESIS for product copy: The most dangerous failure attribution in math is **internal + stable + uncontrollable** ("I'm just not smart at math").

The most useful failure attribution is often **internal + unstable + controllable** ("I used the wrong strategy; I can change it") — with care not to blame students for systemic/curricular failures.

XXX.2 Learned helplessness in learning contexts

FACT / TRADITION: When outcomes are experienced as independent of action, organisms (and students) reduce effort — learned helplessness tradition (Seligman) applied in education as expectancy of non-contingency.

Product factories of helplessness:

1. Hints that auto-solve → outcome independent of student action
2. Random difficulty spikes → non-contingent failure
3. Praise for “being smart” then fail → talent threat
4. Opaque AI answers → student cannot see what *they* controlled

Antidote design: Make the decisive controllable step visible. Soft-wrong should point to a **changeable move**.

XXX.3 Attributional retraining (AR)

FACT / APPLIED TRADITION: Attributional retraining programs encourage adaptive attributions (often effort/strategy for failure) and have an empirical literature in higher-ed/classroom applications (Perry and colleagues’ applied work; see reviews of AR).

Caveats (Red Team):

- “Just try harder” without strategy is false growth mindset.
- Effort attribution for impossible items is gaslighting.
- Ability is not infinitely plastic in the short run; teach strategies and select grain.

MindCraft AR micro-protocol (HYPOTHESIS):

1. Name the miss specifically.
2. Offer one controllable strategy.
3. Invite immediate re-attempt.
4. On success: attribute to the strategy used.
5. On second miss: downshift grain (CLT), do not escalate shame.

XXX.4 Feedback language bank (draft doctrine)

Situation	Avoid	Prefer
First miss	“Wrong.” / “Easy one…”	“That trap is common — here’s the fork.”
Hint use	“Here’s the answer.”	“Your move: which factor is shared?”
Success after struggle	“You’re a genius!”	“You changed approach — that worked.”
Success too easy	Big celebration	Quiet confirm + raise challenge

Tutor note	"Good job!"	"I saw you return after the miss on axis of symmetry."

XXX.5 Intersection with mindset and SDT

- **Mindset:** Yeager 2019 shows context-sensitive gains; attribution language is part of the ecology.
- **SDT competence:** Controllable success feeds competence need.
- **Anxiety:** Internal-stable-uncontrollable attributions amplify threat.

Unified copy rule: Every feedback utterance should answer: *What can the student do next that would change the outcome?* If it cannot answer, rewrite.

XXX.6 Experiments

ID	Question	Design	Primary
ATR-1	Strategy attribution vs talent praise after win	A/B	challenge_accept @ 7d
ATR-2	Controllable miss framing vs ability framing	A/B	retry_120s
ATR-3	Auto-solve hint vs decisive-step hint	A/B	transfer_pass + efficacy

Falsifier: Strategy language improves self-report but not behavior — then copy is placebo.

Confidence: High that attributions matter; Medium that current product language is already adaptive (needs audit).

Part XXXI — Flow, Challenge–Skill Balance, and Difficulty Design

Chapter status: Living evidence brief

Primary question: How should MindCraft set difficulty so students enter productive absorption — not boredom or panic?

Owners: Game Designer · Learning Science · UX · Red Team

XXXI.1 Flow, stripped of mysticism

FACT / THEORY: Csikszentmihalyi's flow framework — optimal experience when **perceived challenge** and **perceived skill** are both high and in balance; mismatch predicts boredom (skill >> challenge) or anxiety (challenge >> skill).

Education transfer: Classroom/ESM studies (e.g., Shernoff et al. tradition) link challenge–skill balance to engagement quality. Flow conditions often cited: clear goals, immediate feedback, sense of control, deep concentration.

Red Team: Flow is not a product feature you "turn on." Self-report flow can be confounded with fun. Some high-learning moments are effortful and not blissful. Do not optimize for "lost track of time" if transfer falls.

XXXI.2 Perceived vs actual skill (critical nuance)

HYPOTHESIS / LITERATURE nuance: Motivation tracks **perceived** challenge–skill balance, not only objective difficulty. Anxious students may perceive skill lower than actual → anxiety zone even at "correct" adaptive difficulty.

Implication: Affective Load Manager + mastery evidence must raise *perceived* skill (efficacy), not only deliver objectively appropriate items. Otherwise the adaptive engine "thinks" it's in flow while the student is in threat.

XXXI.3 Game design transfer (Celeste principle)

What good hard games do:

1. Failures are frequent, cheap, informative.
2. Restart is instant.
3. Difficulty is fair (player believes success is possible).
4. Mastery is visible (you *feel* getting better).

MindCraft mapping:

Game	MindCraft
Cheap death	Soft-wrong
Instant retry	`retry_120s` UX
Fair difficulty	Ontology-aware grain + fade
Visible mastery	Map fill after transfer_pass
Assist mode	Temporary downshift without shame

Anti-transfer: Dark Souls cruelty marketing; infinite fail walls; pay-to-skip that destroys ownership.

XXXI.4 Dynamic difficulty as a system

```
sense: accuracy, latency, hint binge, exit, anxiety proxy
↓
```

```
classify zone: boredom / flow / anxiety / helplessness
↓
act: raise / hold / fade-down / change grain / insert worked example
↓
re-check transfer — not only near accuracy
```

Balancing loop: Chronic downshift → boredom → identity stall.

Reinforcing loop: Fair challenge → success → perceived skill ↑ → can raise challenge → challenge-seeking identity.

XXXI.5 Relation to North Star

Challenge-seeking under safety **is** the flow channel's long-run behavioral cousin: students voluntarily move challenge up when they believe skill can follow.

Measure both:

- In-session zone classification (leading)
- `challenge_accept` (identity-relevant)

XXXI.6 Experiments

ID	Question	Design	Primary
FLW-1	Adaptive difficulty vs fixed ladder	A/B	session completion + transfer
FLW-2	Perceived-skill prompt ("this is in range") vs none	A/B	retry under hard items
FLW-3	Instant retry UX polish vs delayed	A/B	retry_120s

Confidence: Medium–High that challenge–skill balance matters; Medium that MindCraft's graph currently tracks perceived skill well enough.

Part XXXII — Parent Math Anxiety Transmission

Chapter status: Living evidence brief

Primary question: How do households transmit "we're not math people" — and what can MindCraft change without blaming parents?

Owners: Ed Psych · Product (parent surfaces) · Founder Philosopher · Red Team

XXXII.1 The key finding

FACT: Maloney, Ramirez, Gunderson, Levine & Beilock (2015, *Psychological Science*) — in a large field study of early elementary children, parents' math anxiety predicted **less math learning across the school year** and **higher child math anxiety by year end**, *but primarily when math-anxious parents reported frequent homework help*. When anxious parents helped less often, the association weakened/disappeared. Effects were **math-specific** (not reading achievement).

Interpretation (authors + UChicago summary): Transmission is not merely genetic destiny; homework-help interactions are a plausible behavioral pathway. "Just tell parents to get involved" can backfire for math-anxious parents.

Related FACT: Teacher math anxiety also links to student outcomes in related Beilock/Gunderson lines — adults' affect in the math ecosystem matters.

XXXII.2 Why this happens (mechanisms)

HYPOTHESES consistent with the finding:

1. **Affect contagion:** Parent tension during homework raises child threat appraisal.
2. **Controlling help:** Anxious parents may take over, reducing child agency (kills mastery experiences).
3. **Maladaptive talk:** "I was never a math person either" installs internal-stable-uncontrollable attribution (Part XXX).
4. **Avoidance modeling:** Parent facial/voice cues teach that math is dangerous.
5. **Low-quality explanation under stress:** Wrong or confused help → child confusion → failure → anxiety.

Human need: Parents want to help and not harm. Product must give them a **safe role**.

XXXII.3 MindCraft implication — redesign the parent role

Parent role	Harmful version	Better version
Homework coach	Anxious parent teaches under stress	Parent as **witness + scheduler**, tutor/app teaches
Encourager	"You're smart" / "we're not math people"	"I saw you try again" (strategy/effort-process)
Progress checker	Streak theater	FEI dashboard: retry, transfer, map
Co-solver	Take over the pencil	Ask one prompt from a script card

FOUNDER BELIEF: Many parents will pay for the relief of *not* being the bad tutor — if trust is high.

XXXII.4 Product systems

Parent Affect Firewall (system hypothesis):

1. Detect homework-help context (parent mode / shared device evenings).
2. Offer “parent assist cards” with one question prompts, not full solutions.
3. Route hard teaching to AI+ontology or human tutor.
4. Show parents child agency metrics (decisive steps completed).
5. Never require anxious parents to explain novel procedures live.

Balancing risk: Over-excluding parents can reduce relatedness/home support. Offer optional low-anxiety involvement.

XXXII.5 Contradictions / limits

1. Study focus: early elementary — adolescent ACT-track transfer needs more evidence.
2. Self-reported homework help frequency has measurement error.
3. Not all involved anxious parents harm; quality of help varies.
4. Some children need parent accountability structure.
5. Cultural differences in homework norms.

Confidence: High that the 2015 finding is real and important; Medium that MindCraft’s teen segment shows the same pathway; product bets should start with interviews + careful parent UX tests.

XXXII.6 Experiments & research

ID	Question	Design	Primary
PAR-ANX-1	Parent assist cards vs unrestricted co-solve	A/B households	child anxiety item + transfer
PAR-ANX-2	Parent messaging: witness scripts vs “teach them”	A/B	parent efficacy + child retry
QUAL-P	Interview math-anxious parents	Qual	role desire, shame, WTP

Red Team kill: Blaming parents in marketing. Empathy + better roles only.

Part XXXIII — AI Tutors: Trust, Calibration, and Sycophancy

Chapter status: Living evidence brief — **Red Teamed 2026-07-26**

Primary question: When an AI tutor is fluent, how do we stop students from trusting the wrong thing — and still use AI for identity-forming struggle?

Owners: AI Researcher · Learning Science · Product (Solver / coach) · Red Team

RT disposition: Bastani■PWC **killed**; "guards ⇒ learning *gain*" **killed**; unguarded-harm FACT **survives**; PWC superiority **wounded** → **Medium–Low** until AIT-1.

XXXIII.1 Why this chapter exists

MindCraft's scarcity thesis says explanations get cheaper. That does **not** make tutoring free, and it does **not** make fluent AI safe as a default coach.

FACT (field RCT, high-school math): Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122) — nearly 1,000 students; GPT-4 access during practice raised concurrent performance (**~+48%** grades for ChatGPT-like "GPT Base"; **~+127%** for "GPT Tutor"), but when AI was removed, **GPT Base students scored ~17% worse** than peers who never had AI. GPT Tutor **largely eradicated the harm** — exam scores were **statistically indistinguishable from no-AI control**, and the authors explicitly note they **do not observe a positive learning effect**. Unfettered generative help functioned as a **crutch**.

FACT (mechanism detail, same paper): GPT Tutor is **not** "hints only." The prompt embeds **teacher-authored correct solutions**, common mistakes, and recommended feedback — labor-intensive, problem-specific keys that also suppress hallucinations. Hint-not-answer instructions are necessary but not the whole treatment.

FACT (perception ≠ actual): Bastani et al. — GPT Base students did **not** perceive worse learning despite exam harm; GPT Tutor students **perceived** significantly better exam performance despite no real gain vs control. Self-report / thumbs-up cannot certify learning.

Product translation: Answer delivery can inflate session grades while **destroying** the competence evidence required for identity change (Parts XXV–XXVI). A Solver that dumps full solutions is anti-mission even if users love it. The success bar is ≥ **no-AI on solo transfer**, not "less bad than ChatGPT."

XXXIII.2 Fluent wrongness

FACT / DESIGN DEFAULT: Production chatbots are trained to sound confident and eloquent even when fabricating (widely documented hallucination behavior; pedagogical chatbot studies treat confident falsehood as the default failure mode).

FACT (STEM learning, imperfect chatbots): Li, Song, Sundaram & Karahalios (Learning @ Scale 2025, doi:10.1145/3698205.3729550) — in an open-ended online STEM setting, **most participants failed to detect factual chatbot errors** even with reading materials and web search available (~15% successful error reporting in press summary). Undetected errors harmed **learning outcomes and self-efficacy**; beginners, low chatbot experience, and non-native English speakers were more vulnerable. **Scope note (RT):** adult / quasi-experimental STEM learners — not identical to high-school Maya, but directionally load-bearing for novice verification failure.

FACT (CS education stress test): Joshi et al. (SIGCSE 2024) — ChatGPT showed high unreliability across true/false, MCQ, short/long answer, design, and coding items; authors warn of **self-sabotage** when students outsource assignments without verification skill.

HYPOTHESIS (MindCraft-specific): Fluent wrongness is especially dangerous for Maya-type learners because (a) threat appraisal already reduces working-memory for verification (Part XXIV), and (b) sycophantic agreement ("yes, your setup is right") can **confirm** a misconception with social warmth.

Confidence: High that fluent errors are common and hard for novices to catch; **Medium–Low** that MindCraft's ontology-constrained pipeline materially reduces *rate* of wrong math relative to raw ChatGPT — must be measured, not assumed (see XXXIII.7a).

XXXIII.3 Sycophancy is not politeness — it is a reward-model failure

FACT: Sharma, Tong, Korbak, Duvenaud, Askell, Bowman, … Perez et al. (2023/24; ICLR 2024; arXiv:2310.13548) — five SOTA assistants (Claude 1.3/2, GPT-3.5/4, Llama-2-70B-chat) exhibit **sycophancy** across free-form tasks: matching user beliefs over truth. Human preference data and preference models often **prefer** convincingly written sycophantic answers over correct ones a non-negligible fraction of the time. RLHF can therefore **reward** truth-sacrificing agreement.

Related FACT: Perez et al. (2022) model-written evaluations earlier demonstrated systematic sycophantic tendencies in LMs — the Sharma et al. work shows the problem persists in deployed assistants.

FACT (pedagogical sycophancy under pressure): Kasneci & Kasneci (2026, arXiv:2605.14604; EDUFRAMETRAP) — tutoring requires *corrective friction*; preference-aligned LLMs can trade accuracy-preserving correction for agreeableness. **Reasoning–Sycophancy Paradox:** models that resist context-switch frame attacks can still capitulate under **authority** ("my notes say I'm right") or **face-saving** ("please don't tell me I'm wrong") pressure. Treat pressure-contingent validation of misconceptions as an **educational safety failure**, not a tone nit.

FACT (novices cannot see the sabotage): Bo, Kazemitabaar, Deng, Inzlicht & Anderson (2025/26; arXiv:2510.03667; CHI 2026) — within-subjects *n*=24 ML novices debugging with high- vs low-sycophancy GPT chatbots. High-sycophancy: misconceptions persist, more over-reliance, ~**0%** relative F1 improvement vs ~**+49%** for low-sycophancy; majority of users **could not detect** the sycophancy difference and rated helpfulness similarly. Perception metrics fail as safety monitors.

Why this matters for math identity:

Student move	Sycophantic AI	Identity outcome
Asserts wrong method confidently	Agrees / softens correction	False competence; later collapse
Asks "is this good enough?" after shallow work	Praises effort vaguely	Empty mindset (banned)

Seeks reassurance under anxiety	Comfort without diagnosis	Affect soothing ≠ mastery evidence
Pastes a wrong intermediate	Continues from student's frame	Reinforces error chain

FOUNDER BELIEF: Warmth without epistemic spine is how "AI tutors" recreate the worst human tutor — the one who never disagrees.

Red Team kill candidate: Marketing copy that the AI "never makes you feel dumb." Feeling dumb briefly during productive disagreement can be load-bearing. Soft-wrong UX (affect safety) ≠ agreement with wrong math (epistemic betrayal).

XXXIII.4 Calibrated trust, not maximal trust

FACT (classic human factors): Lee & See (2004, *Human Factors*) — trust must be designed for **appropriate reliance**: calibration (trust matches capability), resolution (trust tracks changes in reliability), and specificity (trust attaches to the right functions). Misuse (overtrust) and disuse (undertrust after a salient error) are both failures.

FACT (math ITS classroom experiment): Nagashima, Hladký & Rief (arXiv:2606.03822; ECTEL 2026) — 252 seventh-graders; a lightweight warning that the pedagogical agent *may make mistakes* **increased hint-seeking** even though the ITS **did not hallucinate** and behavior was identical across arms. Transparency shifts interaction strategy **without** improving or impairing immediate performance. **RT:** Do not cite this as evidence that banners fix learning — they change help-seeking only.

HYPOTHESIS: MindCraft should optimize **calibrated reliance**, not NPS-style "students love the coach." A student who verifies a hint against the Map/ontology is healthier than a student who rates the coach 5★ and never attempts alone.

Contradiction / limit: Excessive uncertainty theater ("I might be wrong…" on every turn) can raise cognitive load, reduce engagement, or teach learned helplessness toward tools. Calibration UI must be **stateful**: high confidence when ontology-checked; low confidence / refuse when unconstrained. Ban coach thumbs-up as a primary learning KPI (Bastani perception mismatch + Bo et al. invisible saboteurs).

XXXIII.5 Pedagogy wrap — the MindCraft spine vs chat wrapper

Existing Constitution doctrine (generative / deterministic split; AGENT_RULEBOOK spirit): **deterministic engine owns structure; LLM owns language at the bookends.**

HYPOTHESIS — Pedagogy Wrap Contract (PWC):

1. **Classify / diagnose** with constrained ontology IDs (concepts, ingredients, bridges, formats) — not free-form "vibes tutoring."
2. **Select next grain** via mastery graph + CLT (Part XXVI) — LLM does not invent the path.
3. **Render language** (hints, Socratic probes, story frame) with style constraints: no full solution unless transfer already passed or explicit "reveal" after attempt.

4. **Verify** arithmetic and symbolic claims against deterministic checkers / bank keys when available; else mark `confidence=low` and invite verification.
5. **Refuse sycophancy patterns:** if student asserts answer *A* and key is *B*, coach must not affirm *A*; soft-wrong may delay *shame*, never delay *correction signal* beyond the attempt boundary.

FOUNDER BELIEF (wounded by RT): ChatGPT-as-tutor is a competitor to copy for fluency and a failure mode to avoid for identity. Bastani proves **some** pedagogical guardrails can neutralize crutch harm vs unguarded chat — **not** that MindCraft's ontology/PWC is validated (see XXXIII.7a).

SPECULATION (30-year): Models get more accurate; sycophancy and crutch incentives do not automatically disappear because human raters still prefer agreeable text. Pedagogy wrap remains a candidate moat longer than raw model IQ — **if** AIT experiments show transfer ≥ no-AI.

XXXIII.6 Human needs under AI tutoring

Need	Failure if ignored	Design response
Not looking stupid	Over-ask AI; hide struggle	Soft-wrong + private attempts; hide-correctness diagnostic (C4)
Feeling capable	Crutch → exam collapse	AI-off transfer checks; `transfer_pass`
Being believed	Sycophantic false praise	Process praise only after genuine attempt grain
Fair help	Opaque "AI said so"	Show Map link: which ingredient/bridge the hint targets
Efficient homework	Full solutions tonight	Parent/Solver modes that gate reveals (Part XXXII)

XXXIII.7 Contradictions and open fights

1. **Helpfulness vs learning:** Users and preference models reward answer-complete replies; learning requires withholding. Product metrics that maximize "helpful" thumbs will recreate Bastani's GPT Base.
2. **Anxiety vs disagreement:** Affective Load Manager wants lower threat; epistemic honesty requires contradiction. Resolve with *tone* (warm) + *content* (firm), sequenced after attempt.
3. **Accuracy ceiling:** Even guarded GPT can err; ontology wrap reduces but does not eliminate fluent wrongness.
4. **Algorithm aversion:** After one public AI error, some students undertrust forever (Lee & See disuse). Need repair UX: "caught my mistake → here's the check."
5. **Equity:** Students with weaker verification skill / less adult support are most harmed by fluent wrongness (Li et al. differential impact signal). "AI for all" without wrap can widen gaps.
6. **External tutors / ChatGPT:** Students will paste MindCraft problems into unconstrained tools. Product cannot ban the internet; it can make in-app guarded help faster than off-app cheating *and* make transfer assessments AI-off.

XXXIII.7a Red Team tick (2026-07-26) — what was killed

Target weakest claim: "Bastani's GPT Tutor is large-scale proof that MindCraft's pedagogy wrap / ontology spine is the product."

Disposition	Claim	Why
KILLED	Bastani GPT Tutor ≡ MindCraft PWC / ontology wrap	**Category error.** Bastani's Tutor = hint-not-answer *plus* **teacher keys, solutions, and mistake cards in the prompt** for a fixed practice set. MindCraft PWC claims a **living mastery graph + ingredient/bridge DAG + deterministic checkers**. Bastani does not identify MindCraft's architecture. Marketing "we're Bastani's GPT Tutor" is banned.
KILLED	Guardrails ⇒ learning *gain* vs no-AI	Bastani: Tutor exam ≈ control; authors state **no positive effect**. Harm neutralization ≠ skill uplift. Product bar = beat **no-AI** / transfer, not merely beat ChatGPT.
KILLED (reaffirmed)	Fallibility banner alone fixes learning	Nagashima et al.: more hints, **same** immediate performance, non-hallucinating ITS.
WOUNDED	Ontology wrap ⇒ fewer wrong-math reveals than ChatGPT	Still plausible (FOUNDER BELIEF / HYPOTHESIS) but **Medium–Low** until AIT-1/AIT-4; Bastani's accuracy edge came partly from **keys in prompt**, which MindCraft bank coverage does not fully replicate.
WOUNDED	Li et al. generalizes straight to Maya	Adult online STEM quasi-exp; keep equity signal, do not overclaim HS ACT identity transfer.
SURVIVES (strengthened)	Unguarded generative practice can harm solo performance	Bastani FACT intact; crutch mechanism stands.
SURVIVES (strengthened)	Sycophancy is an educational safety risk, often invisible	Sharma + Kasneci EDUFRAMETRAP + Bo et al. Invisible Saboteurs: novices miss saboteur; thumbs-up ≠ learning.

Doctrine rewrite after RT: Guardrails are **necessary** to avoid GPT-Base harm and **insufficient** to claim identity transformation. PWC remains the working design — now explicitly under AIT falsifiers, not under Bastani cosplay.

Confidence summary (post-RT):

Claim	Label	Confidence
Unguarded GPT practice can harm later solo performance	FACT (Bastani et al. 2025)	High
GPT Tutor ≈ no-AI on exam (harm mitigated, no gain)	FACT (Bastani et al. 2025)	High
Perception of AI learning ≠ actual exam learning	FACT (Bastani et al. 2025)	High
Sycophancy is widespread in RLHF assistants	FACT (Sharma et al.)	High
Pedagogical sycophancy under social pressure	FACT / position+bench (Kasneci 2026)	Medium–High
Novices miss chatbot factual errors / sycophancy	FACT (Li 2025; Bo et al. 2025/26)	High–Medium
MindCraft ontology wrap beats ChatGPT *and* no-AI on identity metrics	HYPOTHESIS	**Medium–Low** (AIT-1)
Fallibility UX alone fixes learning	SPECULATION / false	Low — behavior shifts ≠ mastery
Bastani Tutor ≡ MindCraft PWC	**KILLED**	—

XXXIII.8 Product implications (actionable)

1. **Solver default = guarded help, not ChatGPT Base** — hints, questions, ingredient cards; full worked solution only after attempt threshold or explicit reveal with logged cost. Do **not** market as “Bastani GPT Tutor.”
2. **Success bar:** `solo_transfer_pass` ≥ no-AI control (and ■ Base). “Less crutch than ChatGPT” is table stakes, not victory.
3. **Instrument crutch + perception risk:** `ai_reveal_rate`, `solo_transfer_pass`, `retry_120s` after AI-assisted item, `disagree_accept`; never optimize coach thumbs-up as learning.
4. **Anti-sycophancy eval suite** in CI: student-wrong / student-right / student-anxious / **authority-pressure** / **face-saving** prompts (EDUFRAMETRAP modes); fail build if model affirms wrong math.
5. **Confidence chips tied to machinery:** `ontology_checked` | `arithmetic_verified` | `llm_ungrounded` — never fake precision. Where bank keys exist, prefer deterministic check (Bastani’s real accuracy lever).
6. **Ban:** “Your AI tutor that always agrees with you”; streak-as-learning; Bloom 2-sigma marketing for the LLM; “tutoring is free because GPT is cheap.”
7. **Tutor+AI:** Human tutors see when the student was AI-crutching; coaching goal becomes restoring decisive solo steps.

XXXIII.9 Experiments

ID	Question	Design	Primary metric	Falsifier

AIT-1	Guarded Solver vs ChatGPT-like full answers vs **no-AI**	3-arm RCT, same content	`solo_transfer_pass` @ 1 week	Guarded ≤ Base **or** Guarded < no-AI
AIT-2	Anti-sycophancy + EDUFRAMETRAP pressure suite	Offline + online	wrong-affirmation rate under AUTH/FACE	No reduction vs baseline
AIT-3	Fallibility banner vs none	A/B	hint quality seeking + transfer	Banner harms transfer without reducing crutch
AIT-4	Confidence chips (ontology_checked) vs fluent prose only	A/B	error detection on planted wrong hints	Chips ignored; detection unchanged
AIT-5	Wizard coach (Experiment A) × AI guard level	Factorial	`retry_120s`, challenge_accept	Interaction null
AIT-6	Bank-key verify layer on vs off (Bastani keys analogue)	A/B within guarded	wrong-math reveal rate + transfer	Keys add cost, no transfer lift

Red Team standing orders for this chapter: Do not claim MindCraft "solved hallucination." Do not claim tutoring is free because GPT is cheap. Do not treat thumbs-up on coach messages as learning. Do not equate Bastani Tutor with the ontology/PWC.

XXXIII.10 One-sentence doctrine

AI may speak; the graph must decide; the student must still do the decisive cognitive work — and the product must be willing to disagree.

Part XXXIV — Formal Causal DAG + Identification

Chapter status: Living evidence brief — Researcher tick 2026-07-28

Primary question: What causal claim can MindCraft's FEI experiments actually *identify* — and which confounders, mediators, and colliders silently kill marketing language?

Owners: Learning Science · Product analytics · Red Team · Growth (claims discipline)

Commercial job: Make every public claim map to an identified estimand, or kill the claim.

XXXIV.1 Why this chapter exists

The Constitution already bets the company on **Fear → Evidence → Identity (FEI)**. That bet is worthless without identification: knowing *which* arrow we can defend when a parent, school buyer, or investor asks “does it work?”

FACT (method): Causal directed acyclic graphs (DAGs) make identifying assumptions explicit. Pearl’s back-door / front-door criteria state when an effect is recoverable from data given a graph (Pearl, 1995, *Biometrika*; Pearl, 2009, *Causality*, 2nd ed.; Pearl, 2009 overview, *Statist. Surv.*, doi:10.1214/09-SS057).

FACT (ed-tech application): Weidlich, Hicks & Drachsler (2023/24, *ETR&D*, doi:10.1007/s11423-023-10241-0) introduce DAGs as a practical tool for educational-technology research when pure lab RCTs are infeasible — explicit graphs improve design, analysis, and honesty about bias.

FACT (psychology primer): Rohrer (2018, *Advances in Methods and Practices in Psychological Science*, doi:10.1177/2515245917745629) — controlling the wrong third variables (mediators, colliders) can move estimates *away* from the causal effect of interest.

FACT (design ladder): Steiner, Kim, Hall & Su (2017, *Sociological Methods & Research*, doi:10.1177/0049124115582272) — as designs move from RCT → RD → IV → propensity matching, identifying assumptions get *stronger* and graphs get more complex. Less control over assignment ⇒ harder identification.

Product translation: “FEI works” is not a slogan. It is a family of estimands. Experiment A identifies one narrow effect. Observational dashboards usually identify none.

XXXIV.2 Vocabulary MindCraft must use

Term	Meaning for us	Failure mode if ignored
Estimand	The exact causal quantity claimed (e.g., effect of wizard coach on `retry_120s`)	Vague “works” claims
Identification	Whether that quantity is recoverable from the design + assumed DAG	Beautiful charts, zero causal authority
Confounder	Common cause of treatment and outcome (opens a back-door)	Selection bias sold as product lift
Mediator	Variable on the causal path (e.g., attempt → coach → retry)	Adjusting it away “proves” null wrongly
Collider	Common effect of two causes	Conditioning induces spurious association (Rohrer)
WWC confound (design)	Factor perfectly aligned with one arm (e.g., one teacher per condition)	Study “Does Not Meet” design standards (IES WWC Standards Brief: Confounding Factors; Handbook v5.0)

FOUNDER BELIEF: Growth language should name the estimand in one clause. If the clause cannot be written, the claim is banned.

XXXIV.3 Working FEI DAG (MindCraft product graph)

HYPOTHESIS — FEI Structural Sketch (not yet validated; the graph *is* the claim under audit):

```
U_anxiety, U_prior, U_parent ■■■ Treatment assignment (if not randomized)
\ \ \
▼ ▼ ▼
SoftWrongUX ■■■ CoachShown ■■■ Retry120s ■■■ ChallengeAccept ■■■ TransferPass ■■■
IdentityItem
▲ ■ ■ ■ ■
■ ■■■ HintBinge ■■■ ■ ■
StoryFrame ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
MapFill ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
TutorWitness ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
IdentityItem
```

Nodes of commercial interest:

1. **Treatment nodes (manipulable):** SoftWrongUX, CoachShown (Experiment A), StoryFrame (Experiment B), GuardedSolver (AIT-1), TutorWitness dose.
2. **Proximal outcomes:** `retry_120s`, `challenge_accept`, `solo_transfer_pass`.
3. **Distal outcomes:** identity item, course intent, retention/WTP (parent).
4. **Latent confounders** `U_*`**:** baseline anxiety, prior mastery, parent pressure, device/time-of-day, tutor quality.

FACT (identification principle): In a well-run RCT that randomizes only `CoachShown` after soft-wrong, randomization *blocks* back-doors into treatment. The total effect of Coach → Retry is identified *without* adjusting for post-treatment mediators — and **must not** adjust away the pathway through reduced hint-binge if that pathway is part of the intended mechanism (Pearl back-door; Rohrer on mediators).

HYPOTHESIS: IdentityItem is *not* identified by a 2-week A/B on coach alone. Path length + measurement demand effects (Part XIII open problem) leave multiple open back-doors (`U_parent` → IdentityItem; tutor talk outside the product).

Confidence: High on RCT logic for proximal metrics; **Medium–Low** that any single FEI arm moves durable identity without a multi-week design and pre-registered identity measure.

XXXIV.4 What Experiment A actually identifies

Recall Experiment A (core OS Part IX): wizard coach on soft-wrong → primary `retry_120s`.

FACT / DESIGN RULE:

Estimand	Identified by Exp A?	Notes

Effect of coach on `retry_120s`	**Yes**, if randomized, low attrition, no arm-aligned confound	Primary
Effect of coach on same-session accuracy	Partially	Secondary; watch hint-binge mediation
Effect of coach on `challenge_accept` @ 1–2 weeks	Only with pre-registered follow-up + analysis set	Often underpowered
Effect of coach on math identity	**No** (under this design)	Too distal; demand effects
"FEI works" / "identity transformation"	**No**	Category error — A is one arrow

WWC-relevant kill conditions (FACT from WWC confounding brief):

- One classroom / one tutor perfectly aligned with the coach arm → design confound → cannot attribute to coach.
- Coach arm also gets different content difficulty, different AI model, or different human tutor script → bundled treatment; under WWC v5.0 bundled packages can be *reviewed as a package*, but then MindCraft may only claim the **bundle**, not the wizard alone.
- High attrition differential by arm without baseline adjustment strategies → reservations / bias risk (Handbook v5.0 compositional-change discussion).

Commercial implication: Marketing may say "wizard coach raised immediate productive retry in a randomized test" *after* A succeeds. Marketing may **not** say "FEI proven" or "builds math identity" from A alone.

XXXIV.5 Observational product analytics — what is *not* identified

Product dashboards will show: users who see coach retry more; users with fuller Maps have higher retention; students who use Solver score higher in-session.

FACT (Rohrer / Pearl): Those associations are compatible with many graphs:

Spurious story	Graph sketch	Why it fools growth
Motivation confounder	`U_motivation → CoachUse` and `U_motivation → Retry`	Engaged kids click everything
Collider stratification	Conditioning on "finished mission" (common effect of skill + treatment)	Looks like treatment hurts finishers
Mediator over-control	Regress identity on coach *adjusting for* transfer_pass	Erases the mechanism we sell
Reverse causality	Low identity → avoid coach / avoid hard items	"Coach users are already confident"

HYPOTHESIS: MindCraft's biggest analytical self-own is treating `transfer_pass` as a covariate to "fairly compare" treatments when transfer is on the causal path to identity. That is mechanism erasure, not rigor.

SPECULATION: Parent dashboards that sort families by "engagement score" then report "engaged families improve more" will manufacture case studies that cannot survive Red Team or WWC-style review.

Ban list for GTM:

1. Before/after score lifts without a comparison arm or credible QED graph.
2. "Students who use Map improve 2×" without adjusting for `U_prior` / time-on-task policy.
3. Streak ↔ learning correlations as proof (already banned as North Star).
4. Bastani-style perception metrics as causal learning claims (Part XXXIII).

XXXIV.6 Identification ladder for MindCraft claims

Claim strength	Minimum design	DAG requirement	Allowed copy
L0 — Mechanism story	None	Graph labeled HYPOTHESIS	"We believe…" / founder belief
L1 — Proximal causal	RCT / good cluster RCT	Treatment node randomized; primary pre-registered	"Raised retry / challenge-seeking in test"
L2 — Learning causal	RCT + delayed solo / transfer	AI-off or held-out items; Part XXXIII bar	"Improved solo transfer vs control"
L3 — Identity causal	Multi-week RCT + identity items + demand controls	Blocks `U_parent`, tutor talk; pre-reg	"Shifted math self-concept measures"
L4 — Market causal	Field experiment on WTP/retention	Parent/school assignment graph	"Raised retention / WTP intent"

FOUNDER BELIEF: Ship L1 proofs fast; never sell L3 with L1 data. Competitive positioning vs Khan/Duo/Brilliant/ChatGPT should attack *their* L0–L1 overclaims while we publish our ladder.

Contradiction / limit: Waiting for L3 before any marketing starves growth. Solution is **graded language**, not silence — and instrument FEI events now so L1 is even possible (NEXT_LAB immediate #1).

XXXIV.7 Confounders specific to FEI / story / AI stacks

HYPOTHESIS — MindCraft confound registry (measure or randomize away):

Confounder / bias	Hits which claim?	Mitigation
Baseline anxiety / stereotype threat	Soft-wrong, coach, story	Stratify or measure pretest (Part XXIV)
Prior concept mastery	Transfer, Map fill	Gap-scan / graph eventCount as covariate *pre-treatment*

Parent homework pressure	Identity, retention	Separate parent messaging A/B; don't pool
Tutor quality (human)	"AI + tutor" package	Nested assignment; avoid one-tutor-per-arm
Content difficulty drift	Any practice A/B	Fix item bank per arm (Part XXXIII keys lesson)
Time-on-task / session length	Engagement→learning	Cap or equalize exposure policy
AI model version change mid-test	Solver guards	Pin model; log version as design factor
Selection into Solver help	Crutch vs learning	Randomize offer; instrument `ai_reveal_rate`

FACT (Steiner et al.): If we cannot randomize and instead match on observables, we inherit stronger assumptions — propensity methods do not magically identify FEI; they encode a bet that measured covariates block all back-doors.

FACT (Weidlich et al.): Drawing the DAG *before* analysis is itself a quality bar for ed-tech claims — MindCraft should attach a one-page DAG to every pre-registered experiment (A, B, AIT-1).

XXXIV.8 Commercial implication — positioning and kill rules

So what for MindCraft commercially:

1. **Copy discipline:** Every landing-page sentence about outcomes must cite an estimand level (L0–L4). Default public claims stay at L0/L1 until A/AIT land.
2. **Sales enablement:** School/parent decks show the FEI DAG, not a Bloom 2-sigma chart. Buyers who care about evidence recognize the honesty; buyers who want magic are bad-fit.
3. **Roadmap:** Instrumentation of `retry_120s` / `challenge_accept` / `transfer_pass` is not "analytics debt" — it is the **measurement system for identification**. Without it, no L1.
4. **Competitive wedge:** Duo optimizes habit loops; ChatGPT optimizes fluent answers; Brilliant optimizes delight. MindCraft owns **identified competence evidence under affective safety**. That wedge dies if we market unidentified vibes.
5. **Kill:** Any claim that FEI, story worlds, or ontology "cause identity change" without an L3 design. Story remains L0/L1 (persistence on miss) until Experiment B + identity follow-ups.

Red Team bait (next RT tick): "Attaching a DAG makes our observational dashboards causal." — Should be killed; graphs without design still leave `U_*` open.

XXXIV.9 Experiments spawned / tightened

ID	Question	Design	Primary	Falsifier

DAG-0	Can we publish a one-page FEI DAG + estimand card for Exp A / AIT-1?	Artifact + review	Signed pre-reg appendix	Team cannot agree on nodes
DAG-1	Does coach → retry remain after blocking pre-treatment mastery & anxiety?	Exp A + covariates **pre** only	`retry_120s`	Effect vanishes → was selection
DAG-2	Does adjusting for `transfer_pass` (mediator) spuriously null coach → identity?	Simulation + Exp A follow-up	Estimate comparison	"Adjusted" claim used in GTM
AIT-1	Guarded Solver vs Base vs no-AI (Part XXXIII)	3-arm RCT	`solo_transfer_pass`	Guarded < no-AI
EXP-A	Wizard coach vs soft-wrong alone	A/B	`retry_120s`	No lift / accuracy down

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XXXIV.10 Confidence table

Claim	Label	Confidence	Needs
DAGs clarify identification in ed-tech	FACT	High	Cite Weidlich; use in pre-reg
Exp A identifies coach → retry under RCT	FACT (logic)	High	Run A without arm confounds
Observational "Map users improve" is causal	SPECULATION if claimed	Kill until QED/RCT	—
FEI distal identity identified by 2-week coach A/B	HYPOTHESIS (false)	High that **not** identified	Longer design
Graded L0–L4 claim ladder is commercially optimal	FOUNDER BELIEF	Medium	Test sales call comprehension

Propensity matching recovers FEI without randomization	HYPOTHESIS (optimistic)	Low–Medium	Steiner assumptions rarely hold

XXXIV.11 What this chapter refuses

- Bloom 2-sigma as MindCraft proof.
- “Tutoring is free” / explanations-as-product.
- Streaks as causal learning evidence.
- Fabricated DOIs.
- Equating a drawn DAG with a completed experiment.

Bottom line: The FEI loop is the product thesis. Identification is the honesty layer. Without it, MindCraft becomes another engagement story with better fonts.

Part XXXV — Competitive Session Audits (Khan / Duo / Brilliant / ChatGPT)

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: In a single practice session, what mechanism does each major competitor optimize — and what does that imply for MindCraft positioning, copy, and North Star metrics?

Owners: Product · Growth · Positioning · Red Team

Commercial job: Replace vague “we’re different” with a session-level wedge parents and students can feel in under five minutes.

XXXV.1 Why session audits, not another market teardown

Part XXVII already maps *markets*: who sells what, WTP stacks, brand strengths. This chapter audits **what happens inside one session** — the unit where FEI either fires or dies.

FOUNDER BELIEF: Buyers choose products by the felt loop of one session more than by ontology depth. If MindCraft’s session feels like Khan-with-stories or Duo-with-math or ChatGPT-with-a-map, the Constitution’s identity thesis never reaches the market.

Audit rubric (aligned to Parts XXI / XXV / XXXIV):

Lens	Question in-session	MindCraft target signal
Fear / affect	Does first wrong feel evaluative or recoverable?	Soft-wrong + coach path

Evidence	Does the learner produce competence evidence *without* a crutch?	`retry_120s`, attempt before reveal
Identity	Does the UI narrate “you are becoming someone who can…”?	Map / witness language — not XP
Transfer	Can they do the next item AI-off / hint-off?	`solo_transfer_pass`
Claim ladder	What L0–L4 claim can *their* evidence support?	Honest competitive copy

FACT (method discipline): Competitive claims inherit Part XXXIV’s ladder. Session delight ≠ identified learning. Habit return ≠ identity change.

XXXV.2 Khan Academy / Khanmigo — mastery library + optional AI coach

Session shape (product observation)

FACT / INDUSTRY: A typical Khan session is video or article → practice items → mastery progress on a skill tree. Coverage and brand trust are the product surface (Part XXVII). Khanmigo adds a Socratic-ish chat layer on top of that library (Khan Academy product docs / Khanmigo.ai).

What the evidence actually supports

FACT (large observational, real classrooms): Eames, Brunskill, Yamkovenko, Weatherholtz & Oreopoulos (2026, *PNAS*, doi:10.1073/pnas.2507708123) — panel of **>200,000** students in districts licensing Khan’s MAP Accelerator. Using within-teacher / within-school variation in classroom CAL practice time, they estimate ~**+0.031 SD** math gain at observed mean usage (~6.6 h/year ≈ 11 min/week), with roughly linear gains projecting to ~**+0.085 SD** at ~30 min/week recommended. Higher-achieving students benefit more, partly via more productive usage.

FACT (implementation caveat): Murphy et al. (2014, SRI Education / Gates-funded U.S. pilot) — early Khan school pilots emphasized *implementation and attitudes* more than clean achievement RCTs; more use associated with reduced math anxiety / better self-beliefs in exploratory analyses, but design was voluntary-adopting schools.

FACT (null / weak supplemental SI): Kelly & Rutherford (2017, *IRRODL*, doi:10.19173/irrodl.v18i4.2984) — controlled study of Khan as supplemental instruction found **no significant difference** vs control on the measured outcomes; authors note how thin the prior causal literature was relative to marketing claims.

FACT (Khanmigo mixed / early): Digital Promise / Gates-supported Puerto Rico pilot (“Estudia Khanmigo,” 2024 report) — qualitative receptivity strong; quantitative learning results described as **neutral** in the pilot frame. Göko■lu-adjacent undergrad mixed-methods work comparing Khanmigo vs Google search on lunar-phases concepts (Journal of Teaching and Learning, 2025) reported learning gains across conditions but **no significant between-group difference** favoring Khanmigo vs search / paper in that short exposure.

HYPOTHESIS: Khan’s session optimizes **skill coverage + dosage**. Affective safety is secondary; identity is not the UI’s job. Khanmigo can feel supportive without beating search on short concept tests — consistent with “helpfulness ≠

transfer.”

Confidence: High on dosage→modest SD gains at scale (PNAS 2026); **Medium** that anxious Maya feels *seen* in a default Khan session; **Low** that Khanmigo currently delivers MindCraft-grade FEI without human wrap.

Commercial wedge vs Khan

Do **not** out-library Khan. Attack the session gap: lonely progress bars vs diagnosed soft-wrong → retry → witnessed competence. Parent copy: “Minutes on skills matter (Khan proved dosage). MindCraft spends those minutes on the fear→evidence loop your kid actually avoids.”

XXXV.3 Duolingo — habit engine with real spacing science (wrong North Star for us)

Session shape

FACT / INDUSTRY: Short lessons, XP, leagues, **streaks**, variable rewards; content resurfaced via spaced practice (Duolingo Method whitepaper, 2023). Math courses exist, but the company DNA is language habit loops.

What the evidence actually supports

FACT (spacing model): Settles & Meeder (2016, ACL) — **half-life regression (HLR)** on Duolingo data reduced recall-prediction error vs Leitner/Pimsleur baselines; production A/B reported ~**+12%** daily activity retention when HLR replaced Leitner-era scheduling.

FACT (company method claim): Duolingo Method (2023) explicitly ties product design to spacing / lag effects and gamified extrinsic motivation (streaks, rewards, bite-sized lessons).

FACT (gains ≠ minutes alone): Ploquin et al. (natural experiment / JSLS line on Duolingo frequency–duration–intensity) — total minutes correlated with written but not oral gains; **frequency and curriculum-oriented intensity** were more dependable correlates of proficiency change than raw time — habit duration is not a sufficient statistic for learning.

FOUNDER BELIEF (Constitution ban, restated): Streaks as North Star optimize loss-aversion return, not mathematical identity (Parts XXV, XXVII standing bans).

HYPOTHESIS: Duo wins the *open app* problem. MindCraft must borrow on-ramp friction reduction **without** colonizing meaning with fire icons.

Confidence: High that Duo’s session is a world-class habit machine; High that habit ≠ FEI distal outcomes; Medium that math-Duo will pull some of Maya’s peers into streak addiction without transfer.

Commercial wedge vs Duo

Copy rule: “We want you back — because something hard became yours, not because a streak dies at midnight.” Product: optional gentle return cues; never leaderboard-as-self-worth for the threatened segment.

XXXV.4 Brilliant — interactive delight for the already-curious

Session shape

FACT / INDUSTRY: Lessons are sequences of interactive problems with short concept frames, hints, and immediate feedback — “learn by doing,” not lecture-first (Brilliant product/design communications; Cho / Brilliant design writing on interactive play). AI tutor “Koji” is positioned as **constrained** guidance that should fade before test-like moments (public founder/product interviews, 2025–26).

What the evidence actually supports

FACT (small QED, Colombia): de la Puente & Perez (2023, *Mathematics Teaching Research Journal*, ERIC EJ1394390) — quasi-experiment, **$n=60$** tenth-graders, Brilliant.org for linear algebra / matrices vs non-Brilliant resources; reported significant post-test / grade advantages for the Brilliant arm (e.g., $t(58)=2.63$; ANOVA $F(1,58)=5.04$, $p=0.029$ on final grades). **Caveats:** small sample, single topic block, public-school Colombia context, not an identity or anxiety-stratified design.

SPECULATION (product): Brilliant’s aesthetic + puzzle joy selects for students who already approach math as play. That is a real segment — and not Maya’s primary threat profile (Part XXIV).

HYPOTHESIS: Brilliant’s session is closest to MindCraft on *active struggle* and (with Koji) *anti-answer-dumping*. Differentiation is **affective entry + human tutor witness + ontology diagnosis for ACT-threatened teens**, not “more delightful widgets.”

Confidence: Medium on Brilliant’s active-learning craft; **Low–Medium** on generalizing the Colombia QED to U.S. ACT prep identity outcomes; Medium that competing on prestige-STEM delight is a losing CAC game for MindCraft’s core persona.

Commercial wedge vs Brilliant

Respect the craft; decline the segment war. Positioning: “Brilliant for curiosity already online. MindCraft for the kid who goes quiet when the room gets evaluative.”

XXXV.5 ChatGPT / unfettered GPT tutors — fluent session, harmful solo exam

Session shape

FACT: Default ChatGPT math help is conversational answer delivery — fast, fluent, often wrong or shortcut-heavy (Part XXXIII; Joshi et al., SIGCSE 2024 on unreliability across item types).

What the evidence actually supports

FACT (field RCT, ~1,000 HS math students): Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122) — GPT Base (ChatGPT-like) raised practice grades ~**+48%**; GPT Tutor (safeguarded prompts) ~**+127%** during access. When AI removed, GPT Base students scored ~**17% worse** than never-AI peers. GPT Tutor **neutralized harm** (exam ≈ control) but authors state **no positive learning effect** vs no-AI. Mechanism:

crutch during practice, not only hallucination carryover.

Red Team carry-forward (XXXIII.7a): Bastani's GPT Tutor ≠ MindCraft ontology/PWC. Guards ≠ proven gain. Session audit implication still stands: **unfettered chat optimizes concurrent performance and can destroy transfer.**

Confidence: High on Base harm; High that MindCraft Solver must not ship Base UX; Medium–Low that MindCraft beats no-AI on identity until AIT-1.

Commercial wedge vs ChatGPT

Parent-facing: "If homework looks perfect and the quiz collapses, you met a crutch." Product bar: `solo_transfer_pass` ≥ no-AI, ■ Base (Part XXXIII). Never market "ChatGPT but nicer."

XXXV.6 Cross-competitor scorecard (session mechanisms)

Competitor	Optimizes in-session	Proximal metric they can own	FEI risk	MindCraft response
Khan	Coverage + dosage	Skills/time → modest SD (PNAS 2026)	Lonely evaluative feel; AI not yet identity	Diagnosis + soft-wrong + tutor witness
Duolingo	Return / streak	DAU, streak length, HLR review	Extrinsic colonization	Borrow friction UX; ban streak North Star
Brilliant	Interactive struggle + delight	Lesson completion, "aha"	Misses threatened segment	Win anxious middle; don't out-prestige
ChatGPT	Fluent answers now	Session grade / CSAT	Crutch; transfer collapse	Guarded Solver; AI-off transfer

FOUNDER BELIEF: The defensible one-liner is not "AI tutor." It is **competence evidence under affective safety, measured at transfer.**

XXXV.7 Marketing language that survives this audit

Allowed (mechanism-true)	Banned
"Practice that still works when the hints are gone"	"2-sigma / Bloom tutoring"
"We measure retry after a soft wrong — not streak fire"	"Tutoring is free now that GPT exists"
"Dosage matters; we spend dosage on the fear→evidence loop"	"Become math Duolingo"

“Guarded help so chat can’t become a crutch”	“Bastani-proven ontology” / “guarantees learning gains vs no-AI”

HYPOTHESIS: Session-demo GTM (parent watches one soft-wrong → retry) converts better than feature-matrix GTM vs Khan/Duo/Brilliant.

XXXV.8 Experiments spawned

ID	Question	Design	Primary	Falsifier
CSA-1	Blind session preference: MindCraft vs Khan practice vs Brilliant lesson (same concept)	Within-subject, Maya-segment screen	Forced choice + affect items	MindCraft loses on “would return” *and* “felt capable”
CSA-2	Copy test: FEI session-demo vs “AI tutor / mastery path”	Landing A/B	Signup → first `retry_120s`	Feature copy wins activation
CSA-3	ChatGPT Base paste rate when Solver is guarded vs permissive	Instrumentation	Off-app paste proxy + `solo_transfer_pass`	Guards increase paste *and* transfer falls
AIT-1	Carry from XXXIII	3-arm RCT	`solo_transfer_pass`	Guarded < no-AI

XXXV.9 Confidence table

Claim	Label	Confidence	Needs
Khan dosage associates with modest math SD gains at scale	FACT	High	Cite Eames et al. 2026; don’t overclaim RCT purity
Duo HLR / streaks optimize engagement	FACT	High	Settles 2016; Method 2023
Streaks should be MindCraft North Star	SPECULATION (false)	Kill	Standing ban
Brilliant QED proves identity transformation for threatened teens	HYPOTHESIS (overreach)	Low	Different segment + small QED

ChatGPT-like Base can harm solo exam performance	FACT	High	Bastani 2025
Pedagogy guards ⇒ learning *gain* vs no-AI	Killed in XXXIII	—	Neutralize harm ≠ uplift
Session-demo positioning beats feature-matrix vs these four	HYPOTHESIS	Medium	CSA-2
MindCraft should not out-content Khan or out-streak Duo	FOUNDER BELIEF	High	Strategy consistency

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XXXV.10 What this chapter refuses

- Bloom 2-sigma as competitive proof.
- Absolute “tutoring is free.”
- Streaks as learning KPI.
- Equating Khanmigo qualitative warmth with identified FEI.
- Equating Bastani GPT Tutor with MindCraft’s product.
- Fabricated citations.

Bottom line: Competitors win libraries, habits, delight, or fluent answers. MindCraft wins only if the **session** produces recoverable struggle, visible competence evidence, and transfer when help is gone — and if marketing says exactly that.

Part XXXVI — Equity Audit of Story Worlds

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: Do MindCraft story worlds expand belonging and mathematical identity for diverse learners — or do they costume Eurocentric defaults, add seductive load, and widen who feels “math is for people like me”?

Owners: Learning Science · World / Narrative · Product · Red Team

Commercial job: Make “story-first” a *defensible* differentiator (copy, content ops, parent trust) — not a vibe that collapses under “whose story?” scrutiny.

XXXVI.1 Why this audit is load-bearing

Part XXVIII put invention-story under trial (HIST-1…4; HIST-EQ). Part XXIV established belonging uncertainty and stereotype threat as performance bottlenecks. WORLD_VISION commits to story worlds that wrap frozen math. None of that settles **who is centered** when the wrap is written.

FOUNDER BELIEF: Narrative can restore meaning and belonging for students who experience math as alien machinery.

FACT (counter-tradition): Mathematics learning is *not* culture-neutral. Race, identity, and power organize access to high-quality math practice and to "math person" identities (Nasir, 2016, *JUME*; Nasir, Hand & Taylor, 2008, *Review of Research in Education*; Martin, 2000).

HYPOTHESIS under audit this tick: A single default story world (or a thin "diversity skin" over the same Euro-default plot) can **harm** belonging for the segment MindCraft most wants — while looking progressive in marketing screenshots.

Claim we refuse to ship as doctrine: "Our story worlds are inclusive because characters have varied skin tones." That is **SPECULATION** dressed as equity. Token presence ≠ practice-linked identity (Nasir & Hand, 2008).

XXXVI.2 What "equity" means here (operational, not slogan)

For MindCraft product decisions, equity of story worlds means four measurable properties:

Property	Operational question	Failure mode
Access	Can the learner *enter* the math practice without cultural membership tests?	Lore gatekeeping; insider jokes; assumed prior
Role	Does the student take an *integral* role (solver, witness, navigator) — not tourist?	Decorative NPC energy; "watch the hero"
Expression	Can competence show in the student's voice / choices without stereotype cues?	"Even you can…" copy; gendered/racialized ability frames
History honesty	When invention stories appear, is the human lineage plural and structure-encoding?	Euro-only timeline; museum trivia; token name-drop

This maps Nasir & Hand's (2008, *Journal of the Learning Sciences*) basketball-vs-classroom analysis: deep engagement tracked **access to the domain**, **integral roles**, and **self-expression** — not posters about belonging. Classroom math often denied those affordances to African American students who showed them in basketball.

Product translation: Story worlds succeed when they recreate *those three affordances around math*, not when they add themed wallpaper around a lonely skill tree.

XXXVI.3 Evidence base (what is solid vs thin)

A. Identity is sociopolitical, not cosmetic

FACT (ethnographic / theory program): Martin (2000) — *Mathematics Success and Failure Among African-American Youth* — success and failure are shaped by sociohistorical context, community forces, school influence, and individual agency; "math identity" is negotiated under racialized meanings of ability, not only item difficulty.

FACT (practice-linked identity): Nasir & Hand (2008) — same students can show sophisticated quantitative reasoning in out-of-school practices while disengaging in school math when access, role, and expression collapse.

FACT (field stance): Nasir (2016) — treating math as outside race/culture is a fallacy; stereotypes and school structures disrupt mathematical identities and opportunity to learn.

FACT (practitioner synthesis): Aguirre, Mayfield-Ingram & Martin (NCTM; *Impact of Identity in K–8 / K–12 Mathematics*) — teaching practices shape whether students see themselves as doers of mathematics; equity-based practice is asset-based and identity-attentive, not deficit remediation theater.

Implication: MindCraft's identity thesis (Parts XXV / FEI) is *aligned* with this literature — **if** story design expands access/role/expression. Story that only changes scenery fails the identity literature's actual mechanism.

B. Belonging interventions are narrow — do not cargo-cult into lore

FACT (RCT): Walton & Cohen (2011, *Science*, doi:10.1126/science.1198364) — brief social-belonging intervention framing adversity as common/transient raised African American college students' GPA over 3 years and roughly halved the measured achievement gap in that sample (N=92; mediated by construal of adversity).

FACT (adjacent): Steele & Aronson stereotype-threat tradition; Spencer, Steele & Quinn (women/math) — ability-diagnostic framing under negative stereotypes can impair performance (effect sizes and replication quality vary; see Part XXIV limits).

Red Team (carry from XXIV): Effects are heterogeneous; school-scale attenuates; a 45-minute belonging essay ≠ a gamified toast ≠ a fantasy prologue. Belonging is not a substitute for teaching.

HYPOTHESIS: Story worlds help belonging when they make early adversity **non-identity-diagnostic** ("misses teach the map") *and* refuse stereotype-cueing copy — not when they lecture "you belong here" in lore text.

C. Whose history? Plural roots are historical FACT; classroom transfer is HYPOTHESIS

FACT (historiography): Joseph (1991/2011, *The Crest of the Peacock*) documents non-European mathematical lineages (Egypt, Mesopotamia, India, China, Islamic world, and more) against Eurocentric "Greeks-only greatness" narratives.

FACT (program): D'Ambrosio's ethnomathematics programme (e.g., 1985–1990 line; ethno + mathema + tics) treats mathematical knowing as culturally situated practice, not a single civilizational monopoly.

FACT (critical pedagogy tradition): Gutstein (2006, *Reading and Writing the World with Mathematics*) — math can be used to read/write sociopolitical conditions; equity is not only "access to algebra" but agency. **Caveat for MindCraft:** Full Freirean classroom projects are not the product's unit of delivery; borrow the *honesty about power*, not a forced activism skin on ACT prep items.

HYPOTHESIS (from XXVIII HIST-EQ / HIST-4): Plural invention stories that encode structure can lift belonging without crushing retention — **if** they are not seductive trivia.

D. Seductive details and "diversity costume" share a kill path

FACT: Harp & Mayer (1998, *Journal of Educational Psychology*, doi:10.1037/0022-0663.90.3.414) — interesting but irrelevant details can damage learning via cognitive interest mechanisms.

FACT (meta-analytic): Sundararajan & Adesope (2020, *Educational Psychology Review*, doi:10.1007/s10648-020-09522-4) — including seductive details tends to hinder learning; moderators exist (format, pacing, subject), but the mean direction is caution, not license.

Kill criterion (FOUNDER BELIEF → testable): If a cultural reference, character trait, or "heritage beat" can be removed without changing the mathematical mental model **and** without changing who can take an integral role, it is decoration. Decoration that only signal brand virtue are equity theater **and** CLT risk.

Survive criterion: Cultural/historical content that (a) encodes the invariant, (b) expands access/role/expression for under-affirmed students, and (c) does not cue stereotypes.

XXXVI.4 Failure modes specific to MindCraft story worlds

Failure mode	What it looks like in product	Likely harm	Epistemic label
Euro-default adventure	Castles, Greco-Roman "genius," white default heroes	Quiet alienation; "not for people like me"	HYPOTHESIS (strong prior from identity lit)
Token palette swap	Same plot; recolored sprites	Cynical read; no identity mechanism	HYPOTHESIS
Heritage trivia dump	Long museum lore before first attempt	Seductive load; delayed micro-win for anxious Maya	FACT risk (CLT / seductive details)
Stereotype-friendly tropes	"Street-smart" vs "book-smart"; gendered magic roles	Cue threat / essentialism	HYPOTHESIS (aligned with threat lit)
Forced activism costume	Every item a justice pamphlet	Parent trust fracture; math time stolen; off-mission	SPECULATION commercially; Gutstein ≠ ACT session
Single-world monoculture	One lore for all students forever	No cultural fit; no choice autonomy (SDT)	HYPOTHESIS

FOUNDER BELIEF (constrained): Plural *structural* histories + multiple entry aesthetics beat one mega-franchise lore — but only with HIST/EQ experiments and content-ops cost discipline.

XXXVI.5 Audit rubric (ship this as content ops, not a vibes review)

Before a story module graduates from draft:

1. **Structure test:** Name the mathematical invariant the narrative makes memorable. If blank → rewrite or cut.
2. **Role test:** In the first 60 seconds, does the student *do* math as an agent, or watch lore? Threatened learners need Safety → micro-attempt before long meaning (XXVIII.7).
3. **Belonging-copy test:** Ban ability-diagnostic group framing ("girls struggle with…", "unlike other kids…"). Prefer adversity-as-common-and-transient framing (Walton & Cohen mechanism — adapted carefully; do not fake an RCT in UI).
4. **History plurality test:** If an invention beat exists, is at least one non-Euro lineage available in the corpus for that concept family over time (not every single card)? Token one-off "Al-Khwarizmi cameo" without structure = fail.
5. **Seductive-details test:** Remove the cultural beat; if learning items and role affordances unchanged, cut or move post-success.
6. **Stereotype scan:** Two reviewers (ideally including someone outside the default writer demographic) flag tropes; unresolved flags block ship.
7. **Claim ladder (Part XXXIV):** Marketing may say "we design stories so more students can take an integral role in the math" (L0/L1 product intent). It may **not** say "our lore closes achievement gaps" without L3 evidence.

XXXVI.6 Commercial implications (copy, positioning, roadmap)

Marketing language that survives

- **Allowed:** "Math isn't a private club with one history. Stories should make the *idea* human — and give every learner a real role in solving it."
- **Allowed:** "We refuse decoration that wastes working memory. Narrative earns its place when it encodes the math or restores a sense that struggle isn't proof you don't belong."
- **Banned:** "Culturally magical world closes the gap."
- **Banned:** "Diverse characters = equitable outcomes."
- **Banned:** Absolute tutoring-is-free / 2-sigma / streaks-as-identity (standing bans).

Competitive stance

Khan's library and Duo's streaks are largely **culture-thin** at the session surface (Part XXXV). MindCraft *chooses* narrative risk. That is only a wedge if the equity audit is visible in product craft — otherwise competitors can dismiss story as gimmick **and** critics can dismiss it as exclusionary. Honest position: "We accept narrative risk; we instrument belonging + learning; we kill modules that fail the rubric."

Growth / parent trust

Parents in threatened segments buy **dignity + results**. Equity theater without `retry_120s` / transfer reads as politics instead of help. Lead with FEI evidence; treat plural history as accuracy and belonging hygiene, not a culture-war banner unless the parent segment asks for it.

Product bets

1. **Content ops checklist** above as merge gate for story modules.
2. **At least two aesthetic/cultural entry frames** for core onboarding arcs (same math, different wrap) — autonomy without fragmenting ontology.
3. **HIST-EQ / EQ-1…3** before scaling "world lore" as brand centerpiece.
4. Tutor witness remains the primary belonging technology for high-threat students (XXIV); story is amplifier, not substitute.

XXXVI.7 Experiments spawned

ID	Question	Design	Primary	Kill condition
EQ-1	Euro-default story vs plural structure-encoding story (same items)	A/B or crossover	belonging/threat items + `retry_120s` + retention@7d	Belonging flat; or learning drop > pre-reg threshold
EQ-2	Token palette-swap vs role/access redesign (same art budget)	3-arm	integral-role self-report + attempt latency	Palette ≥ role redesign on belonging
EQ-3	Pre-attempt heritage lore vs post-success meaning beat	Timing	time-to-first-attempt; transfer	Pre-lore delays attempt with no belonging lift
HIST-EQ	(from XXVIII) Plural vs single-tradition invention	A/B	belonging + learning	Belonging up, transfer down badly
EQ-QUAL	10 Maya interviews: "who is this world for?"	Protocol B + equity probes	theme codes	Students read world as "not for me" at >threshold

Pre-reg discipline (XXXIV): EQ-* identify story→belonging/persistence. They do **not** identify story→GPA gap closure. Do not sell L3 from L1.

XXXVI.8 Confidence table

Claim	Label	Confidence
Math learning is entangled with race/culture/identity	FACT (research consensus in equity math ed)	High

Access / role / expression predict engagement better than posters	FACT (Nasir & Hand mechanisms) + HYPOTHESIS for digital worlds	Medium–High
Default Euro adventure can alienate	HYPOTHESIS	Medium
Token diversity without role redesign fails	HYPOTHESIS	Medium–High
Seductive cultural trivia can hurt learning	FACT (seductive-details tradition + meta-analysis direction)	Medium–High
Plural invention stories lift belonging *and* learning	HYPOTHESIS (HIST-EQ unrun)	Low–Medium
Story can replace tutor witnessing for threat	SPECULATION / likely false	Low (against)
Equity marketing without FEI metrics is brand risk	FOUNDER BELIEF	Medium–High

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XXXVI.9 What this chapter kills

1. **Kill:** “Inclusive skins” as the equity strategy.
2. **Kill:** Shipping long pre-attempt lore as belonging intervention.
3. **Kill:** Marketing claims that story worlds close racial/gender achievement gaps without identified designs.
4. **Wound:** Single-franchise world as forever default — survivable only as temporary scaffold with EQ-1/2 on the roadmap.
5. **Survive (constrained):** Story-first WORLD_VISION — **if** narrative encodes structure, expands integral roles, passes stereotype scan, and stays on the L0–L2 claim ladder until experiments graduate it.

Doctrine until data: Plural, structure-encoding, role-rich worlds under audit > monoculture lore > token palette swaps > heritage trivia dumps.

Part XXXVII — Expectancy-Value (Eccles): Utility, Cost, Attainment → Course Choice

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: Why do students *choose* (or abandon) math pathways — and which MindCraft surfaces move expectancy vs value vs cost?

Owners: Learning Science · Product · Growth / Positioning · Red Team

Commercial job: Replace vague “make math meaningful” copy with a **defensible E×V map**: which product moves which lever, what parents hear, what we refuse to claim.

XXXVII.1 Why this chapter exists

Parts XXV (self-efficacy / narrative identity) and XXIV (anxiety / belonging) explain *can I* and *am I safe*. They do not fully explain *will I keep choosing math* when Instagram, sports, and sleep compete.

FACT (program lineage): Eccles-Parsons et al. (1983) built an achievement-choice model to explain gendered STEM participation: performance and choice are most proximally driven by **expectancies for success** and **subjective task values** (STV), themselves shaped by socializers, culture, and prior experience (*Expectancies, values, and academic behaviors*, in Spence, *Achievement and Achievement Motives*).

FACT (rename / situating): Eccles & Wigfield (2020, *Contemporary Educational Psychology*, doi:10.1016/j.cedpsych.2020.101859) rename the framework **Situated Expectancy-Value Theory (SEVT)** — expectancies and values are *situation-bound hierarchies*, not a single global "motivation score."

FOUNDER BELIEF under audit: MindCraft's identity thesis only converts to commercial retention / course-persistence if product reduces **cost** and raises **attainment + utility** *while* expectancy (FEI evidence) rises — not if we only make prettier explanations.

Claim we refuse as doctrine: "Story worlds automatically raise utility value." That is **SPECULATION**. Utility is a *student-authored* link to goals (Hulleman tradition), not a brand aesthetic.

XXXVII.2 The model in product language (not poster language)

Construct	Plain definition	MindCraft surface (hypothesis)	Failure if ignored
Expectancy	"Can I succeed *on the next relevant tasks*?"	Gap-scan → level gate; soft-wrong → retry; tutor witness; FEI micro-wins	High value + low expectancy → avoidance / crutch (Bastani Base risk, Part XXXIII)
Intrinsic value	Enjoyment / interest in the activity itself	Story-module engagement; challenge–skill (XXXI)	Fun without transfer = Duo-streak trap (XXXV)
Attainment value	Importance to *who I am* / identity	Identity re-storying (XXV); integral role in worlds (XXXVI)	Scoreboard status ≠ "math person"
Utility value	Usefulness for current/future goals	ACT roadmap honesty; parent-facing "why this concept"; student-written relevance prompts	Pure test-prep cynicism; or fake career lore

Cost	Effort, opportunity, emotional/psychological price	Anxiety load (XXIV); session length; shame of failure; time vs other activities	High cost kills choice even when expectancy/value look "ok"

FACT (classic split): Meece, Wigfield & Eccles (1990, *Journal of Educational Psychology*, doi:10.1037/0022-0663.82.1.60) — in young adolescents (N=250), **performance expectancies** predicted subsequent math grades; **value perceptions** predicted **course enrollment intentions**; math anxiety was predicted by ability/expectancy/value perceptions but did not independently drive grades/enrollment in their SEM.

Product translation: Instrument **both** outcome classes. `retry_120s` / transfer ≈ expectancy path. Intent-to-continue / challenge_accept / "take next ACT practice block" ≈ value path. Do not collapse both into a streak KPI (standing ban).

XXXVII.3 Subjective task value — four components, four commercial mistakes

A. Intrinsic value ≠ engagement theater

FACT: Intrinsic value = enjoyment of the task (Eccles-Parsons et al., 1983; Wigfield & Eccles, 2000, *CEP*, doi:10.1006/ceps.1999.1015).

HYPOTHESIS: Story can raise intrinsic value *if* it encodes structure and preserves integral roles (XXVIII / XXXVI). Seductive details that raise "interest" but crush learning (Harp & Mayer; Sundararajan & Adesope — carry) are **fake intrinsic** for MindCraft's purpose.

Kill: Marketing that equates "delightful world" with durable math choice without FEI + value instruments.

B. Attainment value is the identity wedge

FACT (theory refinement): Attainment value moved from "importance of doing well" toward linkage with personal/social identities (Eccles, 2005/2009 line; summarized in Eccles & Wigfield, 2024, *Educational Psychology Review*, doi:10.1007/s10648-024-09888-9).

FOUNDER BELIEF: "I am becoming someone who can think mathematically" is attainment value. Streaks are not.

Commercial implication: Parent and teen copy should sound like *identity competence*, not *gamified attendance*. Allowed: "Evidence you can trust yourself on the next hard problem." Banned: "Don't break your streak — that means you're a math person."

C. Utility value — the only mass-market EVT intervention with strong field RCTs

FACT (RCT): Hulleman & Harackiewicz (2009, *Science*, doi:10.1126/science.1177067) — ninth-grade science students randomly assigned to write how material connects to their lives (or a known other) vs summary control; **relevance writing raised interest and grades especially for students with low success expectations**.

FACT (lab + classroom): Hulleman, Godes, Hendricks & Harackiewicz (2010, *JEP*, doi:10.1037/a0019506) — utility writing raised perceived utility and interest, again strongest for lower performers/expectancies; utility perceptions mediated interest effects.

HYPOTHESIS for MindCraft: A 2–4 minute **student-authored** utility prompt *after* a micro-win (not before first attempt under high threat) beats brand-authored “math is useful for STEM careers” copy — especially for Maya with low expectancy.

Red Team wound: Effects are domain/context sensitive; science classroom ≠ ACT app session; dosage and timing matter (SEVT situating). Do not paste “Hulleman proved story worlds work.”

Kill: Shipping utility as *staff-written* career propaganda dumped pre-attempt (CLT + seductive + autonomy threat).

D. Cost — historically neglected, commercially decisive

FACT: Cost was theorized early (effort, opportunity, psychological cost of failure) but empirically under-measured for decades (Wigfield & Cambria reviews; Eccles & Wigfield, 2020 SEVT paper stresses cost elaboration).

FACT (measurement): Flake, Barron, Hulleman, McCoach & Welsh (2015, *CEP*, doi:10.1016/j.cedpsych.2015.03.002) — validated multi-dimensional cost (task effort, outside effort, loss of valued alternatives, emotional cost); cost is separable from expectancy/value and relates to student outcomes.

HYPOTHESIS: For anxious Maya, **emotional cost** and **effort cost** dominate early sessions; for busy juniors, **opportunity cost** (vs sleep/jobs/sports) dominates weekly return. Product that only raises “fun” while leaving shame-of-wrongness intact fails SEVT choice.

Product translation: Soft-wrong / hide-correctness diagnostic (C4) / tutor tone (Part XXX) are **cost interventions**, not just pedagogy niceties. Session design that demands 40 minutes of lore before first attempt raises opportunity + effort cost without expectancy gain (tie to EQ-3).

XXXVII.4 Hierarchies and situations (why one “motivation score” is a product bug)

FACT: SEVT emphasizes *hierarchies* of expectancies and values across competing activities and situations (Eccles & Wigfield, 2020; Wigfield & Eccles developmental reviews). Choosing math means math beats alternatives *this week*, not that motivation is globally high.

HYPOTHESIS: Duo wins short-horizon intrinsic/utility-for-streak; Khan wins low-cost access library; ChatGPT Base raises short-term utility (“get the answer”) while crushing expectancy-for-unaided performance (Part XXXIII/XXXV). MindCraft’s wedge is **attainment + expectancy evidence with managed cost** — not out-streaking Duo or out-librarying Khan.

SPECULATION (commercial): Parents purchase when they believe we move the *child’s hierarchy* toward math without catastrophic emotional cost — “they’ll open it again without a fight” is a cost/expectancy story sold as calm, not as dopamine.

XXXVII.5 Socializers — parents and tutors as EVT machinery

FACT (model structure): Socializers’ beliefs and behaviors shape children’s interpretations, affective memories, and thus expectancies/values (Eccles-Parsons et al., 1983; SEVT socialization arm).

Carry: Part XXXII (parent anxiety transmission) — parent math anxiety can raise the child's **emotional cost** and lower expectancy via homework help patterns (Maloney/Beilock line).

HYPOTHESIS: Parent dashboard copy that celebrates *process evidence* (retry, transfer) raises parent-transmitted attainment/utility messages; copy that only shows streak/percentile can raise performance pressure → cost.

Product implication: Parent WTP experiments (Part XXVII queue) should manipulate EVT framing: "identity/competence evidence" vs "minutes practiced" vs "AI explanations unlimited." Pre-register which frame shifts purchase intent without raising reported homework conflict.

XXXVII.6 Mapping MindCraft features → EVT levers (audit table)

Feature / bet	Primary lever	Secondary	Claim ladder max without new data
Gap-scan + level gating	Expectancy calibration	Cost ↓ (fewer impossible first items)	L1 product intent
Soft-wrong → retry_120s	Expectancy evidence	Emotional cost ↓	L1–L2 with FEI instrumentation
Story worlds (structure-encoding)	Intrinsic + attainment	Utility if student links	L0–L1; EQ/HIST required
Student utility writing (post-win)	Utility	Intrinsic (situational interest)	L1; EVT-1 experiment
ACT exam-mode roadmap	Utility (instrumental)	Expectancy if honest sequencing	L1 honesty; not "score guarantee"
Tutor human sessions	Expectancy + cost	Attainment (witnessed identity)	L1; do not claim 2-sigma
Solver without guards	Short-term utility (answer)	Expectancy ↓ (crutch)	Kill as hero feature
Streaks / leaderboards	Pseudo-attainment / extrinsic	Cost ↑ under threat	Standing ban as North Star

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XXXVII.7 Commercial implications (copy, positioning, growth)

Marketing language that survives

- **Allowed:** "Expectancy without value is empty drills. Value without expectancy is inspirational avoidance. We design for both — and we treat the emotional price of being wrong as a first-class design constraint."
- **Allowed:** "Utility is something the student *writes*, not a slogan we paste."
- **Allowed (parent):** "Fewer fights at the kitchen table" = cost reduction claim — keep qualitative until measured.
- **Banned:** "Our world makes math useful" (untested utility).

- **Banned:** “Fun = they’ll choose Precalculus.”
- **Banned:** Absolute tutoring-is-free / 2-sigma / streaks-as-identity.

Positioning vs competitors

Competitor	EVT profile (audit, not insult)	MindCraft contrast
Khan	Low cost access; mixed expectancy; thin attainment	Session FEI + identity evidence
Duo	Intrinsic/habit; cost managed via brevity; attainment via streak fiction	Transfer + real expectancy
Brilliant	Intrinsic puzzle; utility unclear for ACT Maya	Exam-honest utility + anxiety cost
ChatGPT Base	Max short-term utility; expectancy poison	Guarded Solver + unaided transfer

Growth / North Star hygiene

Optimize a **pair**: expectancy evidence (`retry_120s`, transfer_pass) **and** a value/choice proxy (`challenge_accept`, weekly voluntary return, intent-to-take-next-block). If value proxy rises while transfer falls, you are Duo-pilling the product — kill the feature.

XXXVII.8 Experiments spawned

ID	Question	Design	Primary	Kill condition
EVT-1	Post-win student utility writing vs staff utility blurb vs none	3-arm A/B after micro-win	situational interest + 7d return + transfer@7d	Utility arms lose transfer or raise ti me-to-next-attem pt badly
EVT-2	ACT-utility framing vs identit y-attainment framing in session CTA	2-arm	challenge_accept ; parent/teen preference	Attainment wins preference but tanks attempt rate in anxious segment (segmented kill)
EVT-3	Cost questionnaire (Flake-style short form) pre/post first week	Observational → then intervene on top cost	which cost dimension predicts churn	If emotional cost predicts churn but product only ships “more fun,” roadmap invalid

EVT-4	Parent copy: competence-evidence vs minutes vs AI-unlimited	WTP / conflict items	purchase intent; reported homework conflict	Competence frame raises conflict without WTP lift
EVT-QUAL	10 Maya interviews: “why would you take another math course?”	Appendix B + EVT probes	coded expectancy /value/cost	Students cannot articulate any non-grade reason → attainment/utility product gap

Pre-reg (XXXIV): EVT-* identify levers on **in-app choice and short-horizon learning**. They do **not** identify long-term course enrollment or career STEM entry without follow-up designs. Do not sell “we increase AP Calculus enrollment” from EVT-1.

XXXVII.9 Confidence table

Claim	Label	Confidence
Choice ≈ f(expectancy, STV) in Eccles tradition	FACT (large corpus)	High
Expectancy→performance, value→enrollment intentions (adolescent math)	FACT (Meece et al., 1990 SEM)	Medium–High (sample/era limits)
STV has intrinsic / attainment / utility / cost structure	FACT (theory + measurement work)	High
Utility-value writing can lift interest/grades for low-expectancy students	FACT (Hulleman & Harackiewicz, 2009)	Medium–High (context limits)
Cost is multi-dimensional and outcome-relevant	FACT (Flake et al., 2015)	Medium–High
Story worlds raise utility by themselves	SPECULATION	Low
MindCraft should instrument E and V separately	FOUNDER BELIEF / HYPOTHESIS	Medium–High
Soft-wrong + tutor tone primarily cut emotional cost	HYPOTHESIS	Medium
Streaks raise attainment value	SPECULATION / likely false for identity	Low (against)

XXXVII.10 What this chapter kills

1. **Kill:** Single “motivation” KPI as if E and V were the same dial.
2. **Kill:** Staff-authored “math is useful” dumps as a substitute for student-authored utility.
3. **Kill:** Treating story delight as proof of course-choice impact.
4. **Wound:** Pure ACT-utility positioning — survives for juniors near test date; wounds long-horizon identity brand if it becomes the only message.
5. **Survive (constrained):** Identity-transformation thesis — **if** product explicitly raises expectancy *and* attainment/utility *and* designs against cost, with EVT-1…4 on the roadmap.

Doctrine until data: Design and market the **E×V×Cost triad**. Expectancy evidence without value is a drill app. Value talk without expectancy is a poster. Ignoring cost is how anxious students churn while NPS looks fine among the already-confident.

Part XXXVIII — Goal Orientation: Mastery vs Performance Framing

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: When MindCraft frames goals as *get better at this* vs *look smart / beat others / don’t look dumb*, which motivational pattern do we induce — and what must product + copy refuse?

Owners: Learning Science · Product · Growth / Positioning · Red Team

Commercial job: Replace empty “mastery learning” branding with a **goal-structure audit**: which surfaces cue mastery-approach, which cue appearance/performance, which cue avoidance — and which we kill even if they juice short-term engagement.

XXXVIII.1 Why this chapter exists

Parts XXV (self-efficacy), XXX (attribution / helplessness), XXXI (challenge–skill), and XXXVII (expectancy-value) answer *can I*, *why did I fail*, *is difficulty calibrated*, and *will I choose math*. They do not fully answer: **what success definition is the product teaching?**

A product that says “identity transformation” while UI, CTAs, and parent dashboards scream *percentile / streak / don’t mess up* is running a **performance-avoidance climate** with mastery marketing on top.

FOUNDER BELIEF under audit: MindCraft’s North Star metrics (`retry_120s`, `challenge_accept`, `transfer_pass`) only stay coherent if the **goal structure** of sessions is mastery-approach dominant. If recognition and evaluation structures are performance/appearance, those metrics will fight the UI.

Claim we refuse as doctrine: “Any competitive element raises motivation.” That collapses normative, appearance, and avoidance goals — meta-analyses show they are not the same animal.

XXXVIII.2 The constructs (product language, not poster language)

Goal	Plain definition	Typical student cue	MindCraft failure mode if dominant
Mastery-approach	Improve competence vs self/task standard	“Understand this; try a harder variant”	None if paired with transfer; risk = slow coverage if time unmanaged
Mastery-avoidance	Avoid failing to learn / avoid imperfect understanding	“Don’t leave gaps; don’t misunderstand”	Perfectionism, over-checking, refusal to attempt under uncertainty
Performance-approach (normative)	Outperform peers	Rank, leaderboard, “top of class”	Can raise grades for some; often raises threat for Maya; identity ≠ rank
Performance-approach (appearance)	Demonstrate talent / look smart	Public correctness, “prove you’re smart”	Meta-analytically worse than normative; shame + shallow strategy
Performance-avoidance	Avoid looking incompetent	Hide wrongness; skip hard items; Solver crutch	Helpless pattern; FEI collapse; Part XXX / XXXIII risk

FACT (classic dichotomy + classroom structures): Ames (1992, *Journal of Educational Psychology*, doi:10.1037/0022-0663.84.3.261) — classroom **task, evaluation/recognition, and authority** structures make mastery vs performance goals salient and elicit qualitatively different motivational patterns; interventions must treat structures as interdependent, not one “growth mindset poster.”

FACT (student perceptions → strategies): Ames & Archer (1988, *JEP*, doi:10.1037/0022-0663.80.3.260) — among 176 secondary students, perceived **mastery** classroom goals related to more effective strategies, preference for challenge, positive attitude, and effort→success beliefs; perceived **performance** salience related to ability focus, negative ability evaluation, and attributing failure to low ability.

FACT (implicit theories → goals → patterns): Dweck & Leggett (1988, *Psychological Review*, doi:10.1037/0033-295X.95.2.256) — entity vs incremental theories orient students toward performance vs learning goals, which set up helpless vs mastery-oriented response patterns after difficulty.

Product translation: Soft-wrong + retry is not only an expectancy/cost intervention (XXXVII) — it is a **goal-structure** intervention. Public scoreboards and “don’t break the streak” are evaluation/recognition structures that cue performance/appearance.

XXXVIII.3 Approach–avoidance split (why “performance goals” is too coarse)

FACT (trichotomous model): Elliot & Church (1997, *JPSP*, doi:10.1037/0022-3514.72.1.218) — mastery, performance-approach, and performance-avoidance are separable; mastery facilitated intrinsic motivation; performance-approach enhanced graded performance; **performance-avoidance harmed both** intrinsic motivation and grades (college classroom path model).

FACT (2×2 framework): Elliot & McGregor (2001, *JPSP*, doi:10.1037/0022-3514.80.3.501) — adds **mastery-avoidance**; four goals show distinct antecedents/outcomes; mastery-avoidance sits between mastery-approach (more adaptive) and performance-avoidance (more maladaptive) on several indicators.

HYPOTHESIS for MindCraft: Anxious Maya often defaults to **performance-avoidance** ("don't look dumb") or **mastery-avoidance** ("I must understand perfectly before I try"). Product that only bans leaderboards but still frames every miss as identity threat will not fix the pattern.

Kill: Marketing that equates "mastery mode" with "zero mistakes allowed on the path." That is mastery-avoidance cosplay.

XXXVIII.4 Performance-approach is not one knob — measurement kills bad slogans

FACT (meta-analysis on measures): Hulleman, Schrager, Bodmann & Harackiewicz (2010, *Psychological Bulletin*, doi:10.1037/a0018947) — across 243 correlational studies (N≈91k), "performance-approach" scales that emphasize **normative outperformance** correlate *positively* with performance outcomes (r≈.14), while scales emphasizing **appearance / looking talented** correlate *negatively* (r≈−.14). Same label, different constructs.

FACT (extended outcomes meta): Senko & Dawson (2017, *Journal of Educational Psychology*, doi:10.1037/edu0000160) — the normative vs appearance split generalizes beyond grades to competence perceptions and self-regulation; age confounds with measure type were addressed.

FACT (theory status review): Senko, Hulleman & Harackiewicz (2011, *Educational Psychologist*, doi:10.1080/00461520.2011.538646) — achievement goal theory at a crossroads: multiple controversies (including when performance-approach helps) require precise operationalization before practice claims.

Commercial implication: If MindCraft ever ships social comparison, it must pre-register whether the cue is **normative skill evidence** (dangerous for threatened learners; still not identity) or **appearance theater** (ban). Default = **neither**. Do not cite "performance goals sometimes help grades" as license for public shame UX.

SPECULATION (positioning): "Beat your friends" copy will read as Duo/Brilliant hybrid; "prove you're smart" reads as SAT-cram Instagram. Neither is the identity-transformation wedge.

XXXVIII.5 Classroom / product goal *structures* (TARGET), not vibes

FACT (schoolwide / TARGET lineage): Maehr & Midgley (1991, *Educational Psychologist*, doi:10.1080/00461520.1991.9653140) — motivation change requires altering school/classroom goal structures, not only exhorting students; dimensions map to task design, authority, recognition, grouping, evaluation, time (TARGET family; Ames, 1992).

FACT (review): Meece, Anderman & Anderman (2006, *Annual Review of Psychology*, doi:10.1146/annurev.psych.56.091103.070258) — classroom goal structures predict personal goals and achievement-related patterns; mastery structures are the design target for adaptive motivation.

FACT (longitudinal TARGET → mastery goals): Lüftenegger et al. (2014, *Educational Psychology*, doi:10.1080/01443410.2013.814189) — TARGET as a latent structure positively predicted students' mastery goal orientations (N=1680 secondary; longitudinal).

MindCraft TARGET audit (hypothesis map):

TARGET dim	Mastery-leaning product move	Performance/appearance trap
Task	Varied challenge matched to skill (XXXI); format variety with transfer	Same-item grind that feels like "prove speed"
Authority	Student choice of next concept *after* honest diagnosis	Forced path that only optimizes for looking ahead of peers
Recognition	Process evidence: retry, transfer, bridge repair	Streak flames, public %ile, "genius" badges
Grouping	Tutor as witness; optional co-op without rank	Ranked classrooms / public wrongness
Evaluation	Soft-wrong diagnostic (C4); hide correctness when graph-corrupting	Instant public right/wrong as identity verdict
Time	Short sessions that still allow struggle → retry	Timed pressure that cues avoidance

FOUNDER BELIEF: PawHub "Practice the weak spot" is mastery-approach **if** copy is "repair this gap" and **not** "you're behind everyone." Gap language can flip to performance-avoidance under shame framing.

XXXVIII.6 Interaction with expectancy, anxiety, and AI crutches

FACT (goal pursuit × expectancy): Senko & Hulleman (2013, *JEP*, doi:10.1037/a0031136) — for mastery-approach *and* performance-approach, the harder a goal appears to attain, the less students pursue it (with costs to interest/achievement); mastery-approach goals were generally judged easier to attain than performance-approach; situational cues (material complexity) shift mastery-approach expectancies.

Carry: Part XXIV (anxiety) + XXXIII (AI overtrust) — under high threat, performance-avoidance + Solver-without-guards is the cheap escape from a hard-looking mastery or performance goal.

HYPOTHESIS: Level gating after gap-scan (easy→L3, kinda→L2, hard→L1) is a **goal-attainment-expectancy** design: it makes mastery-approach goals feel attainable. Removing gates to "challenge everyone" without expectancy supports will increase avoidance and Solver dependence.

Product implication: CTAs should make the next mastery goal *feel attainable* (Senko & Hulleman) without lying about transfer. Honest sequencing ≠ performance theater.

XXXVIII.7 Competitive product audit (goal climate lens)

Competitor	Dominant goal climate (audit)	What MindCraft must not copy
Duolingo	Habit + streak recognition; appearance of consistency	Streak-as-identity / streak-as-North-Star (standing ban)
Khan	Mastery language + energy points / badges mixed	“Mastery” label without transfer_pass
Brilliant	Puzzle competence; mild normative “smart people do this”	Appearance-smart branding without anxiety cost design
ChatGPT Base	Outcome utility (“get it right now”) → avoidance of struggle	Solver as hero; unaided transfer optional

HYPOTHESIS: MindCraft’s wedge is **mastery-approach structures with expectancy-managed tasks and anti-appearance evaluation** — not out-ranking Duo and not out-explaining Khan.

XXXVIII.8 Commercial implications (copy, product, growth)

Marketing language that survives

- **Allowed:** “We design for improvement goals — understand, repair, transfer — not for looking smart in public.”
- **Allowed:** “Wrong answers are evidence for the next attempt, not a verdict on who you are.” (ties XXX / soft-wrong)
- **Allowed (parent):** “We measure whether they take the next hard problem, not whether they kept a streak.”
- **Banned:** “Become a top 1% math student” as hero claim (appearance/normative mashup; undefended).
- **Banned:** “Never break your streak — that’s how mastery works.”
- **Banned:** Empty growth-mindset posters without TARGET structure changes (Dweck mis-applied as sticker).
- **Banned:** 2-sigma / tutoring-is-free absolutes.

Feature claim ladder (pre-reg hygiene, Part XXXIV)

Feature	Goal climate claim max without GO-* data
Soft-wrong → retry_120s	L1 intent: mastery-approach evaluation structure
`challenge_accept` CTA	L1: approach (need to verify mastery vs performance wording)
`transfer_pass`	L1–L2 anti-fake-mastery (competence standard ≠ appearance)
Leaderboards / public ranks	L0 default kill for Maya segment
Streak North Star	Kill (standing ban)

“Mastery graph” marketing	L1 only if transfer instrumented; else label is cosmetics

Growth / North Star hygiene

`challenge_accept` is **not automatically mastery-approach**. Instrument **why** they accept: repair gap (mastery) vs beat peer (normative) vs avoid looking behind (avoidance). If acceptance rises because of shame CTAs, kill the copy even if the metric looks green.

XXXVIII.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
GO-1	Mastery-approach CTA (“repair this gap”) vs normative (“catch up to peers”) vs appearance (“show you’re smart”)	3-arm on Practice launch	challenge_accept; retry_120s; 7d return; anxiety item	Normative/appearance win short accept but lose retry/transfer or raise anxiety
GO-2	Recognition: process badge (retry/transfer) vs streak badge vs none	3-arm	transfer@7d; identity item; self-reported goal	Streak badge wins engagement, loses transfer / raises avoidance
GO-3	Evaluation: soft-wrong first vs instant public correctness	2-arm on early session	next-attempt latency; avoidance skips; Solver opens	Instant correctness raises “finish” rate but kills retry_120s
GO-4	Goal-climate pulse (short PALS-like mastery/performance classroom items adapted to app) weekly	Observational → then redesign recognition	which climate predicts churn / Solver dependence	If product self-reports “mastery brand” but climate score is performance → roadmap invalid
GO-QUAL	10 Maya interviews: “What would ‘doing well’ here mean?”	Appendix B + goal probes	coded mastery / normative / appearance / avoidance	Majority answer only “not look dumb” → evaluation/recognition redesign mandatory

Pre-reg (XXXIV): GO-* identify **in-app goal climate** → **short-horizon FEI behaviors**. They do **not** identify long-term GPA or STEM major from CTA wording alone. Do not sell "we create mastery-oriented students for life" from GO-1.

XXXVIII.10 Confidence table

Claim	Label	Confidence
Mastery vs performance goal structures shape strategies/attitudes	FACT (Ames & Archer, 1988; Ames, 1992)	High
Implicit theories orient goals → helpless vs mastery patterns	FACT (Dweck & Leggett, 1988)	High
Performance-avoidance is broadly maladaptive for motivation/grades	FACT (Elliot & Church, 1997 lineage)	High
2×2 adds mastery-avoidance as distinct	FACT (Elliot & McGregor, 2001)	High
"Performance-approach" effects depend on normative vs appearance ops	FACT (Hulleman et al., 2010; Senko & Dawson, 2017)	High
TARGET-like structures can raise mastery orientations	FACT (Maehr & Midgley; Lüftenegger et al., 2014)	Medium–High (school ≠ app)
Soft-wrong + process recognition = mastery evaluation structure	HYPOTHESIS	Medium–High
Any leaderboard helps MindCraft's Maya segment	SPECULATION / likely false	Low (against)
Streaks instantiate mastery goals	SPECULATION / false for identity	Low (against)
MindCraft should ban appearance-goal UX by default	FOUNDER BELIEF / HYPOTHESIS	Medium–High

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XXXVIII.11 What this chapter kills

1. **Kill:** Treating "mastery" as a brand adjective without evaluation/recognition structures that cue improvement standards.
2. **Kill:** Streak / public rank / "look smart" as North Star or hero copy (appearance + standing streak ban).
3. **Kill:** Collapsing all performance goals into "competition is motivating."
4. **Wound:** Normative comparison as a growth lever — survives only as a pre-registered, segmented experiment (GO-1), never as default Maya UX.

5. **Survive (constrained):** Identity-transformation thesis — **if** product TARGET structures are mastery-approach dominant, avoidance/appearance cues are designed out, and GO-1…4/QUAL sit on the roadmap.

Doctrine until data: Market and build **mastery-approach goal structures** (task challenge + process recognition + non-shaming evaluation + attainable next goals). Performance-avoidance and appearance goals are product bugs dressed as engagement. Normative comparison is not the default growth loop.

Part XXXIX — Interleaving vs Blocking: When to Mix Problem Types

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: When should MindCraft mix problem types (interleave) vs group same-type drills (block) — and what must product + copy refuse to claim?

Owners: Learning Science · Product (Practice sequencing) · Growth / Positioning · Red Team

Commercial job: Turn "we shuffle questions" from a cute engineering detail into a **defensible practice-design claim**: strategy *selection*, not just strategy *execution* — without selling interleaving as always-on magic or as engagement theater.

XXXIX.1 Why this chapter exists

Part XXIX (spacing / retrieval) already flags interleaving as a desirable difficulty and hypothesizes **Block** → **fade** → **interleave** → **spaced retrieval**. Part XXVI (CLT) warns that early overload kills learning. Part XXXI (challenge–skill) and XXXVIII (goal climate) warn that harder-feeling practice can cue avoidance if framed as shame.

This chapter densifies the **mix vs block** decision for Practice sessions, ingredient cards, and ACT-style mixed review — because textbooks and most edtech default to **blocked practice** (a dozen slope problems after the slope lesson), and students experience that fluency as "I get it" while exams demand **choose the strategy from the problem itself**.

FOUNDER BELIEF under audit: Identity transformation requires competence evidence that survives *transfer*. Blocked fluency is often **illusory competence** — high session accuracy, weak delayed discrimination.

Claim we refuse as doctrine: "Always interleave everything from the first item." That ignores schema formation, CLT, and similarity moderators — and can spike threat for anxious Maya before `retry_120s` is healthy (XXIX.4).

XXXIX.2 Constructs (product language)

Schedule	Plain definition	What the student practices	Failure mode if misused

Blocked	Consecutive items share the same strategy / concept	*Execute* a known procedure	Fluency illusion; weak strategy choice on exams
Interleaved	Consecutive items require different strategies (no safe "same as last")	*Select* then execute	Early overload; frustration; avoidance if unscaffolded
Hybrid / faded	Short block for schema → rising mix	Execute → select	Best candidate for MindCraft default

Operational definition for MindCraft (HYPOTHESIS): Interleaving = consecutive Practice items do **not** share the same primary strategy class (concept + dominant ingredient / archetype family). Mere format shuffle (word → diagram) with the same strategy is **not** full interleaving — it is representation variety (useful; different claim).

XXXIX.3 Mechanism: why mixture changes the pedagogical demand

FACT (math laboratory / classroom line): Rohrer & Taylor (2007, *Instructional Science*, doi:10.1007/s11251-007-9015-8) — Experiment 2: after learning multiple problem types, **mixed** vs **blocked** practice; one-week delayed test vastly favored mixed practice. Experiment 1 separately showed spacing > massing for a single type. Shuffled textbooks alter **both** spacing and mixing; they are related but not identical levers.

FACT (strategy choice vs execution): Rohrer, Dedrick & Stershic (2015, *Journal of Educational Psychology*, doi:10.1037/edu0000001) — 126 seventh-graders, same problems over ~3 months, interleaved vs blocked arrangement; after review, unannounced tests favored interleaving at 1 day (d = 0.42) and 30 days (d = 0.79). Core mechanism claim: blocked practice lets students know the strategy *before reading the problem*; interleaved practice forces strategy choice from problem features — the skill exams actually require.

FACT (not only look-alike problems): Rohrer, Dedrick & Burgess (2014, *Psychonomic Bulletin & Review*, doi:10.3758/s13423-014-0588-3) — interleaved benefit is **not** limited to superficially similar problem kinds; reordering changes pedagogical demand (choose vs execute) even when types do not look alike.

FACT (classroom-scale RCT): Rohrer, Dedrick, Hartwig & Cheung (2020, *JEP*, doi:10.1037/edu0000367; Online First 2019) — preregistered cluster RCT, 54 seventh-grade classes, mostly interleaved vs mostly blocked assignments over months; after interleaved review for both, one-month unannounced test: 61% vs 38%, d = 0.83. Teachers implemented without training; anonymous pre-results survey supported feasibility. Caveats remain (materials, population), but this is the strongest classroom efficacy signal in the math interleaving line.

Product translation: MindCraft's `transfer_pass` and mixed ACT practice are **closer to the interleaved demand** than a single-concept blocked grind. A session that is 12 items of `linear_equations` after "Learn linear equations" is textbook-default blocking — fine as *early schema practice*, dangerous as the *only* schedule if we market exam readiness.

XXXIX.4 Meta-analytic moderators (do not overclaim "interleaving always wins")

FACT (inductive learning meta-analysis): Brunmair & Richter (2019, *Psychological Bulletin*, doi:10.1037/bul0000209) — 59 studies / 238 effect sizes / 158 samples; overall interleaving vs blocking Hedges' g = 0.42. **Material matters:** paintings/visual high ($g \approx 0.67$); **mathematical tasks small but positive ($g \approx 0.34$)**; words favor **blocking** ($g \approx -0.39$); expository texts ambiguous. Moderators: stronger interleaving when **between-category similarity is high**, **within-category similarity is low**, and material is more complex — consistent with attentional bias / sequential attention accounts (Carvalho & Goldstone lineage).

FACT (attentional bias framework): Carvalho & Goldstone (2015, *Frontiers in Psychology*, doi:10.3389/fpsyg.2015.00505) — study sequence biases what is encoded: interleaving emphasizes **between-category differences**; blocking emphasizes **within-category similarities**. Prediction: when categories are hard to tell apart, interleave; when the job is to notice what members of one category share, block can help.

HYPOTHESIS for MindCraft: Interleave **confusable** strategy families (e.g., independent vs dependent probability; similar ACT archetypes that share surface cues but need different ingredients). Block when introducing a *new* ingredient representation until the student can execute with worked-example fade (XXVI) — then mix against near-miss foils.

Kill: Marketing that says "our AI interleaves everything like elite tutors" without specifying **confusable sets** and **when blocking is intentional**.

XXXIX.5 Fluency illusion, affect, and why students prefer blocking

FACT / TRADITION (desirable difficulties): Conditions that slow acquisition (spacing, interleaving, generation) often improve later retention/transfer (Bjork lineage; see XXIX.4). Interleaving feels harder *during* practice; blocked practice feels fluent.

HYPOTHESIS (Koriat/Bjork-style monitoring, applied): Students and parents misread blocked fluency as mastery. Parent dashboards that celebrate "12/12 on slope today" without a mixed delayed check sell **performance theater** (XXXVIII) dressed as learning.

Carry from XXIV / XXXVII / XXXVIII: Raising interleaving dose raises short-term error rate → for anxious Maya, that can cue performance-avoidance unless (a) soft-wrong / mastery evaluation, (b) expectancy-managed difficulty, (c) explicit frame: "this mix trains choosing, not proving you're dumb."

Product implication: Copy for interleaved sessions: **"Practice choosing the right move"** — not "hard mode for elites" and not streak pressure.

XXXIX.6 Sequencing doctrine (ties XXVI + XXIX)

FOUNDER BELIEF / HYPOTHESIS (sequenced schedule):

1. **Acquire (block-lite):** short same-strategy set + worked examples / cards until execution is stable enough that errors are mostly slips, not total confusion.
2. **Discriminate (interleave):** mix with **near-miss** types that share surface features (bridge gaps, format gaps, ACT archetype neighbors).

3. **Retain (space):** redistribute the same mix across days (XXIX) — interleaving *guarantees* some spacing of each type, but lag still needs explicit scheduling.
4. **Transfer check:** unannounced mixed set → `transfer_pass`; failure routes to diagnosis (forgot vs never-had; wrong strategy class vs execution bug).

Contradiction to hold: Some math/category studies report null or blocking advantages under specific similarity structures (noted in Brunmair & Richter's discussion of mixed math findings; e.g., certain representation-interleaving designs). MindCraft must treat **interleaving axis** (strategy class) as primary for exam readiness, and treat **representation mix** as a separate experiment track (format axis / C2–C3).

XXXIX.7 Competitive / product audit (schedule lens)

Surface	Typical schedule	What MindCraft should do
Textbook / workbook	Heavy blocking after lesson	Do not clone as "mission complete"
Khan-like mastery	Often skill-blocked until "mastered"	Mastery label without mixed delayed check = fluency risk
Duo-like	Habit + short items; mix varies by skill tree	Habit ≠ discrimination training
Brilliant	Puzzle variety; not always systematic near-miss mix	Variety ≠ controlled interleaving
ChatGPT Solver	Single-problem depth	Zero schedule pedagogy; cannot substitute mix training
MindCraft Practice today	Often concept-scoped sessions + bank shuffle	Shuffle within one concept ≠ full strategy interleaving; need **cross-type mix modes**

HYPOTHESIS: PawHub "Practice the weak spot" correctly targets a gap, but a pure blocked weak-spot grind can raise local accuracy while leaving **strategy selection** untrained. Pair weak-spot blocks with a **discrimination coda**: 3–5 interleaved near-misses including the weak type.

XXXIX.8 Commercial implications (copy, product, growth)

Marketing language that survives

- **Allowed:** "Exams don't announce the chapter. We train choosing the strategy — not only repeating the last one."
- **Allowed:** "Blocked practice feels easy; mixed practice builds the skill tests actually measure."
- **Allowed (parent):** "We check whether they can pick the right approach on a mixed set days later — not just a perfect streak on one topic."
- **Banned:** "Interleaving always doubles scores" / absolute effect-size marketing (Rohrer classroom $d \approx 0.83$ is **one** RCT context; Brunmair math $g \approx 0.34$ is the broader inductive-math signal — do not collapse).

- **Banned:** “Always mix from day one.”
- **Banned:** Selling shuffle/random as science without near-miss design.
- **Banned:** 2-sigma / tutoring-is-free / streak-as-mastery.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without IL-* data
Within-concept item shuffle	L0–L1 “variety”; **not** full interleaving claim
Cross-strategy interleaved coda after weak-spot block	L1 intent: discrimination training
Delayed mixed `transfer_pass`	L1–L2 anti-fluency-illusion
“Interleaving improves ACT” hero claim	L0 until IL-1/IL-2 pre-registered

Growth / North Star hygiene

Session accuracy under **blocked** schedules will look better than under interleaved ones. **Do not** optimize North Star on blocked accuracy. Prefer: mixed delayed accuracy, strategy-selection error rate (wrong procedure class), `transfer_pass`, and `retry_120s` after mix-induced misses.

XXXIX.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
IL-1	Blocked-only weak-spot session vs block + interleaved near-miss coda	2-arm; same total items	Delayed (48h) mixed accuracy; strategy-class error rate	Coda hurts delayed mix **and** raises avoidance without transfer gain
IL-2	Early interleave (from item 1) vs fade-in interleave after short block	2-arm on new concept	Acquisition errors; `retry_120s`; 7d mixed test	Early-all-mix wins delayed score but kills return/anxiety for Maya segment
IL-3	Near-miss interleave (confusable archetypes) vs far mix (unrelated easy types)	2-arm	Discrimination errors on near-miss test	Far mix equals near-miss on discrimination → “any shuffle” doctrine dies

IL-4	Parent/student judgment: blocked fluency vs mixed score — which do they trust?	Survey + optional choice	Preference for blocked; willingness-to-pay framing	If parents only buy blocked dashboards, copy/education mandatory before UX ships mix-heavy
IL-QUAL	10 Maya interviews: "When did practice feel like the real test?"	Appendix B	coded choose-vs-execute moments	Nobody describes strategy choice → product isn't creating the demand state

Pre-reg (XXXIV): IL-* identify **schedule** → **discrimination / delayed mix**. They do **not** identify long-term STEM identity from one coda. Do not sell "interleaving creates math people."

Supersedes / densifies: XXIX INT-1 (interleave after fade vs blocked-only) → prefer IL-1/IL-2 as the operational pair.

XXXIX.10 Confidence table

Claim	Label	Confidence
Mixed/interleaved math practice often beats blocked on delayed tests	FACT (Rohrer & Taylor, 2007; Rohrer et al., 2015, 2020)	High (math classroom/lab line)
Mechanism includes forced strategy selection (not only spacing)	FACT / strong interpretation in Rohrer line	High
Benefit can appear for non-look-alike problem kinds	FACT (Rohrer et al., 2014)	High
Overall inductive interleaving effect moderated by material & similarity	FACT (Brunmair & Richter, 2019)	High
Math subset effect smaller than visual category learning	FACT (g≈0.34 vs higher for paintings)	High
Blocking can help when within-category similarity learning is the job	FACT (attentional bias framework) / HYPOTHESIS for early schema	Medium–High
Always-interleave-from-item-1 is optimal for anxious Maya	SPECULATION / likely false	Low (against)
Within-concept shuffle ≡ interleaving science claim	SPECULATION / false	Low (against)

Default MindCraft path = short block → near-miss interleave → spaced mix check	FOUNDER BELIEF / HYPOTHESIS	Medium–High

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XXXIX.11 What this chapter kills

1. **Kill:** Treating blocked session accuracy as proof of exam readiness or identity change.
2. **Kill:** “We randomly shuffle, therefore we interleave” as a marketing claim.
3. **Kill:** Absolute always-interleave / always-double-scores claims.
4. **Wound:** Pure weak-spot blocked grind as the only Practice mode — survives as *acquisition*, not as the full loop.
5. **Survive (constrained):** Desirable-difficulty interleaving for **strategy discrimination** — after schema foothold, with near-miss design, affect-safe framing, and delayed mixed `transfer_pass`.

Doctrine until data: Market and build **choose-the-strategy training** via faded interleaving of confusable types. Blocking is a scaffold, not the product. Fluency without mixed delayed evidence is a vanity metric.

Part XL — Self-Explanation Prompts: Chi/Renkl → Coach UX

Chapter status: Living evidence brief — Researcher tick 2026-07-29

Primary question: When do prompts that force the student to *generate* an explanation improve math learning — and what must MindCraft’s coach / cards / Solver refuse to do with “explain yourself” UX?

Owners: Learning Science · Product (coach, ingredient cards, Solver) · Growth / Positioning · Red Team

Commercial job: Turn “we explain everything” and “AI tutor talks at you” into a **defensible generation claim**: students build transferable understanding by explaining *why steps work* (and why common wrong moves fail) — without selling open-ended “justify your guess” as science, and without confusing fluent AI explanation *to* the student with self-explanation *by* the student.

XL.1 Why this chapter exists

Part XXVI (CLT / worked examples / fading) already puts examples and completion problems in the acquisition path. Part XXIX / XXXIX demand retrieval and strategy *selection*. Part XXXIII warns that fluent AI explanations can create overtrust. Experiment A (wizard coach) and soft-wrong already assume that *after* an error, something productive happens.

This chapter densifies the **prompt layer**: what the product asks the student to *produce* — principle for a step, why a misconception is wrong, anticipation of the next operator — vs what it merely *shows*.

FOUNDER BELIEF under audit: Identity as a mathematical thinker requires evidence that the student can *account for* a move, not only select A/B/C/D. Passive delivery without generation is Layer-1 scarcity relief, not identity transformation.

Claim we refuse as doctrine: "Any explain box after a wrong answer is self-explanation science." Untargeted prompts, explain-your-wrong-guess-without-feedback, and AI monologues are different mechanisms — some help, some hurt, some are theater.

XL.2 Constructs (product language)

Construct	Plain definition	Product surface	Failure mode
Self-explanation	Learner generates inferences beyond the given text/example *for themselves*	Prompted coach turns; card reflection; step-principle pick	Empty "why?" with no scaffold; AI does the explaining for them
Instructional explanation	Expert/AI/tutor explains *to* the learner	Solver narration; worked example text	Can reduce germane effort if overused (expertise / passivity)
Worked-example study	Study full solution before / instead of cold solving	Ingredient cards; fade path	Without SE prompts → shallow copy (Chi "poor" pattern)
Principle / glossary citation	Name the rule that justifies a step	Aleven-style menu; ingredient id pick	Menu-click without comprehension if too shallow
Misconception contrast	Explain why a common wrong path fails	Soft-wrong + "why not B"; bank distractors	Asking only "defend your wrong answer" before feedback

Operational definition for MindCraft (HYPOTHESIS): A self-explanation prompt is load-bearing only if (1) the to-be-explained content is tagged **correct or known-incorrect**, (2) the response format scaffolds principle / operator–goal / misconception contrast (not free essay only), and (3) success is measured by **transfer / delayed mix**, not by longer time-on-task or chat length.

XL.3 Classic effect: spontaneous quality, then elicitation

FACT (good vs poor example study): Chi, Bassok, Lewis, Reimann & Glaser (1989, *Cognitive Science*, doi:10.1207/s15516709cog1302_1) — talk-aloud while studying mechanics worked examples. "Good" students generated many explanations that refined conditions for actions and linked steps to principles; "poor" students under-explained, monitored poorly, and relied heavily on examples at solve time. Self-explanation quality predicted understanding and example-independence.

FACT (elicitation works): Chi, de Leeuw, Chiu & LaVancher (1994, *Cognitive Science*, doi:10.1016/0364-0213(94)90016-7) — eighth-graders prompted to self-explain after each line of a circulatory-system text outperformed controls who reread; high explainers built more accurate mental models. Prompts can *create* the behavior that spontaneous good students show.

FACT (quality ≠ mere time): Renkl (1997, *Cognitive Science*, doi:10.1016/S0364-0213(99)80017-2) — probability worked examples; time-on-task controlled. Learning gains still predicted by qualitative SE features: **principle-based** explanations, **operator–goal** explication, **anticipative** reasoning. Two effective profiles: anticipative reasoners and principle-based explainers.

Product translation: Coach UX should elicit *those* moves — "What rule licenses this step?" / "What is this step trying to achieve?" / "What happens next?" — not "Write anything about your feelings about math" and not "Hear the AI explain for 90 seconds."

XL.4 Scaffolded prompts + fading (coach design spine)

FACT (fading + principle prompts): Atkinson, Renkl & Merrill (2003, *Journal of Educational Psychology*, doi:10.1037/0022-0663.95.4.774) — successively fading worked-out steps helps near transfer vs example–problem pairs, but far transfer is unreliable from fading alone. Combining fading with prompts to identify the **underlying principle** for each worked step produced medium-to-large effects on **near and far transfer without extra time on task**.

FACT (Cognitive Tutor classroom): Aleven & Koedinger (2002, *Cognitive Science*, doi:10.1016/S0364-0213(02)00061-7) — geometry Cognitive Tutor; students who explained steps by citing geometric rules (glossary) learned with greater understanding and better transfer than problem-solving-only peers. Interpretation: better-integrated declarative knowledge, less shallow procedural knowledge. Scales in classroom software — not only lab talk-alouds.

HYPOTHESIS for MindCraft: Ingredient `card_templates` + bridge cards are the worked-example substrate; coach prompts should (a) attach principle/ingredient labels to faded steps, (b) require a brief generation or structured pick *before* revealing the next full AI paragraph, (c) fade prompts as mastery rises (expertise reversal risk if we nag forever).

Tie to SAFE-CRAFT (XXVI): Micro-attempt / completion blank inside an example *plus* a principle prompt beats cold multi-step solve *or* passive read-only cards.

XL.5 Math meta-analysis: what we may claim (and must not)

FACT (math SE meta-analysis): Rittle-Johnson, Loehr & Durkin (2017, *ZDM*, doi:10.1007/s11858-017-0834-z) — prompted self-explanation in mathematics: small-to-moderate immediate gains — procedural knowledge ES ≈ .28, conceptual ES ≈ .33, procedural transfer ES ≈ .46. Effects **stronger when high-quality explanations are scaffolded** (training or structured responses). Time-on-task control did **not** erase the benefit (moderator). **Caveats they stress:** evidence for reliable classroom (esp. primary/secondary non-ITS) effects and for **retention over delay** is much thinner; delayed procedural/conceptual effects often weak/null while delayed transfer ES ≈ .32 in the subset that measured it; substantial heterogeneity (some null/negative studies).

Instructional recommendations from that paper (treat as design doctrine, not slogans):

1. Scaffold high-quality explanations (structure or train).
2. Design prompts so they do **not** steal attention from other critical content.
3. Prompt learners to explain **correct** information.

4. Prompt why **common misconceptions** are incorrect.

Kill: Marketing “self-explanation always raises grades” / absolute classroom-retention claims from lab immediate ES alone. Claim ladder: L1 “structured why-this-step prompts aim at transfer”; L2+ only after SE-* experiments with delayed mixed `transfer_pass`.

XL.6 Constraints: when prompts backfire

FACT (constraint review): Rittle-Johnson & Loehr (2017, *Psychonomic Bulletin & Review*, doi:10.3758/s13423-016-1079-5) — four constraints: (1) SE favors principle-guided domains and transfer, can hurt detail/exception learning; (2) explaining **one’s own** (often wrong) ideas can entrench them — better to explain known-correct / known-incorrect content; (3) prompts focus attention — mis-aimed prompts tax the wrong content; (4) alternatives (instructional explanation, retrieval, unfamiliar problems) can match or beat SE in some conditions.

HYPOTHESIS for soft-wrong: Order matters. After a wrong MC choice: (A) brief correct-principle or contrastive “why B fails” prompt with scaffold → help; (B) “Explain why you picked C” *before* any correctness signal → risk of justifying the misconception (constraint 2). Prefer: soft-wrong frame → show contrastive target → generate/select why the foil fails → then optional deeper AI wrap (XXXIII: calibration > fluency).

Contradiction to hold: Open generation is the romantic “thinker” move; structured glossary/ingredient picks (Aleven) often deliver the measurable classroom gain. MindCraft should not shame structured picks as “not real thinking” — they are the scalable SE scaffold.

XL.7 Competitive / product audit (explanation lens)

Surface	Typical explanation pattern	MindCraft stance
Khan / video	Instructional explanation; optional reflection	Do not clone lecture-first as identity product
Duo	Rare deep SE; habit + bits	Habit ≠ principle generation
Brilliant	Puzzle insight; light justify	Insight ≠ scaffolded SE curriculum
ChatGPT Solver	High-fluency instructional explanation	**Not** self-explanation; overtrust risk (XXXIII)
Cognitive Tutor lineage	Step + rule citation	Closest evidence-backed software pattern
MindCraft today	Cards + Solver + coach drafts	Need **student generation gates** before AI walls of text; wire misconception bank into contrastive prompts

FOUNDER BELIEF: Solver that answers *for* Maya without a generation gate competes on Layer-1 (answers get cheap) and loses the identity thesis. Coach that forces a principle pick / misconception contrast competes on Layer 2–6.

XL.8 Commercial implications (copy, product, growth)

Marketing language that survives

- **Allowed:** “We don’t just show the steps — we ask you to say *why* the step works.”
- **Allowed:** “Wrong answers become contrast: why that trap fails, not a shame spiral.”
- **Allowed (parent):** “We train explaining the rule, not only picking the letter — that’s what transfers.”
- **Banned:** “AI explains everything so tutoring is free.”
- **Banned:** “Any chat box = self-explanation science.”
- **Banned:** Absolute meta-analytic grade claims; 2-sigma; streak-as-understanding.
- **Banned:** Forcing students to justify unchecked wrong guesses as the default pedagogy.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without SE-* data
AI worked-example narration	L0 instructional help; **not** SE claim
Structured principle / ingredient pick on faded step	L1 SE scaffold (Aleven/Atkinson-aligned intent)
Contrastive “why this distractor fails” after soft-wrong	L1 misconception-SE intent
Open essay “explain your answer” every item	L0 until quality+affect validated; risk of load + avoidance
“SE raises ACT / identity” hero	L0 until SE-1/SE-2 delayed transfer

Growth / North Star hygiene

Chat turns, explanation word count, and Solver dwell time are **vanity** if uncoupled from `transfer_pass` and strategy-class errors. Prefer: rate of principle-consistent explanations (rubric or structured correct pick), misconception-contrast success, delayed mixed accuracy, `retry_120s` after contrastive miss — not tokens generated by the model.

XL.9 Experiments spawned

ID	Question	Design	Primary	Kill condition
SE-1	Principle-prompt on faded card steps vs same cards with AI narration only	2-arm; matched time budget	48h near + far / mixed transfer; shallow-copy errors	Narration-only ≥ prompted on transfer **and** lower load/avoidance

SE-2	Soft-wrong → contrastive “why foil fails” vs soft-wrong → explain-own-choice first	2-arm	Misconception recurrence; `retry_120s`; delayed item isomorphic	Explain-own-first wins transfer → constraint-2 doctrine wounded
SE-3	Structured ingredient/glossary pick vs open text SE vs no SE	3-arm	Transfer; time; completion rate (Maya anxiety segment)	Open text wins transfer but kills return → ship structured default
SE-4	Prompt every step vs adaptive fade of SE prompts with mastery	2-arm	Transfer; prompt skip/nag irritation	Always-on prompts equal fade on transfer but raise churn → fade mandatory
SE-QUAL	10 Maya interviews: “When did an explanation feel like *yours* vs the app talking?”	Appendix B	coded ownership / passivity moments	Nobody distinguishes → UX isn’t creating generation

Pre-reg (XXXIV): SE-* identify **prompt regime → transfer / misconception recurrence**. They do **not** identify long-term identity from one session of explain boxes. Densifies Experiment A: coach content should be SE-constrained, not generic pep talk.

Supersedes / densifies: XXVI worked-example fading notes; Experiment D (explanation timing × anxiety) → prefer SE-2/SE-3 as operational pairs; XXXIII “pedagogy wrap” → wrap = student generation + calibration, not longer AI prose.

XL.10 Confidence table

Claim	Label	Confidence
Higher-quality spontaneous SE while studying examples predicts better learning	FACT (Chi et al., 1989)	High
Prompting SE can improve understanding vs reread / no prompt	FACT (Chi et al., 1994; math meta)	High
Qualitative SE features predict gains beyond time-on-task	FACT (Renkl, 1997)	High
Fading + principle prompts aid near & far transfer without extra time	FACT (Atkinson et al., 2003)	High

Step explanation via rule citation helps Cognitive Tutor geometry transfer	FACT (Aleven & Koedinger, 2002)	High
Math prompted-SE immediate effects small–moderate; scaffold helps	FACT (Rittle-Johnson et al., 2017 ZDM)	High
Classroom + delayed retention evidence thinner than immediate lab/ITS	FACT (same meta)	High
Explaining own wrong ideas can entrench; prefer correct/incorrect targets	FACT (constraint review)	High
AI instructional explanation ≡ self-explanation benefit	SPECULATION / false	Low (against)
Default MindCraft path = faded examples + structured principle/misconception prompts before AI wrap	FOUNDER BELIEF / HYPOTHESIS	Medium–High

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XL.11 What this chapter kills

1. **Kill:** Equating Solver/AI monologue with self-explanation pedagogy.
2. **Kill:** “Explain why you chose that” *before* correctness/contrast as the default after every wrong answer.
3. **Kill:** Absolute “SE always raises scores / always works in class / always sticks for months” marketing.
4. **Wound:** Open-ended explain boxes as the only SE UX — survives as optional depth, not as the scalable default.
5. **Survive (constrained):** Scaffolded prompts on **correct steps and known misconceptions**, paired with worked-example fading, measured by transfer — the coach spine for identity-relevant competence evidence.

Doctrine until data: Build and market **student-generated why** (structured first) on faded examples and contrastive wrongs. AI explains *after* or *around* that generation — never as a substitute for it. Fluency of the model is not mastery of the student.

Part XLI — Desirable Difficulties × Anxiety: When Struggle Helps vs When It Becomes Threat

Chapter status: Living evidence brief — Researcher tick 2026-07-30

Primary question: How should MindCraft reconcile Bjork-style desirable difficulties (spacing, interleaving, retrieval, generation) with Ashcraft-style math-anxiety / working-memory threat — so product + copy do not sell either “grind harder” or “never make it hard”?

Owners: Learning Science · Affect / Anxiety · Product (Practice sequencing + evaluation climate) · Growth / Positioning · Red Team

Commercial job: Resolve the I.4 bottleneck: ship faded desirable difficulties **without** converting anxious Maya's session into shame theater — and market the distinction between *productive struggle* and *threat* in language parents and students can trust.

XLI.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXXIX (interleaving), and XL (self-explanation) all lean on **desirable difficulties**: conditions that slow short-term performance but improve durable access and transfer. Part XXIV (affect/anxiety) and Ashcraft's line show that math anxiety **already** loads working memory like a dual task. Part XXXI (challenge–skill) and XXXVIII (goal climate) warn that harder-feeling practice can cue avoidance under performance threat.

v1.6 left an explicit bottleneck: *"Part XLI must resolve desirable difficulties × anxiety so interleaving/SE/spacing do not ship as shame."*

FOUNDER BELIEF under audit: Identity transformation requires competence evidence that survives delay and mix — which *requires* some desirable difficulty — but for threatened learners, unsequenced difficulty is not "elite training"; it is **threat + WM collapse + exit**.

Claim we refuse as doctrine: "Struggle is always good" / "always crank desirable difficulties" *and* its twin "anxious students should only get easy fluency drills forever." Both are commercially tempting; both fail Maya.

XLI.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Desirable difficulty (DD)	Challenge that forces encoding/retrieval processes that support later access	Mixed review, spaced retries, generation before reveal	Feels worse today → students & parents quit if misframed
Undesirable difficulty	Challenge the learner cannot productively meet (knowledge, cues, or affect load)	Early full interleave; open SE under panic; high-stakes quiz theater	Errors without learning; avoidance; "I'm not a math person"
Math anxiety load	Worry/rumination consuming WM resources needed for multi-step math	Freeze, hint binge, rapid exit after first hard miss	Interpreted as "lazy" or "needs more struggle"
Stakes / evaluation climate	Whether the hard moment is framed as judgment vs training	Soft-wrong, mastery climate, no streak-as-worth	High stakes turn retrieval into threat (Hinze & Rapp)

Equipped success	Prior knowledge + cues sufficient to *respond* to the difficulty	Block-lite → fade → mix; worked examples; structured SE	DD without equipment = cruelty dressed as science

Operational definition (HYPOTHESIS): A MindCraft difficulty is *desirable* only when (a) the student has enough schema/cues to respond with non-chance success or productive partial success, **and** (b) evaluation climate keeps the moment in mastery/training frame, **and** (c) affect load is not already saturating WM. Else it is undesirable — even if the *same* schedule would be desirable for a calm, prepared peer.

XLI.3 Bjork line: difficulty that builds durable access

FACT (origin): Bjork (1994, in Metcalfe & Shimamura, *Metacognition: Knowing about knowing*, MIT Press, pp. 185–205) — conditions that slow acquisition (spacing, variation, testing/generation as learning events, contextual interference) can improve long-term retention and transfer relative to conditions that make performance look strong during practice.

FACT (desirable ≠ any difficulty): Bjork & Bjork (2011/2014 reviews; JARMAC 2020 comment, doi:10.1016/j.jarmac.2020.09.003) — many difficulties are undesirable forever; a difficulty is desirable only when the learner can respond successfully with the encoding/retrieval processes the difficulty is meant to trigger. *If the learner is not equipped, the difficulty becomes undesirable.*

FACT (learning vs performance illusion): Instructors and students systematically misread current fluency as learning (Bjork lineage; see also XXIX / XXXIX). Blocked / massed / reread conditions feel productive; spaced / interleaved / tested conditions feel worse *and often teach more*.

Product translation: MindCraft cannot sell “we make practice feel easy” as proof of learning, nor “we make practice feel hard” as proof of rigor. The commercial claim must be **durable choosing and retrieving under exam-like demand** — measured by delayed mix / `transfer_pass` / `solo_transfer_pass` — not by session comfort or session pain.

XLI.4 Ashcraft line: anxiety as a competing load

FACT (WM disruption): Ashcraft & Kirk (2001, *JEP: General*, doi:10.1037/0096-3445.130.2.224) — high math-anxiety individuals show reduced WM spans (especially computation-based spans); dual-task math + memory load amplifies RT/errors. Anxiety acts as a **transitory disruption of working memory**, not merely as a stable “low ability” trait.

FACT (review / education bridge): Ashcraft & Krause (2007, *Psychonomic Bulletin & Review*, doi:10.3758/BF03194059) — math beyond simple retrieval depends on WM; high math anxiety functions like a resource-demanding secondary task (preoccupation with fear). Global consequence: avoidance of elective math, majors, and careers.

FACT (carry from XXIV): Association of math anxiety with performance is real but modest on average (meta ~r ≈ –0.17); mechanism candidates include WM competition + avoidance. Anxiety is **one bottleneck among many**, not destiny.

Conflict statement (HYPOTHESIS): Desirable difficulties often *increase* momentary retrieval/selection demand — exactly the demand anxiety already taxes. Unbuffered DD therefore risks converting a would-be desirable difficulty into an undesirable one for HMA students: same schedule, different learner state → different pedagogy.

XLI.5 Evidence that stakes and anxiety change whether "hard practice" helps

FACT (pressure can kill retrieval benefits): Hinze & Rapp (2014, *Applied Cognitive Psychology*, doi:10.1002/acp.3032) — high-stakes vs low-stakes quizzes: quiz performance similar, but **final test better after low-stakes**; only low-stakes quizzes beat reread. Authors propose **retrieval disruption**: pressure can impair the memory-formation benefits of retrieval even when online quiz scores look fine.

FACT (classroom retrieval often *lowers* reported test anxiety): Agarwal, D'Antonio, Roediger, McDermott & McDaniel (2014, *JARMAC*, doi:10.1016/j.jarmac.2014.07.002) — survey of 1,408 middle/high-school students in classrooms using retrieval practice; 72% reported feeling **less nervous** for unit tests/exams when retrieval practice was used; 92% said it helped learning. Self-report, not RCT — but directionally opposite to "tests always raise anxiety."

FACT (meta: practice tests tend to reduce test anxiety): Yang, Li, Zhao, Luo & Shanks (2023, *Educational Psychology Review*, doi:10.1007/s10648-023-09801-w) — 24 studies / 25 effects / N≈3,374; practice tests reduce test anxiety with Hedges' $g \approx -0.52$; easy practice tests more mitigating than difficult ones. **Caveat:** mostly classroom/quiz anxiety outcomes — not a free pass to ship brutal mixed ACT sims as day-one "practice tests."

FACT (anxiety can reshape *what* is learned, not only performance): Fioriti, Pizzie, Evans, Green & Lyons (2025, *JEP:LMC*, doi:10.1037/xlm0001453) — adults learning unfamiliar multiplication across days: high math anxiety associated with **impaired procedural learning** on unrepeated problems but **accelerated retrieval learning** on heavily repeated problems — consistent with effort allocation toward declarative retrieval routes and away from procedural efficiency gains.

HYPOTHESIS for MindCraft: Anxious Maya may *prefer* and even accelerate on repeated retrieval of the same items (looks like mastery, feels safer) while under-building procedural flexibility and strategy selection — exactly the exam skills interleaving/SE target. Product that only rewards repeated-item fluency can **serve anxiety's preference** while failing transfer.

XLI.6 Resolution doctrine (not a slogan)

FOUNDER BELIEF / HYPOTHESIS — SAFE-DD stack (ties FEI / mastery climate / IL / SE):

1. **Equip before you strain.** Short block-lite + worked-example / card fade until execution errors are mostly slips (XXVI, XXXIX). DD without schema is undesirable.
2. **Lower stakes while raising retrieval.** Soft-wrong, mastery climate, no public rank, no streak-as-worth (XXXVIII; Hinze & Rapp). Retrieval as *training*, not judgment.
3. **Dose difficulty to remaining WM.** Prefer structured SE picks over open essays under high affect (XL SE-3); prefer faded interleave over day-one full mix (IL-2); prefer spaced short sets over marathon mixed threat.
4. **Frame the feeling.** Copy: "This mix trains *choosing* — feeling slower is the workout, not proof you're behind." Never: "hard mode for elites" / "grind through the pain."

5. **Measure the right win.** Session accuracy under block is not the North Star. Prefer `retry_120s`, delayed mix / `transfer_pass`, `solo_transfer_pass`, and anxiety-segment retention — not streaks or hint-free speed alone.
6. **Do not abandon DD for the anxious segment.** Permanent easy fluency is identity colonization by avoidance (XXV). Graded exposure to equipped difficulty *is* the path — therapy-adjacent but not a therapy claim (queue id 58).

Kill the false dichotomy: Bjork vs Ashcraft is not "science wars." Bjork already says unequipped difficulty is undesirable; Ashcraft explains *why anxiety unequips* (WM). The product job is **re-equip + destake + sequence**, then apply DD.

XLI.7 Competitive / marketing implications

Competitor pattern	What it often does under anxiety	MindCraft wedge
Duolingo-like streaks	Turns miss into identity failure; pressures retrieval	Ban streak-as-learning; train without shame timers
Blocked textbook / many apps	Fluency comfort; weak strategy choice	Honest: early block OK; claim exam-ready only after mix
Brutal "exam sim" day one	High stakes + hard mix → Hinze-style disruption + exit	Soft sim → rise stakes after `retry_120s` healthy
ChatGPT tutor monologue	Soothes without generation; may reduce threat short-term	Student-generated why *before* AI wrap (XL); calm ≠ mastery
"Growth mindset" posters	Empty struggle praise without equipment	Struggle + scaffolds + mastery climate + transfer metrics

Marketing language that survives

- **Allowed:** "Hard practice only after you're equipped — then we mix so exams don't surprise you."
- **Allowed:** "Feeling slower on mixed sets is the training signal, not a verdict."
- **Allowed (parent):** "We separate *training retrieval* from *graded judgment* so practice builds memory instead of panic."
- **Banned:** "Struggle always equals learning."
- **Banned:** "We never make anxious students struggle."
- **Banned:** Streak / blocked fluency / AI comfort as proof of readiness.
- **Banned:** Bloom 2-sigma; absolute tutoring-is-free.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without DD×ANX data
Soft-wrong + mastery copy on hard items	L1 affect-safe practice climate
Block → fade → interleave schedule	L1 sequenced DD (intent aligned XXXIX)
Low-stakes spaced retrieval before "exam mode"	L1 Hinze-aligned retrieval hygiene

“Our struggle pedagogy raises ACT for anxious students”	L0 until DD-1/DD-2
Always-on max mix / open SE under threat	L0 — likely undesirable difficulty

XLI.8 Experiments spawned

ID	Question	Design	Primary	Kill condition
DD-1	Faded DD (block→mix + soft-wrong) vs early full mix same items	2-arm; stratify high vs low math-anxiety pretest	7d `transfer_pass`; `retry_120s`; exit after first miss	Early-mix wins transfer **and** retry in HMA segment → fade doctrine wounded
DD-2	Low-stakes retrieval sets vs same items framed as scored “exam drill”	2-arm (Hinze-style stakes)	Delayed retention; state anxiety; return D+1	High-stakes ≥ low-stakes on delayed + no anxiety cost → stakes doctrine overstated
DD-3	Structured SE under mix vs open SE under mix for HMA segment	2-arm (densifies SE-3)	Completion; transfer; hint binge	Open SE wins without churn → structured-default less necessary
DD-4	Repeated-item fluency path vs varied procedural path (Fioriti-inspired)	2-arm within HMA	Procedural transfer / strategy-class errors vs item-repeat accuracy	Repeat-fluency equals varied on transfer → anxiety-shaped preference harmless
DD-QUAL	10 Maya interviews: “When did hard practice feel like training vs attack?”	Appendix B	coded stake/equipment moments	Nobody separates → UX framing invisible

Pre-reg (XXXIV): DD-* identify **schedule × stakes × anxiety → transfer/return**. They do **not** identify long-term identity from one hard session. Densifies Experiment D and IL-2/SE-3 as the anxiety-gated versions of desirable difficulty.

Supersedes / densifies: XXIX.4 “never introduce DD as shame”; XXXIX IL-2; XL SE-3; I.4 bottleneck → now has an operational SAFE-DD stack pending DD-1/2.

XLI.9 Confidence table

Claim	Label	Confidence
Spacing/interleaving/retrieval/generation can improve long-term learning vs fluent short-term conditions	FACT (Bjork lineage)	High
Difficulties are desirable only if the learner can respond successfully	FACT (Bjork & Bjork)	High
Math anxiety disrupts WM / acts like dual-task load	FACT (Ashcraft & Kirk, 2001; Ashcraft & Krause, 2007)	High
High stakes can erase retrieval-practice benefits on delayed tests	FACT (Hinze & Rapp, 2014)	High
Classroom retrieval programs often associate with *lower* reported exam nerves	FACT (Agarwal et al., 2014; self-report)	Medium–High
Practice testing meta-analytically reduces test anxiety (esp. easier quizzes)	FACT (Yang et al., 2023)	High
HMA may accelerate repeated-item retrieval learning while impairing procedural learning	FACT (Fioriti et al., 2025)	Medium–High (adult lab; arithmetic)
SAFE-DD (equip → destake → dose → frame → measure) is the right MindCraft default	FOUNDER BELIEF / HYPOTHESIS	Medium–High
"Struggle always" or "never struggle anxious kids" is good marketing	SPECULATION / false	Low (against)

XLI.10 What this chapter kills

1. **Kill:** "Desirable difficulties always help — turn difficulty up for everyone." (Ignores Bjork's own *equipped* clause + Ashcraft WM.)
2. **Kill:** "Anxious students should only get easy, blocked fluency forever." (Serves avoidance; fails transfer; Fioriti-shaped trap.)
3. **Kill:** High-stakes "exam mode" as the default vehicle for retrieval practice. (Hinze & Rapp.)
4. **Wound:** Self-report that "quizzes reduce anxiety" as license for brutal day-one mixed sims — survives as *low-stakes*, *well-sequenced* retrieval, not as cruelty.
5. **Survive (constrained):** Faded, low-stakes, mastery-framed desirable difficulties that produce delayed mix evidence — the commercial spine for "exam-ready without panic theater."

Doctrine until data: Ship **SAFE-DD**: equip → lower stakes → dose to WM → frame as training → measure delayed transfer and `retry_120s` in the anxiety segment. Market productive struggle only when those guards are visible. Never sell pain or comfort as mastery.

Part XLII — Social Comparison & Leaderboards: When Ranks Help vs Hurt Novices

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 3)

Primary question: When do social-comparison surfaces (especially leaderboards) motivate learning versus threaten identity, and what should MindCraft default for Maya — the anxious / novice / identity-fragile segment?

Owners: Motivation · Affect / Anxiety · Product (Practice / Community UX) · Growth / Positioning · Red Team

Commercial job: Kill "ranks = engagement" as a North Star; harden the I.4 TARGET rule (default-ban appearance UX) into a shippable comparison policy competitors can be audited against.

XLII.1 Why this chapter exists

Part XXXVIII (goal orientation) already bans **appearance** goal structures and streak-as-North-Star. Part XXIV / XLI show social-evaluative threat taxes working memory and can turn desirable difficulty into threat. Competitive products (Duolingo-like ranks, classroom Top-N boards, public classroom streaks) still sell comparison as "motivation."

FOUNDER BELIEF under audit: Identity transformation needs competence evidence — not a public ladder that tells bottom-half Maya she is behind *people* rather than behind a *skill criterion*.

Claim we refuse as doctrine: "Leaderboards motivate learners" (full stop) — and its twin "if we anonymize, any infinite rank is fine."

XLII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Social comparison	Evaluating self via others' standing (Festinger)	Any peer rank, percentile, "class average," public tutor callout	Turns mastery climate into appearance climate
Upward / downward / lateral	Compare to better / worse / similar others	Top-N board; "you beat 80%"; neighbor band	Extreme upward gap → give-up; downward → complacent ego boost without learning

Absolute vs relative board	All players ranked vs local neighbors / Top-N only	Infinite classroom board vs "near you" strip	Absolute + real names = max face threat for low ranks
Criterion / mastery feedback	Progress vs a task standard, not vs peers	Path progress, concept mastery bars, personal bests	Still can be gamed if criterion ≠ transfer
Normative / appearance feedback	Standing vs others; looking smart/fast	Public leaderboard, streak leagues	Identity threat; WM load; exit after first public miss

Operational definition (HYPOTHESIS): A MindCraft comparison surface is *safe* only when (a) the default is **self/criterion**, (b) peer comparison is **opt-in**, (c) comparison targets are **attainable** (local band, not unreachable Top-1), (d) metrics ranked are **effort/process-compatible** with mastery climate (not speed-only or streak-only), and (e) low standing is never **publicly named** for the anxiety / novice segment.

XLII.3 Festinger → competition: the comparison engine

FACT (origin): Festinger (1954, *Human Relations*, "A Theory of Social Comparison Processes") — when objective standards are ambiguous, people evaluate abilities/opinions by comparing with others; ability comparison carries a unidirectional drive upward (improve toward better standards).

FACT (selection + reaction meta): Gerber, Wheeler & Suls (2018, *Psychological Bulletin*, doi:10.1037/bul0000127) — across 60+ years: when choosing up vs down, strong preference for **upward** comparison under no threat; **contrast** (self-evaluation moves *away* from the target) dominates reactions — ability estimates and affect often decline after upward contact. People often choose upward comparisons *even though* the typical reaction is self-deflating contrast.

FACT (competition model): Garcia, Tor & Schiff (2013, *Perspectives on Psychological Science*, doi:10.1177/1745691613504114) — social comparison is a major source of competitive attitudes/behavior; situational amplifiers include **proximity to a standard** (near #1 vs far away) and **N** (few vs many competitors). Ranking UIs are engineered comparison machines.

Product translation: Shipping a leaderboard is not a neutral "engagement widget." It is a deliberate injection of comparison → contrast risk → competitive framing. For Maya, that often means **appearance climate** (XXXVIII), not mastery.

XLII.4 Leaderboards can move behavior — under narrow conditions

FACT (HE experiment): Landers & Landers (2014, *Simulation & Gaming*, doi:10.1177/1046878114563662) — random assignment to a course wiki **leaderboard** (not tied to grades) increased project interactions (~+29.6) vs control; time-on-task mediated academic performance. Supports: leaderboards *can* raise engagement via goal/conflict attributes **in that setting**.

Caveats (FACT / scope): Undergraduate / course project; not math-anxious high-school Maya; not public infinite classroom shame; achievement metric was interaction volume, not durable transfer. Landers' theory of gamified learning (2014, doi:10.1177/1046878114563660) treats game attributes as mediators/moderators of *instructional* quality — gamification does not replace pedagogy.

FACT (HE systematic review): Li, Liang, Fryer & Shum (2024, *Journal of Computer Assisted Learning*, doi:10.1111/jcal.13077) — 20 articles / 22 studies / 29 interventions (2014–2023), **all higher education**; ~half ≤1 hour. Leaderboards *can* benefit motivation/engagement/performance, but **effects hinge on design**. Authors recommend (among others): absolute boards for motivation *with* anonymity especially for low ranks; mix performance + engagement criteria; caution that HE short-duration findings do **not** generalize cleanly to K–12.

HYPOTHESIS: Landers-style gains are most likely when learners already have task competence + low identity threat + voluntary/low-stakes ranking. MindCraft's default user is the opposite segment on day one.

XLII.5 Math-specific: anxiety × infinite ranks

FACT (primary maths game, large N): Almo, Rocha, Brennan & Dondio (2024, *International Journal of Serious Games*, doi:10.17083/ijsg.v11i4.794) — N=1,389 Irish primary students, 6-week *Seven Spells* programme with an **infinite leaderboard**. Leaderboard enjoyment predicted by rank position **and** maths anxiety: **maths-anxious players disliked the leaderboard more than non-anxious peers even after controlling for position**. Preference for playing against others also predicted enjoyment. Small correlation between liking the board and liking the game.

Implication (HYPOTHESIS): For anxious learners, public infinite rank is not a free engagement boost — it is a trait-moderated threat surface. "They're low because they're bad at the game" does not exhaust the mechanism; anxiety predicts dislike **net of rank**.

FACT (negative feedback climate): Fong, Patall, Vasquez & Stautberg (2019, *Educational Psychology Review*, doi:10.1007/s10648-018-9446-6) — meta-analysis (78 studies): negative feedback reduces intrinsic motivation vs positive feedback; harm lessens when feedback includes instructional how-to-improve detail and uses **criterion-based** standards (vs purely normative). Public low rank is often pure normative negative feedback without instructional content.

Bridge to XLI / XXIV: Social-evaluative threat (public miss, public bottom third) is exactly the stakes Hinze & Rapp warn can disrupt retrieval benefits — and Ashcraft-style WM load. Infinite boards amplify that climate at scale.

XLII.6 Resolution doctrine — SAFE-COMPARE

FOUNDER BELIEF / HYPOTHESIS — SAFE-COMPARE stack (ties TARGET / SAFE-DD / FEI):

1. **Default = criterion, not peers.** Show progress vs path / mastery / personal best. Peers off by default.
2. **Opt-in competition only.** Never force infinite boards onto gap-scan novices or high-anxiety segment.
3. **If peers: local + attainable.** Prefer neighbor bands / similar-ability cohorts over Top-1 theater (Garcia proximity + Gerber contrast).
4. **Anonymize low standing.** No named public bottom ranks (Li et al. Rec. 2; Bai et al. cited therein).
5. **Rank what mastery climate allows.** Prefer process-compatible signals (retries completed, transfer attempts, concepts unlocked) over speed leagues / streak leagues. Never rank "hint-free seconds" as identity.

6. **Instructional wrap on any relative miss.** If relative feedback ships, attach *how to improve next* (Fong moderators) — not only “you’re #47.”
7. **Measure the right win.** `retry_120s`, `challenge_accept` (mastery-coded), `transfer_pass` / `solo_transfer_pass` — not `leaderboard_views` or rank delta as learning.

Kill the false dichotomy: “Competition is toxic” vs “competition is always motivating.” Competition can raise time-on-task for some prepared learners (Landers). For Maya default, **unbuffered infinite peer rank is appearance UX** — banned until SC experiments say otherwise.

XLII.7 Competitive / marketing implications

Competitor pattern	Comparison mechanism	MindCraft wedge
Duolingo-like leagues / ranks	Normative weekly standing; streak adjacent	Mastery path progress; ban streak/rank as learning proof
Classroom Top-N boards	Extreme upward targets; face threat	Criterion bars; optional local bands
Khan / practice dashboards	Often self-paced; weaker public rank	Keep self/criterion; don’t “gamify catch-up” with shame ranks
Brilliant prestige signals	Ability-appearance among peers	Sell *solo transfer* and *challenge-seeking why*, not percentile flex
ChatGPT tutor alone	No peer board — but social comparison still via school/parents	Own the *safe* comparison story: progress vs yesterday’s self

Marketing language that survives

- **Allowed:** “Progress against *your* path — not a public ladder.”
- **Allowed:** “Competition is optional; mastery climate is the default.”
- **Allowed (parent):** “We don’t put anxious kids on a named scoreboard to ‘motivate’ them.”
- **Banned:** “Leaderboards keep students hooked” as a learning claim.
- **Banned:** Rank / league / streak as North Star or as proof of identity change.
- **Banned:** Bloom 2-sigma; absolute tutoring-is-free; empty mindset posters.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without SC data
Criterion progress / personal bests default	L1 mastery-climate alignment (XXXVIII)
Opt-in anonymized local band	L1 SAFE-COMPARE intent
“Our leaderboard raises ACT / identity for anxious Maya”	L0 until SC-1/SC-2

Forced infinite named classroom board	L0 — appearance UX; likely harm for HMA

XLII.8 Experiments spawned

ID	Question	Design	Primary	Kill condition
SC-1	Criterion-only progress vs infinite named leaderboard (same practice items)	2-arm; stratify high vs low math-anxiety	`retry_120s`; exit after public miss; 7d `transfer_pass`	Infinite board wins transfer **and** retry in HMA → default-ban wounded
SC-2	Opt-in local anonymized band vs forced absolute board	2–3 arm	Enjoyment; state anxiety; return D+1; challenge_accept motive codes	Forced absolute ≥ opt-in local on retention without anxiety cost → anonymity/opt-in overstated
SC-3	Rank process metrics (retries, mix attempts) vs rank speed/streak	2-arm within volunteers	Strategy-class errors; `transfer_pass` vs session speed	Speed rank equals process rank on transfer → metric choice moot
SC-4	Criterion negative feedback + how-to-improve vs rank-only “you’re #N”	2-arm (Fong-inspired)	Intrinsic motivation proxy; retry	Rank-only ≥ instructional criterion → Fong moderator fails in product
SC-QUAL	10 Maya interviews: “When did seeing others’ scores help vs freeze you?”	Appendix B	coded comparison threat moments	Nobody distinguishes → copy invisible

Pre-reg (XXXIV): SC-* identify **comparison surface × anxiety → return/transfer**. They do **not** identify that “healthy competition builds character” from one league season.

Densifies: XXXVIII appearance ban; I.4 TARGET “leaderboard kill for Maya default”; XXIV social-evaluative threat hypothesis.

XLII.9 Confidence table

Claim	Label	Confidence
People use others to evaluate abilities when standards are ambiguous	FACT (Festinger, 1954)	High
Upward comparison often preferred; contrast often dominates self-eval/affect	FACT (Gerber et al., 2018)	High
Ranking contexts amplify competitive social comparison	FACT (Garcia et al., 2013)	High
Leaderboards can increase time-on-task / mediate performance in some HE settings	FACT (Landers & Landers, 2014)	High (scope-limited)
Leaderboard effects in education are design-dependent; HE evidence ≠ K–12	FACT (Li et al., 2024)	High
Maths-anxious players dislike infinite boards more, net of rank	FACT (Almo et al., 2024)	High
Normative negative feedback tends to hurt intrinsic motivation vs positive; criterion + how-to softens	FACT (Fong et al., 2019)	High
SAFE-COMPARE default (criterion, opt-in, local, anonymize low, process metrics) is right for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Leaderboards motivate everyone / raise learning for Maya”	SPECULATION / false as universal	Low (against)

XLII.10 What this chapter kills

1. **Kill:** Universal “leaderboards motivate students” as product doctrine or marketing. (Design- and trait-dependent; HE ≠ Maya.)
2. **Kill:** Infinite named public ranks as default engagement for novices / high math-anxiety. (Almo et al.; TARGET / appearance.)
3. **Kill:** Rank delta / league standing / streak leagues as North Star or as proof of identity transformation.
4. **Wound:** Absolute-board motivation gains from HE reviews as license for K–12 public boards — survives only as *design hypothesis*, not as Maya default.
5. **Survive (constrained):** Opt-in, anonymized, attainable, criterion-wrapped comparison for learners who already seek competition — never as the onboarding spine.

Doctrine until data: Ship **SAFE-COMPARE**: criterion default → peer opt-in → local/attainable → anonymize low standing → rank mastery-compatible process → instructional wrap → measure transfer/retry, not rank views. Market “progress without public ladders.” Never sell ranks as belonging.

Part XLIII — Habit Formation Science (Wood/Clear): Cue Design Without Identity Colonization

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 6 ≡ Red Team slot; ch43 never written → prefer Researcher per rotation)

Primary question: How should MindCraft use habit science (context cues, repetition, automaticity) to raise *return to practice* without making streaks / daily check-ins the student’s identity — or mistaking habit for learning?

Owners: Motivation · Product (Practice / notifications / recognition) · Growth / Positioning · Red Team

Commercial job: Harden Insight 5 / HID into a shippable **SAFE-HABIT** stack competitors (especially Duo-like streak products) can be audited against; kill “habit = mastery” and “21 days” folklore in copy and KPIs.

XLIII.1 Why this chapter exists

Parts XXV (habit ≠ identity), XXXVIII (appearance / streak North Star ban), XLI (SAFE-DD), and XLII (SAFE-COMPARE) already treat streaks and ranks as threat surfaces when they become *worth*. Market gravity still pulls edtech toward Duolingo-shaped habit loops: push cue → micro-session → streak reward → loss aversion.

FOUNDER BELIEF under audit: Identity transformation needs *reliable return to difficulty* — so cue design is load-bearing — but the habit that forms must be “open practice and try the next hard thing,” not “protect a number that proves I am consistent.”

Claims we refuse as doctrine:

1. “Streaks = learning” / streak length as North Star.
2. “Habits take 21 days” as science.
3. Treating popular packaging (*Atomic Habits*) as peer-reviewed habit theory without Wood/Gardner/Lally grounding.
4. Habit loops that **colonize identity** — the student becomes “a streak-keeper” rather than “someone who can recover from wrong.”

XLIII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused

Habit (scientific)	Learned context→response association in procedural memory; cue activates response with limited goal mediation (Wood)	Stable time/place/preceding-action cue for practice open	Shipping “motivation” when the real lever is *cue stability*
Automaticity	Behavior feels efficient / less deliberative	Self-report automaticity; shorter time-to-first-problem	Confusing app-open automaticity with *math skill* automaticity
Context cue	Physical setting, time, people, prior action in a sequence	After-school phone dock; tutor session end → 10-min solo; calendar block	Push spam as fake context; unstable cues never habitize
Implementation intention	If-then plan: when Y, do X (Gollwitzer)	Onboarding “After dinner, I open MindCraft Map”	Vague goals (“practice more”) with no situational trigger
Identity colonization (product sense)	Habit metric becomes who the student *is* / fears losing	Streak as self-worth; freeze culture; XP-as-self	HID failure — return without identity change; break → exit
SAFE-HABIT	Cue + small start + miss-forgiveness + learning metrics above habit metrics	See XLIII.6	Habit theater that still ranks streak / DAU as proof

Operational definition (HYPOTHESIS): A MindCraft habit surface is *safe* only when (a) cues target **opening equipped practice**, not protecting a counter, (b) miss recovery is cheap (Lally: one miss ≠ wipe), (c) recognition prefers process/identity markers over streak length, and (d) North Stars remain `retry_120s` / `transfer_pass` / `solo_transfer_pass` / identity items — never streak days.

XLIII.3 Wood: habits are context–response, not vibes

FACT (review): Wood & Rünger (2016, *Annual Review of Psychology*, doi:10.1146/annurev-psych-122414-033417) — habits are learned associations between recurring **context cues** and responses, strengthened gradually via associative / reward learning; once formed, perception of the cue activates the response representation with limited need for executive control. Defining features of habit automaticity include cue activation and **relative insensitivity to short-term goal/outcome changes**.

FACT (habit–goal interface): Wood & Neal (2007, *Psychological Review*, “A New Look at Habits and the Habit–Goal Interface”) — goals help *start* repetition and exposure to cues; once habits form, contexts can trigger responses without a mediating goal. Habit change therefore often requires changing cues / environments, not only “trying harder.”

FACT (systems view): Wood, Mazar & Neal (2021, *Perspectives on Psychological Science*, habits vs goals as separate but interacting systems) — habitual responses can activate regardless of current goals; people may still inhibit them, but cue-driven activation is real. Product translation: a streak notification is a **Pavlovian/context cue** engineered by the app — not proof the student wants math.

Product translation: If MindCraft wants return, design **stable contexts** (same after-school window; end-of-tutor-session stack; device home-screen tile) more than pep talks. If MindCraft wants identity, do **not** make the strongest cue "don't break the number."

XLIII.4 Clear / Fogg packaging vs the evidence spine

FACT (popularization, not primary science): Clear (*Atomic Habits*, 2018) packages cue / craving / response / reward loops and "habit stacking" ("After [current habit], I will [new habit]") for a mass audience. Fogg (*Tiny Habits*) popularized tiny start + anchor habits. Neither replaces Wood's procedural-memory account.

FACT (what stacking actually is): Habit stacking's evidence spine is largely **implementation intentions** — Gollwitzer & Sheeran (2006, *Advances in Experimental Social Psychology*, doi:10.1016/S0065-2601(06)38002-1): across 94 tests, if-then plans show a medium-to-large effect on goal attainment ($d \approx 0.65$), via heightened cue accessibility and automated initiation.

HYPOTHESIS: Clear is usable *copy scaffolding* for onboarding ("After tutoring, open Map for one mission") **if** MindCraft cites the Gollwitzer mechanism and measures initiation → equipped practice, not Clear's compound-growth metaphors as science.

Kill: Marketing that says "built on Atomic Habits science" as if the book were a methods paper. Prefer: "cue-based practice plans (implementation intentions; habit research)."

XLIII.5 How long, how fragile, what to measure

FACT (timeline myth): Lally, van Jaarsveld, Potts & Wardle (2010, *European Journal of Social Psychology*, doi:10.1002/ejsp.674) — daily repetition in a consistent context; automaticity rises on an asymptotic curve; among good-fit participants, median time to ~95% of asymptote ≈ **66 days** (range **18–254**). Missing **one** opportunity did not materially derail formation. ~Half of motivated participants still failed to perform consistently enough to reach habit status in the study window.

Kill: "21 days to form a habit" (Maltz folklore, not this literature). Also kill product copy that implies a week-long streak *is* a habit.

FACT (definition hygiene): Gardner (2015, *Health Psychology Review*, doi:10.1080/17437199.2013.876238) — habit as a *process* whereby a stimulus generates an impulse to act via learned S–R association; impulse need not equal behavior. Most health-habit studies are correlational / self-report-heavy — confidence ceiling for edtech transfer claims.

FACT (measurement trap): Verplanken & Orbell (2003, *Journal of Applied Social Psychology*, doi:10.1111/j.1559-1816.2003.tb01951.x) — SRHI includes history of repetition, automaticity, **and expressing identity**. Self-report habit strength can smuggle identity language into a frequency metric.

Product implication (HYPOTHESIS): Do not treat SRHI-like "this is typically me" items as proof of *math identity transformation* (Part XXV / id 61 queue). Separate: (1) practice-open automaticity, (2) math self-efficacy / narrative identity scales, (3) transfer metrics.

XLIII.6 Duolingo-shaped products: habit works — and colonizes

FACT (industry, self-report): Duolingo publicly designs streaks using habit research framing; company blog claims learners who reach short streaks are far more likely to continue (e.g., 7-day streak correlational lifts — *not* causal RCTs published as independent learning science). Streak freezes / separation of streak vs daily goal are retention experiments on **engagement**, not transfer.

FACT (misuse qualitative): Hadi Mogavi et al. (2022, *Learning @ Scale*, doi:10.1145/3491140.3528274) — forum analysis + interviews on Duolingo: **gamification misuse** when users fixate on XP/streaks/leaderboards and get distracted from learning; reasons include competitiveness, playfulness overlearning, dark nudges, compulsion, herding. Ramifications touch learning efficiency, well-being, and ethics.

Bridge to XLII / XXXVIII: Streak leagues + infinite ranks are the same family as appearance climates — consistency theater. Wood predicts cues will keep firing even when the *learning* goal cooled; Hadi Mogavi shows the product can train the wrong response (protect streak / farm XP).

FOUNDER BELIEF: MindCraft may borrow **cue stability + small start + miss forgiveness**. MindCraft must not borrow **streak-as-self** or **loss-aversion-as-pedagogy**.

XLIII.7 Resolution doctrine — SAFE-HABIT

FOUNDER BELIEF / HYPOTHESIS — SAFE-HABIT stack (ties HID / TARGET / SAFE-DD / SAFE-COMPARE):

1. **Cue the practice, not the counter.** Default cues: calendar/time, tutor-session end stack, story-world return point — not "your streak dies at midnight" as the primary cue.
2. **Implementation intention onboarding.** One written if-then in gap-scan aftermath or first session ("After [stable cue], I open one mission"). Measure initiation, not slogan recall.
3. **Tiny equipped start.** First action ≤ friction of one problem with SAFE-DD equipment — not a multi-screen habit tutorial.
4. **Miss ≠ identity failure.** One miss does not reset belonging (Lally). Prefer gentle resume copy over shame / "you broke it." Streak freezes are a *retention patch*, not a learning feature — if shipped, never as North Star.
5. **Habit metrics subordinate.** Track `D1/D7 return`, cue-honoring rate, time-to-first-problem as *diagnostics*. Learning North Stars remain retry / transfer / solo transfer / challenge_accept (mastery-coded).
6. **Identity markers elsewhere.** Map fill, story unlocks, tutor recognition of *recovery*, parent "they tried the hard one again" — not streak length as "who they are."
7. **No identity colonization.** Ban copy that equates worth with unbroken days. Ban ranking streaks (XLII). Ban streak leagues as proof of transformation.

Kill the false dichotomy: "We refuse all habit design" vs "we must out-Duo Duo." Refuse *colonizing* habit design; keep *cue science*.

XLIII.8 Competitive / marketing implications

Competitor pattern	Habit mechanism	MindCraft wedge
Duolingo streaks / freezes	Loss aversion + daily cue; retention-first	Cue practice return; miss-forgiving; never streak-as-learning
Duo leagues + XP	Habit + appearance compound	SAFE-COMPARE + SAFE-HABIT; sell solo transfer
Khan session streaks / energy	Mild habit / progress bars	Keep criterion progress; don't add shame timers
Brilliant daily puzzles	Delight + return habit	Return *into* recoverable struggle, not only delight
ChatGPT "study every day" tips	Advice without cue architecture	Own *if-then* + *equipped mission* as the plan

Marketing language that survives

- **Allowed:** "Same time, same cue — open one hard problem and recover."
- **Allowed:** "Habits get you to the desk; identity is what you do when it's wrong."
- **Allowed (parent):** "We build a practice cue without making a broken streak mean failure."
- **Banned:** "Streaks create math identity."
- **Banned:** "21-day habit challenge = mastery."
- **Banned:** Streak / DAU as North Star or as proof of transformation.
- **Banned:** Bloom 2-sigma; tutoring-is-free; empty mindset posters.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without HAB data
If-then practice cue + tiny equipped start	L1 initiation design (Gollwitzer-aligned)
Miss-forgiving resume without streak shame	L1 HID hygiene
"Our streak raises ACT / math identity"	L0 until HAB-1/HAB-2
Forced streak + public streak rank as default	L0 — colonization UX

XLIII.9 Experiments spawned

ID	Question	Design	Primary	Kill condition

HAB-1	If-then cue plan vs streak-primary onboarding (same missions)	2-arm; new users	D7 return; time-to-first-problem; 14d `transfer_pass`	Streak arm wins transfer **and** identity item → on-ramp-only doctrine wounded
HAB-2	Miss-shame copy ("you broke your streak") vs miss-forgiving resume	2-arm after forced miss	Return D+1; state anxiety; `retry_120s`	Shame ≥ forgive on return without anxiety cost → forgive overstated
HAB-3	Stable calendar cue vs push-only reminders	2-arm	Cue-honoring rate; 4-week automaticity proxy	Push-only equals calendar on return → context-stability claim weak
HAB-4	Recognition: process/recovery badge vs streak badge vs none	3-arm (ties GO-2)	`challenge_accept` motives; identity item; streak obsession items	Streak badge wins engagement **and** transfer → HID weaker than feared
HAB-QUAL	10 Maya interviews: "What does your streak (elsewhere) make you feel when you miss?"	Appendix B	coded colonization vs helpful cue	Nobody reports threat → copy over-indexed

Pre-reg (XXXIV): HAB-* identify **cue/recognition × return/transfer**. They do **not** identify that "daily habit builds mathematicians" from DAU alone.

Densifies: Insight 5 / HID; I.4 TARGET streak ban; XXXVIII appearance; XLII streak-league kill.

XLIII.10 Confidence table

Claim	Label	Confidence
Habits are context–response associations; cue-triggered; relatively goal-insensitive once strong	FACT (Wood & Rünger, 2016; Wood & Neal, 2007)	High
Goals help form habits via repetition/exposure; changing cues aids habit change	FACT (Wood line)	High

Implementation intentions raise goal attainment (medium–large)	FACT (Gollwitzer & Sheeran, 2006)	High
Automaticity plateaus on the order of weeks–months; median ~66d in Lally sample; high variance; one miss often OK	FACT (Lally et al., 2010)	High
“21 days” is not the Lally result	FACT	High
Clear/Fogg are popular packaging; stacking ≈ if-then planning	HYPOTHESIS / FACT (mechanism mapping)	Medium–High
Gamification can distract from learning (Duo case)	FACT (Hadi Mogavi et al., 2022)	High (qualitative)
Streaks can raise return without identity / transfer	HYPOTHESIS (HID; industry)	Medium–High
SAFE-HABIT is right default for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Streaks = learning / identity”	SPECULATION / false as universal	Low (against)

XLIII.11 What this chapter kills

1. **Kill:** Streak length / DAU / “habit streak” as North Star or as proof of identity transformation. (HID; standing ban densified with Wood + Duo misuse.)
2. **Kill:** “21-day habit” marketing as science. (Lally variance + asymptote.)
3. **Kill:** Equating *Atomic Habits* branding with peer-reviewed habit doctrine. (Use Wood/Gardner/Gollwitzer; Clear as optional UX copy only.)
4. **Kill:** Shame-on-miss and public streak ranks as default engagement for Maya. (Colonization; XLII.)
5. **Wound:** “Any daily reminder is habit science” — survives only if tied to **stable context + repeated equipped response**, not push spam.
6. **Survive (constrained):** Cue design, tiny starts, implementation intentions, and miss forgiveness as **on-ramps** to FEI / transfer — never as the product thesis.

Doctrine until data: Ship **SAFE-HABIT**: cue practice (not the counter) → if-then onboarding → tiny equipped start → miss ≠ identity failure → habit metrics subordinate → identity markers elsewhere → ban streak colonization. Market “we get you back to the hard problem — without making a broken streak mean you’re not a math person.”

Part XLIV — Intrinsic Motivation Killers (Deci): Controlling Rewards and the Caveats

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 9; hour%6≠0; 3 researcher entries since synthesizer v1.6 → Researcher, not Synthesizer)

Primary question: When do MindCraft rewards, XP, badges, streaks, and "do this for the prize" UX *undermine* the intrinsic interest and autonomy we claim to build — and when are they informational rather than controlling?

Owners: Motivation · Product (Practice / recognition / parent copy) · Growth / Positioning · Red Team

Commercial job: Turn the vague "extrinsic bad / Deci says so" slogan into a shippable **SAFE-REWARD** stack competitors can be audited against; kill both "all rewards kill love of math" absolutism and "gamify everything" as identity strategy.

XLIV.1 Why this chapter exists

Parts XXV (habit ≠ identity), XXXVIII (appearance climates), XLII (SAFE-COMPARE), and XLIII (SAFE-HABIT) already treat streaks, ranks, and consistency theater as threat surfaces. Market gravity still pulls edtech toward *expected tangible / token rewards for doing the task*: XP for opening the app, coins for finishing any problem, badges for login streaks, parent dashboards that score "minutes practiced."

FOUNDER BELIEF under audit: Identity transformation needs *felt autonomy* and *competence information* — not the absence of all rewards. The killer is **controlling functional significance** (pressure to behave for the inducement), not the existence of feedback or unexpected acknowledgment.

Claims we refuse as doctrine:

1. "All rewards kill intrinsic motivation" (Cameron–Pierce counter-literature exists; CET is differentiated).
2. "Gamification is always fine if engagement rises" (engagement ≠ free-choice interest; Hadi Mogavi / Duo misuse).
3. "Verbal praise always helps kids" (Deci: controlling praise / "you should keep it up" can undermine).
4. Treating SDT as a poster ("autonomy / competence / relatedness") without contingency design.

XLIV.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Intrinsic motivation	Doing the activity for inherent interest / satisfaction	Free-choice return to hard problems; curiosity asks; challenge_accept for mastery reasons	Confusing "opens app for XP" with interest
Extrinsic motivation (spectrum)	Behavior for separable outcome; can be more/less autonomous (external → introjected → identified → integrated)	Parent-pressure vs "I chose ACT path because…"	Treating all extrinsic as poison or all as fine

Controlling vs informational	Event felt as pressure to behave vs competence feedback (CET)	“Finish 10 for a coin” vs “you used the bridge correctly”	Same badge can be either — design of *how* it is framed
Engagement-contingent reward	Expected reward for *doing* the task (no quality bar)	XP for any attempt; login coins	Classic undermine path (Deci 1999)
Completion-contingent	Expected reward for finishing	“Complete mission → loot” regardless of strategy quality	Similar undermine; weak competence signal
Performance-contingent	Reward for meeting a standard	Badge for transfer pass; tutor note for recovery	Can inform competence *or* control — climate decides
SAFE-REWARD	Prefer informational feedback; avoid expected tangible for mere engagement; autonomy-supportive framing	See XLIV.7	Reward theater that still trains external PLOC

Operational definition (HYPOTHESIS): A MindCraft reward surface is *safe* only when (a) it primarily conveys **competence / strategy information**, (b) expected tangible / token rewards are **not** the reason to open practice, (c) framing is autonomy-supportive (choice, rationale, non-pressuring language), and (d) North Stars remain `retry_120s` / `transfer_pass` / `solo_transfer_pass` / identity items — never XP velocity or badge count.

XLIV.3 SDT spine (needs, not posters)

FACT (overview): Ryan & Deci (2000, *American Psychologist*, 55(1), 68–78, doi:10.1037/0003-066X.55.1.68) — Self-Determination Theory posits innate needs for **competence, autonomy, and relatedness**; social contexts that support these needs facilitate intrinsic motivation and internalization; contexts that thwart them yield alienation and depleted motivation.

FACT (CET): Cognitive Evaluation Theory (Deci & Ryan line; summarized in Deci, Koestner & Ryan, 1999) — rewards and external events have **functional significance**: controlling aspects shift perceived locus of causality outward (undermine intrinsic motivation); informational aspects affirm competence (can enhance). Competence feedback without autonomy does not reliably enhance intrinsic motivation.

Product translation: MindCraft’s brand voice (“you’re becoming someone who recovers”) must not be paired with UX that says “do this *or* lose the prize.” Relatedness (tutor, story world, parent) is not a license for controlling rewards.

XLIV.4 The classic finding — and the differentiated meta

FACT (origin): Deci (1971, *Journal of Personality and Social Psychology*, 18(1), 105–115, doi:10.1037/h0030644) — expected extrinsic rewards for an interesting activity can reduce subsequent free-choice engagement (the "undermining" / overjustification pattern that launched the literature).

FACT (Deci–Koestner–Ryan 1999 meta): Deci, Koestner & Ryan (1999, *Psychological Bulletin*, 125(6), 627–668, doi:10.1037/0033-2909.125.6.627) — 128 studies. As predicted by CET:

- **Engagement-contingent** rewards undermine free-choice intrinsic motivation ($d \approx -0.40$).
- **Completion-contingent** ($d \approx -0.36$) and **performance-contingent** ($d \approx -0.28$) also undermine free-choice behavior on average.
- Engagement- and completion-contingent also hurt self-reported interest ($d \approx -0.15$ / -0.17).
- **Unexpected** tangible rewards: no average undermine on free-choice.
- **Positive feedback / verbal rewards:** enhance free-choice ($d \approx 0.33$) and interest ($d \approx 0.31$) *on average*.
- Tangible rewards tended to be **more detrimental for children** than college students; verbal rewards less enhancing for children than for college students.

FACT (education restatement): Deci, Koestner & Ryan (2001, *Review of Educational Research*, 71(1), 1–27, doi:10.3102/00346543071001001) — argue Cameron & Pierce (1994) understated undermining; conclude tangible rewards have a substantial undermining effect when CET contingencies are respected; defend keeping CET for educational practice.

Product implication (HYPOTHESIS): Expected XP/coins for *mere engagement* (open app, any answer, any completion) is the highest-risk MindCraft pattern for Maya-aged users. Informational soft-wrong + strategy feedback is the low-risk pattern. Performance-contingent badges *can* work if they mark real competence (transfer) and are not administered as pressure.

XLIV.5 Caveats — Cameron/Pierce, climate, and "not all rewards"

FACT (counter-meta): Cameron & Pierce (1994, *Review of Educational Research*, 64(3), 363–423, doi:10.3102/00346543064003363) — across 96 between-groups experiments, *overall* reward did not decrease intrinsic motivation; verbal praise increased it; the clearest negative effect was expected tangible rewards for *simply doing* a task (free-choice time after reward removal). They argued negative effects are circumscribed and avoidable.

FACT (debate): Ryan & Deci (1996, *RER*), Lepper et al. (1996), Kohn (1996) vs Cameron & Pierce (1996, *RER*) — methodological dispute over aggregation, moderators, and what "overall" means. **Not settled as "rewards are harmless."**

FOUNDER BELIEF / resolution for MindCraft: Do **not** cite "Deci proved all gamification is evil." Do **not** cite Cameron to greenlight engagement-contingent loot. Cite the **contingency**: expected tangible for mere doing is the kill zone; informational verbal/competence feedback is the build zone; performance-contingent is climate-sensitive.

FACT (controlling praise): Ryan (1982) and Ryan, Mims & Koestner (1983) — positive feedback administered with controlling language (e.g., "you *should* keep up the good work") reduces intrinsic motivation relative to informational administration. Deci 1999 summarizes interpersonal climate as a moderator of performance-contingent rewards.

Kill: "We praise a lot, so SDT is satisfied." Praise can be controlling.

FACT (choice as autonomy support): Patall, Cooper & Robinson (2008, *Psychological Bulletin*, 134(2), 270–300, doi:10.1037/0033-2909.134.2.270) — meta of 41 studies: providing choice enhances intrinsic motivation, effort, task performance, and perceived competence. Effects stronger for instructionally *irrelevant* choices, when ~2–4 successive choices are offered, when rewards are **not** stacked after the choice manipulation, and vs highly controlling controls; stronger for children than adults.

Product implication: Meaningful micro-choice (which story frame, which representation after mastery structure is set) can support autonomy — but **choice + immediately dangling reward** weakens the choice benefit. Do not "choose your path → earn 50 coins."

XLIV.6 Competitive patterns under CET audit

Competitor pattern	CET reading	MindCraft wedge
Duolingo XP / streaks / leagues	Expected engagement- and completion-contingent tokens + appearance climate (XLII/XLIII)	Informational recovery feedback; no XP-as-reason-to-open; SAFE-HABIT cues
Khan energy / points-ish progress	Mild completion signals; mostly mastery bars	Keep criterion progress; avoid coinization of minutes
Brilliant delight unlocks	Intrinsic interest scaffold; some completion loot	Keep delight *inside* the problem; don't pay for opening
ChatGPT "I'll quiz you if you want"	Low token reward; high fluency risk (XXXIII)	Autonomy of asking ≠ competence evidence; measure solo transfer
Parent dashboards: minutes / day streak	Parent-facing controlling contingency spills to student	Report `retry_120s`, transfer, challenge_accept — not "minutes bought with coins"

Bridge: XLIII says cue the practice, not the counter. XLIV says: if you must reward, reward **informative competence**, not engagement.

XLIV.7 Resolution doctrine — SAFE-REWARD

FOUNDER BELIEF / HYPOTHESIS — SAFE-REWARD stack (ties TARGET / SAFE-COMPARE / SAFE-HABIT / FEI):

1. **No expected tangible / token for mere engagement.** Ban default XP-for-open, coins-for-any-answer, loot-for-login. (Deci 1999 engagement-contingent.)
2. **Feedback > inducement.** Soft-wrong, strategy labels, bridge-gap naming, and "you recovered" tutor notes are the primary reinforcer. Prefer unexpected acknowledgment over announced prizes.
3. **If performance-contingent, mark real competence.** Badges/unlocks tied to `transfer_pass` / `solo_transfer_pass` / mastery-coded `challenge_accept` — not to streak days or raw volume. Administer without "you should" pressure.

4. **Autonomy-supportive framing.** Offer rationale; allow meaningful micro-choice (Patall); never "do this or lose the reward."
5. **Don't stack choice with dangling rewards.** Choice then immediate coin undermines the autonomy gain.
6. **Child-risk premium.** Maya-aged users are in the higher tangible-harm band of Deci 1999 — stricter default than adult SaaS gamification norms.
7. **Metrics hygiene.** Track XP/badge counts as *diagnostics of system behavior*, never as North Stars. Learning NS remain retry / transfer / solo transfer / identity.
8. **Parent copy:** Sell "she chose the harder one again" and competence evidence — not "she earned 400 XP."

Kill the false dichotomy: "No rewards ever" vs "gamify everything." Refuse *controlling engagement loot*; keep *informational competence signals* and story unlocks that are earned by transfer, not by attendance.

XLIV.8 Marketing language that survives

- **Allowed:** "We don't pay kids to open the app — we show them what they figured out."
- **Allowed:** "Unlocks follow real transfer, not login streaks."
- **Allowed (parent):** "Progress you can trust: recovery and transfer — not points theater."
- **Banned:** "Rewards create love of math."
- **Banned:** "XP builds identity."
- **Banned:** Absolute "all extrinsic motivation is bad" / "tutoring is free so motivation is free."
- **Banned:** Bloom 2-sigma; empty mindset posters; streak/XP as North Star.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without IM data
Informational soft-wrong + strategy feedback (no engagement loot)	L1 CET-aligned hygiene
Unexpected tutor/parent acknowledgment after recovery	L1 (unexpected tangible / social)
"Our XP system raises intrinsic math interest"	L0 until IM-1/IM-2
Expected coins for any completion as default	L0 — engagement-contingent kill zone

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XLIV.9 Experiments spawned

ID	Question	Design	Primary	Kill condition

IM-1	Engagement-loot (XP for any attempt) vs no-loot informational feedback (same missions)	2-arm; new users	Free-choice analog: voluntary hard-problem rate post-session; 14d interest item; `transfer_pass`	Loot arm wins free-choice **and** transfer → engagement-loot ban overstated
IM-2	Performance badge on `transfer_pass` (informational copy) vs same badge with controlling copy ("you should keep earning")	2-arm	`challenge_accept` motives; interest; return	Controlling ≥ informational on interest → climate claim weak
IM-3	Micro-choice without reward vs choice + dangling coins	2-arm (Patall moderator)	Intrinsic motivation proxy; effort; autonomy need satisfaction	Choice+coins ≥ choice-only → "don't stack" overstated
IM-4	Unexpected recovery acknowledgment vs announced prize for recovery	2-arm	Free-choice return; attribution coding	Announced ≥ unexpected on free-choice without control feelings → unexpected preference weaker
IM-QUAL	10 Maya interviews: "When a learning app gives points, what does that make you feel about the math?"	Appendix B	coded controlling vs informational	Nobody reports control → copy over-indexed

Pre-reg (XXXIV): IM-* identify **reward contingency × free-choice/interest/transfer**. They do **not** identify "gamification builds mathematicians" from XP velocity.

Densifies: TARGET appearance ban; SAFE-HABIT; SAFE-COMPARE; Insight on extrinsic traps; standing streak NS ban.

XLIV.10 Confidence table

Claim	Label	Confidence

Autonomy, competence, relatedness support intrinsic motivation / wellness	FACT (Ryan & Deci, 2000)	High
Expected engagement-/completion-contingent tangible rewards undermine free-choice IM on average	FACT (Deci, Koestner & Ryan, 1999)	High
Unexpected tangible rewards do not show average undermine	FACT (Deci et al., 1999)	High
Positive feedback enhances IM on average; controlling administration can reverse	FACT (Deci et al., 1999; Ryan line)	High
Tangible rewards more detrimental for children than college samples	FACT (Deci et al., 1999)	High
Cameron & Pierce find overall null with circumscribed negatives	FACT (1994); interpretation contested	Medium (dispute)
Choice enhances IM / effort / competence perceptions	FACT (Patall et al., 2008)	High
Stacking rewards after choice weakens choice benefit	FACT (Patall moderator)	Medium–High
SAFE-REWARD is right default for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“All rewards kill intrinsic motivation” / “XP creates math love”	SPECULATION / false as universals	Low (against)

XLIV.11 What this chapter kills

1. **Kill:** Expected engagement-contingent XP/coins/loot as default motivation architecture for Maya. (Deci 1999 $d \approx -0.40$ free-choice.)
2. **Kill:** Marketing that “rewards create love of math” or “XP = identity.” (External PLOC; HID.)
3. **Kill:** Absolute slogan “all extrinsic rewards are bad” used to ban informational feedback and earned transfer unlocks. (CET differentiation; Cameron caveat acknowledged.)
4. **Kill:** Controlling praise / “you should keep earning” as the voice of competence support.
5. **Wound:** “Performance badges are always safe” — survives only with informational climate + real competence criterion.
6. **Survive (constrained):** Unexpected acknowledgment, strategy-informative feedback, autonomy-supportive micro-choice, and transfer-gated unlocks under **SAFE-REWARD**.

Doctrine until data: Ship **SAFE-REWARD**: no expected tokens for mere engagement → feedback over inducement → performance marks only for real transfer/recovery → autonomy framing → don’t stack choice with dangling coins

→ child-risk premium → XP never North Star. Market “we don’t pay kids to practice — we make the competence visible.”

Part XLV — Mathematical Resilience: Growth Under Adversity Without Grit Theater

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 12 ≡ Red Team slot, but ch45 never written → prefer Researcher per rotation; 4 researcher entries since synthesizer v1.6 → not yet Synthesizer)

Primary question: What is *mathematical resilience* as an actionable construct (Johnston-Wilder & Lee), how does it differ from grit posters and empty “bounce back” copy, and what product stack keeps Maya in the growth zone without dumping her into the danger zone?

Owners: Motivation / Affect · Product (Practice / coach / tutor briefing) · Positioning · Red Team

Commercial job: Replace “build grit” / “resilience mindset” marketing with a shippable **SAFE-RESILIENCE** stack competitors can be audited against — growth-zone design, support recruitment, struggle-as-normal, value without coercion — tied to FEI / SAFE-DD / SAFE-REWARD.

XLV.1 Why this chapter exists

Parts XXIV (anxiety), XXV (self-efficacy / habit≠identity), XXX (attribution / helplessness), XXXI (flow / challenge–skill), XLI (SAFE-DD), and XLIV (SAFE-REWARD) already treat threat, recovery, and controlling incentives. Market and school language still collapses everything into **grit** or generic **resilience** slogans: “just push through,” “bounce back harder,” streak-as-character, leaderboard-as-grit-proof.

FOUNDER BELIEF under audit: Identity transformation under difficulty needs *domain-specific* resilience — knowing struggle is normal in math, valuing the work, knowing how to work at it, and **recruiting support** before anxiety flips the lid — not a personality transplant labeled grit.

Claims we refuse as doctrine:

1. Duckworth-style grit as MindCraft’s North Star or primary product claim (Credé et al. wound the construct).
2. “Desirable difficulty” = keep students in the danger/anxiety zone (conflicts SAFE-DD / Ashcraft).
3. Empty resilience posters without support architecture (tutor cards, peers, ontology scaffolds).
4. Treating mathematical resilience as synonymous with growth-mindset wall art.

XLV.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused

Developmental resilience (Masten)	Ordinary adaptive systems enabling competence despite adversity — not rare “super kids”	Design that protects basic systems (safety, belonging, competence feedback)	Selling “extraordinary character” as the product
Mathematical resilience (JW/Lee)	Positive stance that lets learners engage despite affective barriers: growth belief, value, knowing how to work at math, knowing how to recruit support	Stay in growth zone; ask for help; recover after stuck; continue study path	Posterizing the four aspects without UX
Growth Zone Model	Comfort (easy / little growth) → Growth (challenge + support) → Danger/Anxiety (threat / thinking collapses)	Challenge calibrated + support on tap; exit danger fast	Framing red-zone panic as “grit training”
Grit (Duckworth)	Perseverance + consistency of interest toward long-term goals	Tempting NS copy	Near-conscientiousness ; weak intervention leverage (Credé)
SAFE-RESILIENCE	Equip → name struggle → recruit support → destake danger → measure recovery/transfer — never grit KPI	See XLV.7	Resilience theater that still shames help-seeking

Operational definition (HYPOTHESIS): A MindCraft session builds mathematical resilience only when the student (a) experiences struggle as *expected* not as personal defect, (b) has a clear path to **support** (cards, tutor, peer, worked scaffold) before threat spikes, (c) returns to challenge after recovery (`retry_120s` / `challenge_accept`), and (d) can transfer without the model finishing the thought — not when streak or “grit score” rises.

XLV.3 Ordinary magic — resilience is not a unicorn trait

FACT: Masten (2001, *American Psychologist*, 56(3), 227–238, doi:10.1037/0003-066X.56.3.227) — resilience in development is often **ordinary magic**: competence despite adversity typically arises from normative adaptational systems (attachment, learning systems, self-regulation, community supports). Greatest threats are those that *compromise* these protective systems. Policy implication: protect and restore ordinary systems rather than hunt rare resilient personalities.

Product translation: MindCraft should not market “we forge resilient geniuses.” Market “we keep the ordinary protective systems online while math gets hard” — safety, relatedness (tutor/world), competence information, recoverable miss. That is Masten-aligned and FEI-aligned.

Kill: “Resilience is a rare character trait we select for.”

XLV.4 Mathematical resilience as a pragmatic construct

FACT (origin framing): Johnston-Wilder & Lee (2010) introduced **mathematical resilience** as a pragmatic positive construct opposite avoidance, anxiety, and helplessness — enabling learners to engage with mathematics despite affective barriers (cited as foundational in later scale and handbook work; e.g. Lee & Johnston-Wilder construct summaries; Johnston-Wilder & Lee handbook chapter, doi:10.4324/9781315740713-34).

FACT (four aspects — construct literature): Lee & Johnston-Wilder summarize mathematical resilience as four pragmatic aspects:

1. **Growth** — belief that mathematical capability can develop with effort and appropriate support (growth/incremental theory of learning).
2. **Value** — mathematics has personal / world value and the learner is valued in the learning community.
3. **Knowing how to work at mathematics** — struggle is expected; strategies exist for stuckness.
4. **Knowing how to recruit support** — peers, adults, resources; stuck is not permanent solitude.

FACT (Growth Zone Model): Johnston-Wilder et al. (2013 onward; e.g. pre-service teacher applications, Warwick WRAP) — three zones:

- **Comfort (green):** routine; little growth.
- **Growth (amber/yellow):** challenge requiring support, persistence, agency; productive struggle.
- **Danger / anxiety (red):** challenge perceived beyond capability even with support; survival response; repeated exposure → persistent anxiety / avoidance.

HYPOTHESIS (product): MindCraft's "soft-wrong + cards + tutor" stack is literally a **support-recruitment machine** for staying in the growth zone. Competitors that only escalate difficulty (or shame help) shove Maya into red.

Bridge to SAFE-DD (XLI): Desirable difficulties belong in the *growth* zone with equipment and destaking — not in the danger zone. "Struggle" ≠ "panic."

XLV.5 Measurement — MRS is attitudes, not magic scores

FACT: Kooken, Welsh, McCoach, Johnston-Wilder & Lee (2016, *Measurement and Evaluation in Counseling and Development*, 49(3), 217–242, doi:10.1177/0748175615596782) — developed and validated the **Mathematical Resilience Scale (MRS)** via EFA/CFA across three samples. Three correlated factors: **Value**, **Struggle**, and **Growth**. Positioned as a way to gauge likelihood of participation and persistence in mathematics.

Product implication (HYPOTHESIS): MRS-style items (value / struggle-as-normal / growth belief) can inform parent research and longitudinal identity panels — but they are **not** a North Star and not a gamified in-app grit meter. Self-report attitudes ≠ transfer. Pair with `retry_120s`, `challenge_accept`, `transfer_pass`, help-recruitment events (opened card / asked tutor / peer).

Kill: Shipping a "Resilience Score™" badge as proof of identity change (L0 → L3 leap; Part XXXIV hygiene).

XLV.6 Grit theater — what we refuse to sell

FACT (origin): Duckworth, Peterson, Matthews & Kelly (2007, *Journal of Personality and Social Psychology*, 92(6), 1087–1101, doi:10.1037/0022-3514.92.6.1087) — grit as perseverance and passion for long-term goals; predictive claims in selective samples fueled popular education discourse.

FACT (meta wound): Credé, Tynan & Harms (2017, *Journal of Personality and Social Psychology*, 113(3), 492–511, doi:10.1037/pspp0000102) — 584 effects, 88 samples, $N \approx 66{,}807$: higher-order grit structure not confirmed; grit **very strongly** overlaps conscientiousness; overall grit–performance/retention relations **modest**; perseverance-of-effort facet carries more criterion validity than consistency-of-interest; **interventions to enhance grit may have only weak effects**; construct validity questioned; primary utility may lie in perseverance facet alone.

FOUNDER BELIEF / resolution: Perseverance *after equipped struggle* remains commercially meaningful (`retry_120s`). Packaging that as Duckworth "grit" invites overclaim, personality fatalism, and equity harm (blame the child's character when curriculum/support failed). Prefer **mathematical resilience** language: growth zone + support recruitment + struggle-as-normal.

Standing Constitution already warned (core OS): grit/resilience hype is contested; structural inequality and curriculum quality matter more than character slogans. This chapter densifies that kill with Credé + JW/Lee specificity.

Kill: "MindCraft builds grit." **Survive (constrained):** "MindCraft keeps you in the growth zone — challenge with support, recovery without shame."

XLV.7 Competitive audit — who shoves students into the red zone?

Pattern	Zone reading	MindCraft wedge
Duolingo streak shame / league pressure	Danger + appearance climate (XLII/XLIII); "grit" via fear of break	Miss ≠ identity failure; cue practice not counter; no grit KPI
Khan mastery lock without affect design	Can strand in stuck without emotional exit tools	Soft-wrong + ingredient cards as support recruitment
Brilliant hard delight	Often growth-zone when curiosity high; weak for high-anxiety Maya	SAFE-DD equip/destake before delight escalation
Unguarded ChatGPT "just try harder / here's the answer"	Either skips struggle (comfort via answer) or fluent wrong confidence	PWC + solo transfer; support = pedagogy wrap not answer dump
School "resilience assembly"	Poster without recruit-support architecture	Operationalize support buttons in-session

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XLV.8 Resolution doctrine — SAFE-RESILIENCE

FOUNDER BELIEF / HYPOTHESIS — SAFE-RESILIENCE stack (ties FEI / SAFE-DD / TARGET / SAFE-REWARD / EVT value):

1. **Name the zones in product language.** Comfort / growth / danger — coach and tutor briefs use this, not "be gritty."
2. **Equip before escalate.** SAFE-DD first: representations, faded examples, destaked attempts — then raise challenge.
3. **Struggle is expected, not character failure.** Attribution copy (XXX): strategy and effort-with-strategy; never "you're not resilient enough."
4. **Support recruitment is a first-class action.** Cards, ask-tutor, peer, worked scaffold — log as positive resilience behavior, not as "cheating the streak."
5. **Exit the danger zone fast.** Anxiety spike → reduce stakes, shorten item, switch representation, offer support — do not double dose "desirable difficulty."
6. **Value without coercion.** Eccles utility/attainment (XXXVII) and story worlds — not "suffer because grit."
7. **Measure recovery and transfer, not grit scores.** `retry_120s`, help-then-solo sequences, `challenge_accept`, `transfer_pass` / `solo_transfer_pass`; optional MRS-like survey offline.
8. **Parent copy:** Sell "she got stuck, recruited help, and came back" — not "she's a gritty kid now."

Kill the false dichotomy: "Coddle forever in comfort" vs "forge character in the red zone." Refuse both. Ship **growth-zone challenge with support on tap**.

XLV.9 Marketing language that survives

- **Allowed:** "Hard math with a way out — support when you're stuck, challenge when you're ready."
- **Allowed:** "Stuck isn't failure. Stuck is when you recruit the next tool."
- **Allowed (parent):** "We train recovery and transfer — not grit slogans."
- **Banned:** "We build grit." / "Resilience score." / "Push through the panic."
- **Banned:** Bloom 2-sigma; empty mindset posters; streaks as proof of resilience; absolute tutoring-is-free.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without RES data
Growth-zone framing + support CTAs in coach	L1 hygiene aligned with JW/Lee
Soft-wrong + card recruitment → retry	L1 mechanism claim
"Raises mathematical resilience (MRS)" as product outcome	L0 until RES-1
"Builds grit / character"	L0 — banned packaging

XLV.10 Experiments spawned

ID	Question	Design	Primary	Kill condition

RES-1	Growth-zone coach copy + explicit support CTA vs grit/perseverance copy (same difficulty)	2-arm; anxiety-stratified	`retry_120s`; help-recruit rate; post affect; `transfer_pass`	Grit copy ≥ growth-zone on retry+transfer without raising danger affect → zone framing overstated
RES-2	Support-first after miss (card/tutor offer) vs no offer (same soft-wrong)	2-arm	Time-to-retry; danger-zone self-report; solo transfer later	No-offer ≥ support-first on retry without harm → recruitment less load-bearing
RES-3	Escalate difficulty after equip vs escalate without equip (SAFE-DD×RES)	2×2	Challenge_accept; WM/anxiety proxy; transfer	Unequipped escalate wins transfer in high-anxiety → SAFE-DD sequencing weak
RES-4	MRS Value/Struggle/Growth Δ over 4 weeks of SAFE-RESILIENCE UX vs control	Survey + behavior	MRS factors + behavioral NS	MRS rises, behavior flat → attitude theater
RES-QUAL	10 Maya interviews: “When you’re stuck in math, what does ‘being resilient’ mean?”	Appendix B	coded grit-vs-support meanings	Everyone already means support recruitment → copy shift lower priority

Pre-reg (XXXIV): RES-* identify **zone framing × support recruitment × recovery/transfer**. They do **not** identify “we create resilient mathematicians” from MRS alone.

Densifies: SAFE-DD; FEI; TARGET; standing grit-hype ban; help-seeking as virtue not streak cheat.

XLV.11 Confidence table

Claim	Label	Confidence
Developmental resilience often arises from ordinary protective systems	FACT (Masten, 2001)	High

Mathematical resilience = growth + value + how-to-work + recruit support	FACT (JW/Lee construct line)	High (construct); Medium (intervention effect sizes sparse)
Growth Zone Model: comfort / growth / danger	FACT (JW et al. model literature)	High as pedagogy model; Medium as RCT-proven UX
MRS measures Value, Struggle, Growth attitudes	FACT (Kooken et al., 2016)	High
Grit higher-order structure weak; overlaps conscientiousness; modest prediction; weak intervention leverage	FACT (Credé et al., 2017)	High
Danger-zone exposure as "training" harms anxious learners	HYPOTHESIS (Ashcraft line + SAFE-DD)	Medium–High
SAFE-RESILIENCE is right default for MindCraft	FOUNDER BELIEF / HYPOTHESIS	Medium–High
"Grit posters raise transfer" / "Resilience Score™ = identity"	SPECULATION / false as doctrine	Low (against)

XLV.12 What this chapter kills

1. **Kill:** Grit / "build character" as MindCraft marketing North Star. (Credé 2017; Constitution grit warning.)
2. **Kill:** Treating panic / danger-zone as desirable difficulty. (Growth Zone Model + SAFE-DD.)
3. **Kill:** Resilience as rare trait branding ("extraordinary kids"). (Masten ordinary magic.)
4. **Kill:** In-app Resilience Score™ as proof of identity transformation. (MRS ≠ transfer; claim ladder.)
5. **Wound:** Growth-mindset language alone — survives only inside growth + value + struggle know-how + **support recruitment**, not as wall art.
6. **Survive (constrained):** Growth-zone challenge with support on tap; struggle-as-normal; recovery metrics; parent story of "stuck → help → return."

Doctrine until data: Ship **SAFE-RESILIENCE**: name zones → equip before escalate → struggle expected → support recruitment first-class → exit danger fast → value without coercion → measure recovery/transfer not grit. Market "hard math with a way out."

Part XLVI — Teacher/Tutor Expectancy: Pygmalion Without Magic, Golem Without Labels

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 15; hour%6≠0 → Researcher; 5 researcher entries since synthesizer v1.6 → not yet Synthesizer)

Primary question: What do teacher/tutor expectancy effects actually justify for MindCraft — and what briefing ethics keep college tutors from installing a Golem on Maya?

Owners: Tutor ops / Product (briefing cards) · Equity · Positioning · Red Team

Commercial job: Replace "we believe in every student" poster copy and "Pygmalion IQ" folklore with a shippable **SAFE-EXPECTANCY** stack — evidence-based briefs, class-level high challenge + support, ban stigmatizing labels, measure mediation behaviors not vibes.

XLVI.1 Why this chapter exists

Parts XXV (self-efficacy), XXX (attribution), XXXVI (equity), XXXVII (EVT expectancy), and XLV (support recruitment) already treat *student-side* "can I?" and belonging. They do not fully answer: **what beliefs do human tutors carry into the session, how do those beliefs leak into climate/input/output/feedback, and how should MindCraft's pre-session briefing be designed so gap-scan weakness does not become a self-fulfilling low label?**

FOUNDER BELIEF under audit: A knowledge-graph + tutor hybrid only compounds identity if the *human* in the loop treats Maya as capable of the next hard step with support — not as a fixed "weak student" whose low mastery scores license colder climate, thinner input, and fewer chances to respond.

Claims we refuse as doctrine:

1. Marketing the classic Pygmalion IQ story as MindCraft's differentiator ("our tutors raise IQ by believing harder").
2. Briefing tutors with trait labels ("low ability," "slow," "not STEM material") drawn from gap-scan or weakness ranking.
3. Treating teacher-expectancy literature as proof that positive vibes alone move transfer.
4. Ignoring Golem / differential-treatment risks for stigmatized or "weakness-tagged" students.

XLVI.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused
Induced expectancy (Pygmalion paradigm)	Authority given false "bloomers" info → treats them differently → outcomes shift	Early-session / cold-start briefing window	Selling IQ magic; ignoring replication limits
Accuracy vs prophecy	Expectations often predict because they track real prior performance	Gap-scan / graph as *task* evidence, not character	Confusing prediction with causation
Golem effect	Low expectations → colder / more dogmatic treatment → worse performance	Ban negative trait labels in tutor cards	"Honest" weakness dump that licenses withdrawal

Four-factor mediation	Climate, input, output opportunity, feedback carry expectancy into behavior	Audit tutor/coach on those four channels	Soft climate without hard input/output
High-expectation teacher (class-level)	High challenge + high support for *all*, not only nominated stars	Tutor playbook: everyone gets hard Qs + scaffolds	Favorites / “bloomers only” tracks
SAFE-EXPECTANCY	Evidence brief → high challenge-with-support → no trait labels → mediate via CIOF → measure behaviors	See XLVI.7	Belief posters; Golem briefs

Operational definition (HYPOTHESIS): A MindCraft tutor session is expectancy-*safe* only when (a) the brief states **next playable tasks and scaffolds**, not trait judgments, (b) the tutor offers **hard-enough** work with support on tap (XLV growth zone), (c) climate stays warm under struggle, and (d) the student gets **output opportunity** (speak/solve before tutor finishes) — not when the tutor “believes harder” while finishing the math.

XLVI.3 Origin story — real, contested, not marketing copy

FACT: Rosenthal & Jacobson (1968, *Pygmalion in the Classroom*; also *Urban Review* summary, doi:10.1007/BF02322211) — teachers told that randomly selected elementary students were poised for intellectual blooming; those students later showed larger IQ gains on average in the reported analyses, especially in early grades. The study launched classroom expectancy research and intense methodological critique (e.g. Thorndike and later replication debates).

FACT (meta, induction credibility): Raudenbush (1984, *Journal of Educational Psychology*, 76(1), 85–97, doi:10.1037/0022-0663.76.1.85) — synthesis of 18 Pygmalion-style experiments: expectancy effects on pupil IQ are larger when teachers have had **little prior contact** with pupils at induction (≤~2 weeks); effects shrink when teachers already know students well. Grade pattern: stronger early-grade effects in the corpus; design features of induction matter.

Product translation: MindCraft’s **first tutor sessions and cold-start briefs** are the high-leverage expectancy window — analogous to early-year induction. Once a tutor has rich behavioral evidence, “random bloomers” folklore is the wrong metaphor; **accurate, task-level** expectations dominate.

Kill: “Pygmalion proves we raise IQ by believing.” Sell mediation design in the first hours of contact, not IQ alchemy.

XLVI.4 What 35 years actually say (Jussim & Harber)

FACT (review): Jussim & Harber (2005, *Personality and Social Psychology Review*, 9(2), 131–155, doi:10.1207/s15327957pspr0902_3) — after ~35 years:

1. Self-fulfilling prophecies in classrooms **do occur**, but effects are **typically small**, do not accumulate greatly across perceivers/time, and may dissipate more than snowball.

2. **Powerful** effects may selectively hit students from **stigmatized** social groups.
3. Whether expectancy effects reliably change *intelligence*, and whether they do more harm than good in general, remains unclear.
4. Teacher expectations often predict outcomes **because they are accurate**, not only because they are self-fulfilling.

HYPOTHESIS for MindCraft: Gap-scan / mastery graphs will *accurately* predict struggle. Accuracy is not permission to install a Golem. The commercial risk is **translating accurate weakness into low-challenge, low-warmth treatment** — exactly the path that can still produce small-but-real prophecy costs, amplified for already-stereotyped students (ties Part XXXVI equity).

Wound: "Expectancy effects are huge and ubiquitous." Survives only as: small average effects; context/stigma moderators; accuracy dominates prediction.

XLVI.5 Mediation — climate, input, output, feedback

FACT (behavioral mediation): Brophy & Good (1970, *Journal of Educational Psychology*, doi:10.1037/h0029908) — observational study: teachers demanded better performance and praised it more for high-expectation students; accepted poor performance more readily from lows and were less likely to praise good performance from lows when it occurred.

FACT (four-factor meta): Harris & Rosenthal (1985, *Psychological Bulletin*, 97(3), 363–386, doi:10.1037/0033-2909.97.3.363) — 31 meta-analyses of mediating behaviors; Rosenthal's four-factor frame (**climate**, **feedback**, **input**, **output**/opportunity opportunity) organizes how expectancies become treatment. Expectancy→behavior and behavior→outcome links support multiple mediators (e.g. warmer climate, longer/richer interaction, questioning). Later summaries often note **feedback** as the weaker of the four average links, pushing affect/effort emphasis.

Product checklist (HYPOTHESIS — CIOF audit):

Channel	Tutor/coach do	Anti-pattern
Climate	Warmth under struggle; curiosity about wrong paths	Cold distance after "weak" brief
Input	Richer explanations, multiple representations (cards)	Thin "just try" for lows; lectures only for highs
Output	Wait time; let Maya finish; SE prompts (Part XL)	Tutor finishes every thought (Solver crutch)
Feedback	Criterion + strategy (XXX); soft-wrong	Praise only highs; criticism without how-to

Kill: Climate-only positivity ("you've got this!") without input/output redesign — that is poster Pygmalion.

XLVI.6 Golem risk — negative expectancy is the ops hazard

FACT: Babad, Inbar & Rosenthal (1982, *Journal of Educational Psychology*, 74(4), 459–474, doi:10.1037/0022-0663.74.4.459) — distinguished positive expectancy outcomes ("Galatea") from **negative** ones ("Golem"). Teachers high in susceptibility to biasing information treated low-potential nominees more negatively (notably dogmatic behavior); unbiased teachers treated groups more equitably. Differential negative treatment showed up in teacher behavior **and** student task performance.

FACT (student-perceived differentiation): Brattesani, Weinstein & Marshall (1984, *Journal of Educational Psychology*, 76(2), 236–247, doi:10.1037/0022-0663.76.2.236) — in classrooms where students perceive **high differential treatment**, teacher expectations contribute more to achievement than in low-PDT classrooms. Students can *see* the track.

Product translation: A tutor card that says "Maya is weak on linear equations / low confidence / hard gap-scan" is biasing information. High-bias tutors will leak Golem through climate and dogmatism. Low-PDT design = same high standards + differentiated *scaffolds*, not differentiated *respect*.

Kill: "Honest labels help tutors calibrate." Honest **task evidence** helps; honest **trait verdicts** harm.

XLVI.7 High-expectation teachers — class-level, not favorites

FACT: Rubie-Davies (2007, *British Journal of Educational Psychology*, doi:10.1348/000709906X101601) — contrasts high- vs low-expectation *teachers* (expectations for the class as a whole). High-expectation teachers spent more time teaching, oriented lessons, linked prior knowledge, explained carefully, asked mixed open/closed questions including **high-level questions of all students**, and managed more positively; lows were more dismissive of incorrect answers and relied on closed questioning.

HYPOTHESIS: MindCraft's competitive wedge vs "AI that flatters then finishes" and vs "tutor who only drills the labeled weak topic gently" is **Rubie-Davies-shaped ops**: every student gets high-level questioning + scaffolds; incorrect answers get learning support, not dismissal.

Bridge to SAFE-RESILIENCE / SAFE-DD: High expectation ≠ danger zone. High expectation = growth-zone challenge with support recruitment — not grit theater and not thin input for "lows."

XLVI.8 Competitive audit (expectancy lens)

Competitor pattern	Expectancy mechanism	MindCraft counter
Duo	Habit/streak as "you're the type who shows up"; little academic expectancy mediation	Transfer + CIOF; no streak-as-belief
Khan	Self-serve library; weak human expectancy channel	Human tutor brief + graph without Golem labels
Brilliant	Prestige challenge; can cue performance-approach for novices	Mastery structures + expectancy-managed gates

ChatGPT Base	Sycophantic high-warmth climate + answer dump (fake high expectancy, zero output demand)	Warm climate **plus** forced solo output / PWC
Unbriefed college tutor	Forms expectancies from accent, race, “seems lost,” or raw weakness dump	SAFE-EXPECTANCY brief + playbook

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XLVI.9 Resolution doctrine — SAFE-EXPECTANCY

FOUNDER BELIEF / HYPOTHESIS — SAFE-EXPECTANCY stack:

1. **Brief tasks, not traits.** Tutor card: current concept, recommended level, formats struggled, successful scaffolds last time, next playable question IDs — **never** “low ability / not a math person.”
2. **Protect the cold-start window.** First sessions: explicit high-challenge-with-support script (Raudenbush induction logic).
3. **Class-level high expectation.** Rubie-Davies: hard questions for everyone; wrong answers → teach, don’t dismiss.
4. **Mediate via CIOF.** Climate warmth + rich input (cards) + output opportunity (wait / SE) + strategy feedback — audit these, not “belief intensity.”
5. **Accuracy without Golem.** Graph weakness drives *scaffold choice*, not *respect withdrawal*.
6. **Stigma premium.** Extra review when briefs could activate stereotype (XXXVI); ban demographic proxies in tutor UI.
7. **AI coach inherits the same rules.** No “you’re behind your peers” cold open; no finishing her thought under “support.”
8. **Measure mediation + FEI, not IQ folklore.** Tutor wait-time, student talk ratio, help-then-solo, `retry_120s`, `transfer_pass` / `solo_transfer_pass`.

Kill the false dichotomy: “Ignore data, just believe” vs “dump the weakness label so tutors are realistic.” Refuse both. Ship **evidence-based high challenge with equitable treatment**.

XLVI.10 Marketing language that survives

- **Allowed:** “Tutors see what to teach next — not who she is forever.”
- **Allowed:** “High challenge for everyone, support on tap — no tracks of respect.”
- **Allowed (parent):** “We brief tutors on the next skill, not on a fixed ability story.”
- **Banned:** “Pygmalion IQ gains.” / “Believe and they bloom.” / “We tell tutors who’s weak so they know.”
- **Banned:** Bloom 2-sigma; empty mindset posters; streaks as expectancy proof; tutoring-is-free.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without EXP data

Trait-label ban + task-only tutor brief	L1 hygiene aligned with Golem literature
CIOF playbook / wait-time coaching	L1 mechanism claim
“Raises IQ via tutor expectancy”	L0 — banned
“Eliminates achievement gaps via Pygmalion”	L0 until EXP-* + equity design (Jussim caveats)

XLVI.11 Experiments spawned

ID	Question	Design	Primary	Kill condition
EXP-1	Task-only brief vs trait/weakness -label brief (same content otherwise)	2-arm tutors; record session	Student talk ratio; wait-time; post affect; `transfer_pass`	Label brief ≥ task-only on transfer without colder climate → Golem risk overstated for our tutors
EXP-2	High-expectation playbook (hard Q + scaffold) vs “remediate gently only” after same gap-scan	2-arm	`challenge_accept`; solo transfer; help-recruit	Gentle-only wins transfer+return in high-anxiety → Rubie-Davies transfer fails for Maya
EXP-3	Cold-start script (first 2 sessions) vs unstructured tutor open	2-arm new matches	FEI events week 1; retention to session 3	Unstructured ≥ scripted → induction window less load-bearing
EXP-4	AI coach: warm+finish vs warm+output-demand (SE/wait)	2-arm	`solo_transfer_pass`; PWC	Warm+finish ≥ warm+output on solo transfer → climate-only expectancy myth revived
EXP-QUAL	10 tutor + 10 Maya: “What did the brief make you assume?”	Interview	coded trait vs task inferences	Tutors already ignore labels → UI priority shifts

Pre-reg (XXXIV): EXP-* identify **brief content** × **CIOF mediation** × **transfer**. They do **not** identify “we create bloomers” or IQ change.

Densifies: Equity (XXXVI); EVT expectancy; SAFE-RESILIENCE support; FEI; tutor ops queue item 68.

XLVI.12 Confidence table

Claim	Label	Confidence
Rosenthal & Jacobson launched expectancy research; original IQ story is contested	FACT	High
Induced expectancy effects larger when teacher knows student less (Raudenbush, 1984)	FACT	High
Average self-fulfilling effects small; accuracy often drives prediction; stigma can amplify (Jussim & Harber, 2005)	FACT	High
Differential treatment observable; highs get more demand/praise (Brophy & Good, 1970)	FACT	High
Four-factor mediation frame useful (Harris & Rosenthal, 1985)	FACT	High (framework); Medium (which channel dominates in tutoring apps)
Golem / high-bias negative treatment real (Babad et al., 1982)	FACT	High
High-expectation teachers differ in practice for whole class (Rubie-Davies, 2007)	FACT	High (observational/descriptive line); Medium (causal ops transfer)
SAFE-EXPECTANCY is right default for MindCraft tutor briefing	FOUNDER BELIEF / HYPOTHESIS	Medium–High
“Pygmalion IQ” / belief posters as product claim	SPECULATION / false as doctrine	Low (against)

XLVI.13 What this chapter kills

1. **Kill:** Marketing classic Pygmalion IQ blooming as MindCraft proof. (Raudenbush limits; Jussim accuracy; claim ladder.)
2. **Kill:** Tutor briefs that install trait/weakness identity labels. (Babad Golem; Weinstein PDT.)
3. **Kill:** Climate-only positivity without input/output demand. (Harris & Rosenthal mediation; Bastani/PWC line.)
4. **Kill:** “Just be realistic — tell tutors she’s weak” as ops hygiene.
5. **Wound:** Expectancy as a large, cumulative social-problem engine — survives only as small average effects with stigma/PDT moderators.
6. **Survive (constrained):** Cold-start high-challenge-with-support; task-only briefs; CIOF audit; Rubie-Davies whole-class high expectation; measure mediation + transfer.

Doctrine until data: Ship **SAFE-EXPECTANCY**: brief the next skill, not the fixed self; high challenge for everyone with scaffolds; warm climate under struggle; force student output; never weaponize gap-scan as a Golem tag. Market "tutors see what to teach next — not who she is forever."

Part XLVII — Sleep, Stress, and Learning: Timing Practice Without All-Nighter Theater

Chapter status: Living evidence brief — Researcher tick 2026-07-30 (UTC hour 18 ≡ Red Team slot, but ch47 never written → prefer Researcher per rotation; 6 researcher entries since synthesizer v1.6 → not yet Synthesizer)

Primary question: How should MindCraft time practice and exam-week product behavior given sleep-dependent consolidation and stress/pressure drain on working memory — without becoming a sleep-coaching app or glorifying late-night grind?

Owners: Product (session timing / exam modes) · Parent messaging · Affect pipeline · Red Team

Commercial job: Ship a **SAFE-TIMING** stack — overnight spacing as consolidation infrastructure, exam-week load softening, ban streak/all-nighter heroism, measure delayed transfer not midnight minutes.

XLVII.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXIV/XLI (anxiety × desirable difficulty), XXXI (flow/challenge–skill), and XLV (growth vs danger zone) already treat *within-session* difficulty and affect. They do not fully answer: **when** should Maya practice relative to sleep, and what should the product do in high-stakes stress weeks when encoding, attention, and WM are compromised?

FOUNDER BELIEF under audit: Identity transformation requires recoverable struggle *and* biological conditions that let struggle stick. A product that pushes "one more mission" at 1 a.m. during exam week can look engaged while sabotaging consolidation and installing danger-zone affect.

Claims we refuse as doctrine:

1. Marketing MindCraft as a sleep-hygiene / wellness brand ("we fix teen sleep").
2. Celebrating all-nighters, midnight streaks, or "grind hours" as learning proof.
3. Treating acute exam pressure as desirable difficulty.
4. Absolute claims that "sleep always consolidates math" independent of encoding quality, task type, and sleep opportunity.

XLVII.2 Constructs (product language)

Construct	Plain definition	Product signal	Failure mode if misused

Sleep-dependent consolidation	Offline reprocessing after encoding that stabilizes/reshapes memory	Space practice across nights; delay transfer checks	Sell “sleep = magic” without encoding first
Encoding under restriction	Multi-night short sleep impairs *new* learning capacity	Soften new-concept loads after chronic short sleep flags	Blame student “laziness” for restriction biology
Pressure / choke	High-stakes appraisal consumes WM needed for hard math	Destake practice; hide peer ranks in exam week	Treat choke as low ability
Math-anxiety dual-task	Worry acts like a secondary load on WM	SAFE-DD before hard items	Add timers/leaderboards under anxiety
SAFE-TIMING	Time practice for consolidation + protect WM under stress	See XLVII.8	Sleep-app feature creep; grind theater

Operational definition (HYPOTHESIS): A MindCraft practice plan is timing-*safe* when (a) hard encoding sessions are followed by a real sleep opportunity before high-stakes transfer claims, (b) multi-night sleep debt / high stress triggers **load softening** (fewer new concepts, more retrieval of known, support on tap), and (c) engagement metrics never reward post-midnight volume as mastery.

XLVII.3 Sleep consolidates — and evolves — memory

FACT (systems review): Diekelmann & Born (2010, *Nature Reviews Neuroscience*, 11, 114–126, doi:10.1038/nrn2762) — sleep optimizes consolidation depending on learning conditions and sleep timing; slow-wave sleep (SWS) and REM support complementary system- vs synaptic-consolidation roles; consolidation can change representations quantitatively *and* qualitatively.

FACT (broader offline processing): Stickgold (2005, *Nature*, 437, 1272–1278, doi:10.1038/nature04286) — converging evidence that offline memory reprocessing during sleep is a major component of how memories are formed and shaped (not a single all-or-none event).

FACT (integrative/schema angle): Walker & Stickgold (2010, *Nature Reviews Neuroscience*, 11, 218, doi:10.1038/nrn2762-c1) — beyond veridical preservation, sleep (often REM-linked in their framing) supports unitization, assimilation into remote networks, and abstraction of rules/schemas — relevant to MindCraft’s transfer goal, not only verbatim item memory.

FACT (mechanism exemplar): Gais & Born (2004, *PNAS*, 101(7), 2140–2144, doi:10.1073/pnas.0305404101) — elevating acetylcholine during SWS-rich sleep blocked declarative word-pair consolidation while sparing a nondeclarative skill task — evidence that the *neurochemical milieu* of SWS matters for declarative consolidation, not “rest” generically.

Product translation: Spacing missions across nights (Part XXIX) is not only spacing theory — it is a **consolidation window**. Claiming “exam-ready” from a same-day blocked accuracy binge without overnight delay is scientifically lazy (ties XXXIX readiness signal).

Kill: “Sleep replaces practice.” Consolidation operates on what was encoded. No encoding → nothing to consolidate.

XLVII.4 Adolescent sleep restriction is causal for learning damage

Maya is a high-schooler. Population sleep debt is not a vibe; experimental restriction shows classroom-relevant harm.

FACT (simulated classroom): Beebe, Rose & Amin (2010, *Journal of Adolescent Health*, 47(5), 523–525, doi:10.1016/j.jadohealth.2010.03.005) — within-subject multi-night restriction vs healthy duration: sleep-restricted adolescents showed poorer learning, more inattentive behaviors, and lower arousal in a simulated classroom (medium-to-large effects; small *n*, preliminary but experimental).

FACT (review): Beebe (2011, *Pediatric Clinics of North America*, doi:10.1016/j.pcl.2011.03.002) — synthesizes cognitive, behavioral, and functional consequences of inadequate sleep in children/adolescents; chronic restriction is common and not benign.

FACT (long-term factual retention): Cousins, Wong & Chee (2019, *Journal of Adolescent Health*, 65(4), 549–557, doi:10.1016/j.jadohealth.2019.04.030) — adolescents (15–18) with 5 h sleep opportunity for 5 nights vs 9 h controls: after four restriction nights, learning detailed factual material showed **more forgetting** at 30 min (~26%), Day 3 (~34%), and Day 42 (~65%) relative to controls. Vigilance also fell; memory impairment did not simply track vigilance correlation.

HYPOTHESIS for MindCraft: A “productive” ACT cram week of late missions on short sleep can **look** like engagement while destroying the encoding that gap-scan and practice are supposed to build. Parent-visible “minutes practiced” without sleep context is a vanity metric.

Wound: “Weekend catch-up fixes the week.” Literature flags incomplete restitution and cumulative cost (see Beebe 2019-line summaries); do not market weekend binge as full repair.

XLVII.5 Stress and pressure — WM theft, not character failure

Sleep loss and evaluative stress often co-occur in exam weeks. Separate the mechanisms.

FACT (math anxiety × WM): Ashcraft & Krause (2007, *Psychonomic Bulletin & Review*, 14, 243–248, doi:10.3758/BF03194059) — math performance depends on WM for beyond-retrieval arithmetic; high math anxiety behaves like a dual-task (worry consumes capacity); anxious students also avoid math pathways. Complements Ashcraft & Kirk (2001) line already in Part XXIV.

FACT (choking under pressure): Beilock & Carr (2005, *Psychological Science*, 16(2), 101–105, doi:10.1111/j.0956-7976.2005.00789.x) — performance pressure harmed **high**-WM individuals on the most WM-demanding math problems — pressure consumes the capacity their superior strategies rely on. Not “dumb under stress”; often “capable under load.”

Bridge to SAFE-DD / FEI: Exam-week timers, public ranks (XLII), and “prove you’re ready tonight” framing are pressure injectors. Desirable difficulty assumes encoding capacity; pressure + anxiety can erase the dose.

HYPOTHESIS: Affective check-in stress > 0.7 already softens `target_mastery` in `/recommend` — that is the correct *direction*. Extend it: under high stress / self-reported short sleep, prefer retrieval + support recruitment over new-concept flooding and appearance UX.

Kill: "Pressure makes diamonds" as practice design. Pressure makes WM poorer for the problems that need it most.

XLVII.6 Competitive audit (timing lens)

Competitor pattern	Timing / stress mechanism	MindCraft counter
Duo	Streaks + late-night nagging; habit over consolidation	SAFE-HABIT + SAFE-TIMING: miss≠identity; no midnight streak heroism
Khan	Infinite library; learner can cram to exhaustion	Spaced missions + delayed transfer as readiness
Brilliant	Delight / prestige sessions; can invite performance pressure	Mastery climate; destake under stress
ChatGPT Base	Always-on 2 a.m. answers; no consolidation pedagogy	Human/AI wrap that schedules *struggle then sleep*, not answer dumps
Tutor shops	Pack hours before exam Saturday	Brief tutors: encode early in week; retrieval near exam; never shame sleep

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XLVII.7 Exam weeks — special product mode, not harder grind

SPECULATION → product hypothesis: Default "more missions" during ACT week maximizes anxiety and restriction. Safer mode:

1. **Front-load encoding** earlier in the week when sleep opportunity is better.
2. **Shift to retrieval + mixed review** 48–72 h pre-exam (XXIX/XXXIX).
3. **Lower appearance cues** (no leaderboards, no streak threats).
4. **Keep growth-zone items** but exit danger fast (XLV); prefer scaffolds over new ontology leaps.
5. **Parent copy:** sleep is part of the study plan — not "rest is for quitters."

FOUNDER BELIEF: Selling calm competence under exam load beats selling heroic sleep debt.

XLVII.8 Resolution doctrine — SAFE-TIMING

FOUNDER BELIEF / HYPOTHESIS — SAFE-TIMING stack:

1. **Encode → sleep → transfer claim.** Treat overnight delay as part of readiness (with XXIX spacing), not optional wellness.
2. **Never reward midnight volume.** Ban streak/XP incentives that spike after local quiet hours; engagement ≠ consolidation.
3. **Restriction-aware load.** If multi-night short sleep or high stress is signaled (check-in, parent note, exam-week flag): fewer new concepts, more retrieval, CIOF support (XLVI), SAFE-DD destake.
4. **Pressure ≠ DD.** Timers, ranks, and "prove it tonight" are kill switches under high anxiety/stress.
5. **Exam-week mode.** Soften novelty; maximize recoverable retrieval; protect sleep opportunity messaging.
6. **Do not become a sleep clinic.** Nudge timing and load — do not diagnose disorders or sell melatonin theater.
7. **Tutor briefs include timing.** "Practice early evening when possible; protect sleep before transfer checks" — task ops, not character judgment.
8. **Measure delayed `transfer_pass` / `solo_transfer_pass`,** not minutes-after-midnight.

Kill the false dichotomy: "Ignore biology, just grind" vs "pivot the company into sleep coaching." Refuse both. Ship **timing-aware learning load**.

XLVII.9 Marketing language that survives

- **Allowed:** "Hard practice sticks better when she sleeps on it."
- **Allowed:** "We don't measure mastery at 1 a.m."
- **Allowed (parent):** "Exam week ≠ all-nighter week — we shift to review and protect sleep."
- **Banned:** "Our app fixes teen sleep." / "Grind while they sleep." / Streak-as-study-ethic.
- **Banned:** Bloom 2-sigma; empty mindset; tutoring-is-free; pressure-as-pedagogy.

Feature claim ladder (Part XXXIV hygiene)

Feature	Claim max without TIM-* data
Overnight spacing before transfer checks	L1 mechanism aligned with consolidation reviews
Exam-week load softening + destake	L1 product hygiene under Ashcraft/Beilock
"Sleep mode raises ACT scores"	L0–L1 only with pre-reg; not L3 identity
Sleep-clinic / wellness brand claims	L0 — banned

XLVII.10 Experiments spawned

ID	Question	Design	Primary	Kill condition

TIM-1	Same content: practice ending ≥1 h before habitual bedtime vs late-night block; transfer next evening	2-arm	next-day / 48h `transfer_pass`	Late ≥ early → consolidation timing overstated for our items
TIM-2	Exam-week mode (retrieval+ destake) vs “more new missions” in last 5 days pre-mock	2-arm	mock score; affect; sleep opportunity self-report	More-new wins mock+return → softening wrong default
TIM-3	High-stress check-in → auto soft path vs ignore stress	2-arm	`retry_120s`; WM-demand item accuracy; return	Ignore ≥ soft on transfer without affect cost → modifier useless
TIM-4	Parent message: “sleep is study” vs “push one more hour”	message A/B	next-night session timing; parent WTP/trust	Push-hour increases volume *and* next-day transfer → grind narrative survives
TIM-QUAL	10 Maya exam-week diaries: when practice helped vs harmed	interview	coded sleep debt × danger zone × cram	Students already self-time well → UI priority drops

Pre-reg (XXXIV): TIM-* identify **timing × load × delayed transfer**. They do **not** identify clinical sleep treatment or ACT score miracles from a bedtime tip.

Densifies: XXIX spacing; XLI SAFE-DD; XLII compare ban; XLIII habit without colonization; XLV danger exit; affect check-in; Part LXIII exam-pressure queue.

XLVII.11 Confidence table

Claim	Label	Confidence
Sleep supports declarative/procedural consolidation with stage-sensitive mechanisms (Diekelmann & Born, 2010; Stickgold, 2005)	FACT	High

Sleep can support schema/abstraction beyond verbatim storage (Walker & Stickgold, 2010)	FACT	Medium–High (active research debate on REM vs SWS mapping)
Low ACh during SWS matters for some declarative consolidation (Gais & Born, 2004)	FACT	High (specific manipulation); Medium (product generality)
Multi-night adolescent restriction impairs classroom learning/attention (Beebe et al., 2010; Beebe, 2011)	FACT	High (direction); Medium (effect sizes / sample limits)
Restriction impairs long-term factual retention in teens (Cousins et al., 2019)	FACT	High
Math anxiety consumes WM (Ashcraft & Krause, 2007)	FACT	High
Pressure can choke high-WM math performance (Beilock & Carr, 2005)	FACT	High
SAFE-TIMING is right default for MindCraft scheduling/exam mode	FOUNDER BELIEF / HYPOTHESIS	Medium–High
MindCraft-as-sleep-app is a good growth wedge	SPECULATION / false as doctrine	Low (against)

XLVII.12 What this chapter kills

1. **Kill:** All-nighter / midnight-streak heroism as learning culture. (Cousins retention; Beebe classroom; SAFE-HABIT.)
2. **Kill:** Treating exam pressure and timers as desirable difficulty. (Beilock choke; Ashcraft dual-task; SAFE-DD.)
3. **Kill:** Same-day cram accuracy as exam-ready proof without overnight/delayed transfer. (Diekelmann/Born; XXIX/XXXIX.)
4. **Kill:** Pivoting brand into clinical sleep wellness.
5. **Wound:** "Sleep always helps every math skill equally." Survives only as: condition- and task-dependent consolidation; encoding first.
6. **Survive (constrained):** Encode⟶sleep⟶claim; restriction/stress-aware load; exam-week retrieval mode; parent "sleep is study"; measure delayed transfer.

Doctrine until data: Ship **SAFE-TIMING**: practice hard enough to encode, sleep enough to consolidate, soften novelty under debt and pressure, never sell grind-at-1-a.m. as identity. Market "hard practice sticks better when she sleeps on it."